CS-663
Digital Image Processing (DIP)

Principal Component Analysis

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Background Expected

- Linear Algebra (MA 106)
  - Vectors
  - Basis
  - Matrices:
    real,
    orthogonal,
    symmetric,
    spectral theorem,
    quadratic forms,
    determinants
  - Eigen decomposition

Description	Vectors in R^n, linear independence and dependence, linear span of a set of vectors, vector subspaces of R^n , basis of a vector subspace. Systems of linear equations, matrices and Gauss elimination, row space, null space, and column space, rank of a matrix. Determinants and rank of a matrix in terms of determinants. Abstract vector spaces, linear transformations, matrix of a linear transformation, change of basis and similarity, rank-nullity theorem. Inner pro duct spaces, Gram-Schmidt process, orthonormal bases, projections and least squares approximation. Eigenvalues and eigenvectors, characteristic polynomials, eigenvalues of special matrices (orthogonal, unitary, hermitian, symmetric, skew-symmetric, normal), algebraic and geometric multiplicity, diagonalization by similarity transformations, spectral theorem for real symmetric matrices, application to quadratic forms.
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Background Expected

- Data Analysis and Interpretation (CS 215)
  - Singular value decomposition (SVD)
  - Vector random variables
  - Transformation of random variables
  - Covariance matrix and its properties
  - Multivariate Gaussian
  - Maximum-likelihood estimation

Background Expected

- Artificial Intelligence and Machine Learning (CS 217)
  - Supervised learning
  - Regression
  - Classification
  - Clustering
  - Bayes theorem
  - Regularization
  - Cross validation
  - Neural networks

Description	(1) Search: Uninformed search, A* search, adversarial search, local search.(2) Planning: Markov Decision Problems, Value Iteration and Policy Iteration.(3) Probabilistic reasoning: Bayes nets, conditional independence, exact and approximate inference.(4) Supervised learning: Linear methods for classification and regression, regularisation, cross-validation, decision trees, neural networks, ensemble methods.(5) Unsupervised leaming: k-means clustering.(6) Selected topics from natural language processing, robotics, computer vision, multi-agent systems
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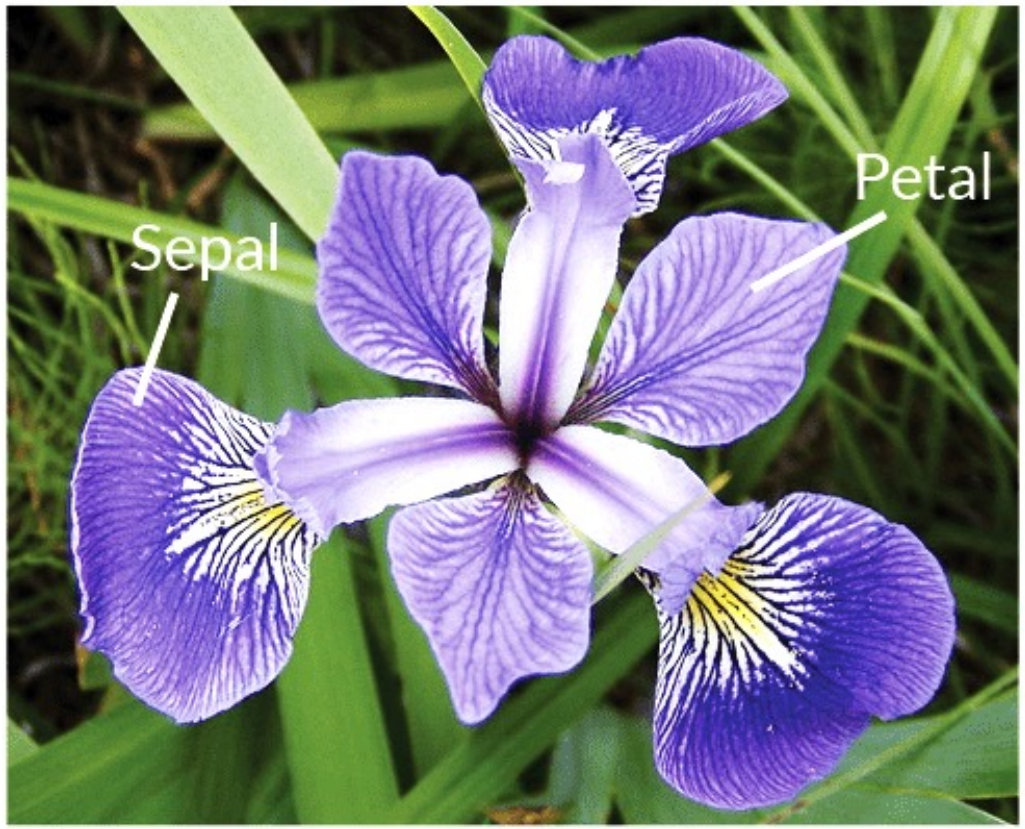
Principal Component Analysis (PCA)

- Iris plant: 3 different species
- Iris flower: 4 different measurements – length & width of sepal & petal



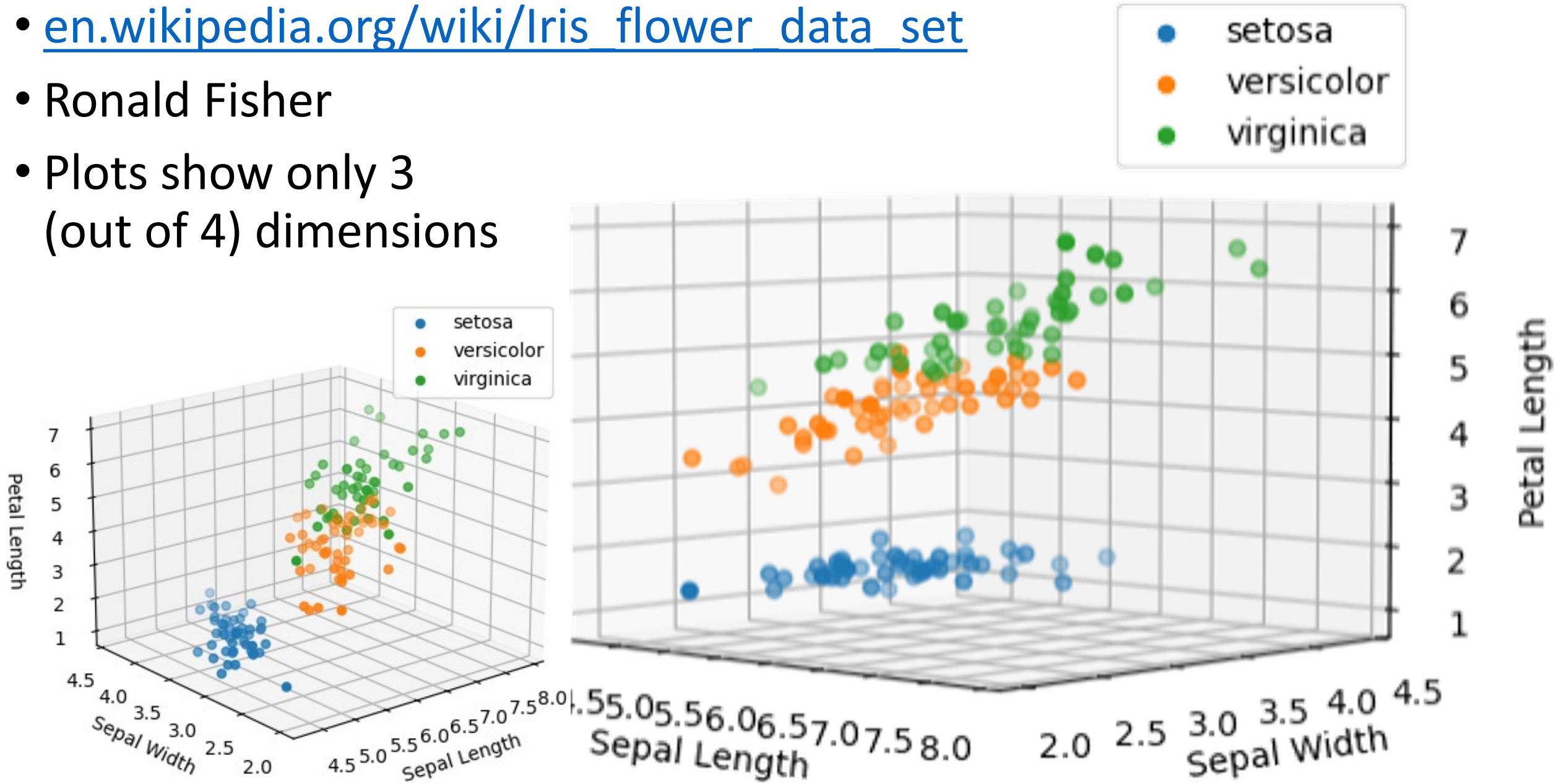
Iris Versicolor

Iris Setosa

Iris Virginica

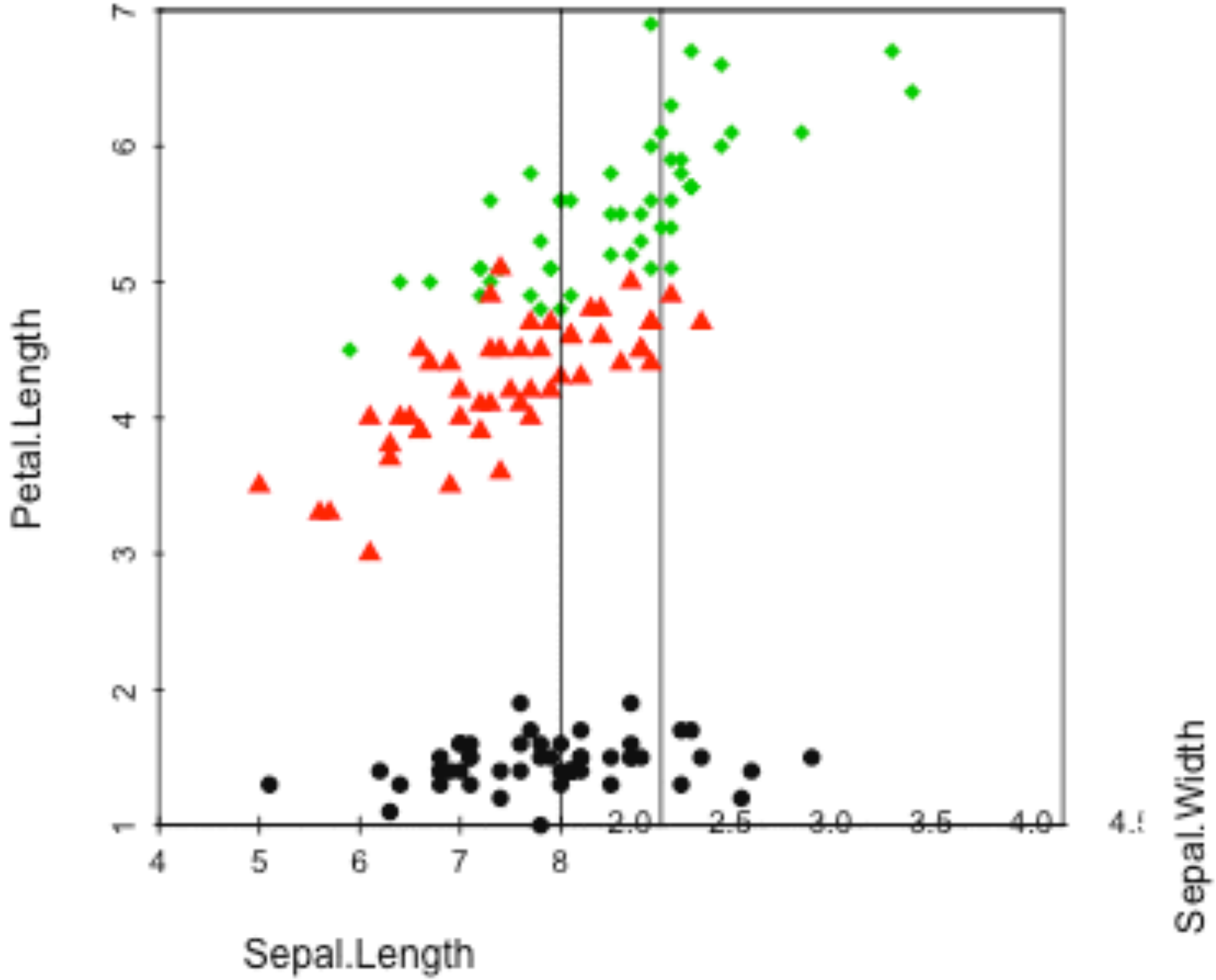
Principal Component Analysis (PCA)

- en.wikipedia.org/wiki/Iris_flower_data_set
- Ronald Fisher
- Plots show only 3 (out of 4) dimensions



PCA

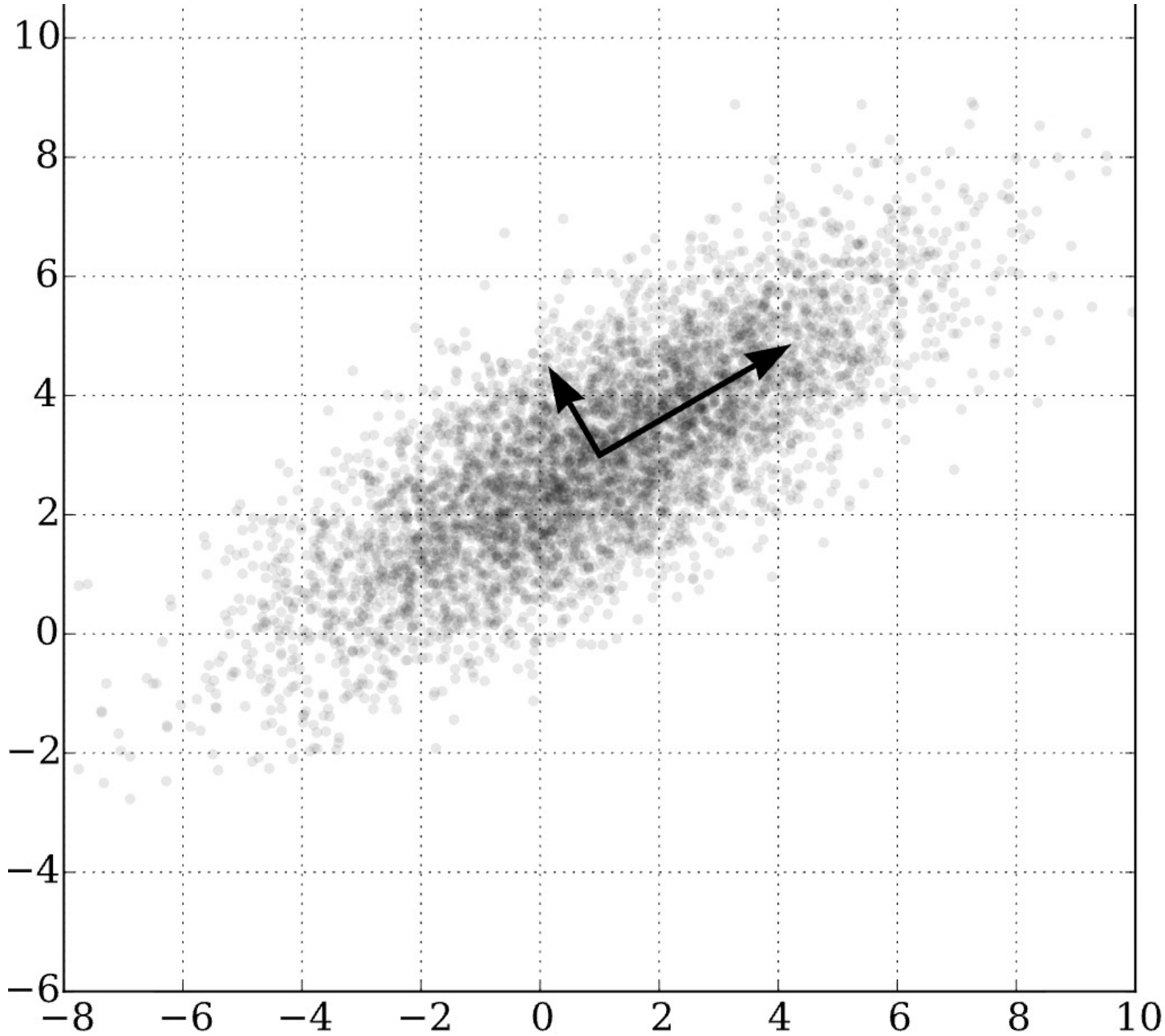
- Actual data is 4D
- Only 3 dimensions displayed



Principal Component Analysis (PCA)

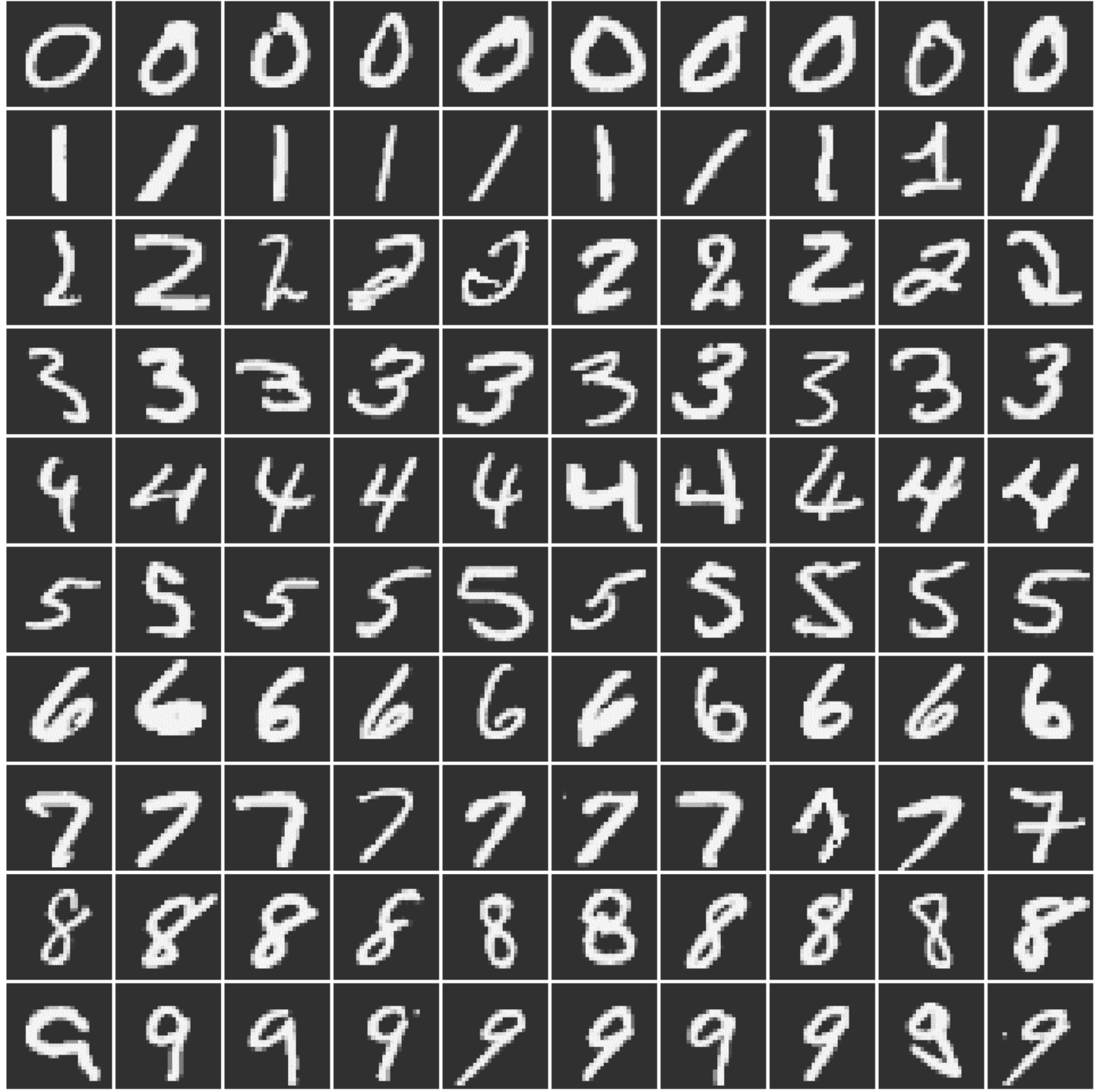
- Data scatter plot;
  can be N-dimensional

1) Estimate "mean"
2) Estimate
   N independent
   "components (directions)
   of variability"
   around mean
3) Estimate
   "amount of variability"
   along each of N
   components/directions



PCA

- For each digit, find principal modes of variation across set of images
- en.wikipedia.org/wiki/MNIST_database
- Grayscale handwritten digits
- Modified NIST database
  - National Institute of Standards and Technology (USA)



Multivariate Gaussian – Definition

- Consider a vector random variable $X := [X_1; X_2; \ldots; X_D]$
  - Column vector of length D

Definition: The RV X has a multivariate (jointly) Gaussian PDF if $\exists$ a finite set of i.i.d. univariate standard-normal RVs $W_1, \cdots, W_N$ (with $D \leq N$) such that each X_d can be expressed as $X_d = \mu_d + \sum_n A_{dn} W_n$ (i.e., $X = AW + \mu$).

Multivariate Gaussian – Identity A

- Consider a vector random variable $X := [X_1; X_2; \ldots; X_D]$
  - Column vector of length D
- Model $X := A\,W + \mu$
- A = identity
- μ = zero

• Example 1 (Zero-Mean + Isotropic / Spherical Gaussian): The case of independent standard-normal RVs $W_1, \cdots, W_D$ with $A := I_{D\times D}$ and $\mu := 0$, i.e. $X = W$

Then, the Gaussian PDF is $p(w) = \prod_d p(w_d) = \frac{1}{(2\pi)^{D/2}} \exp(-0.5 w^\top w)$

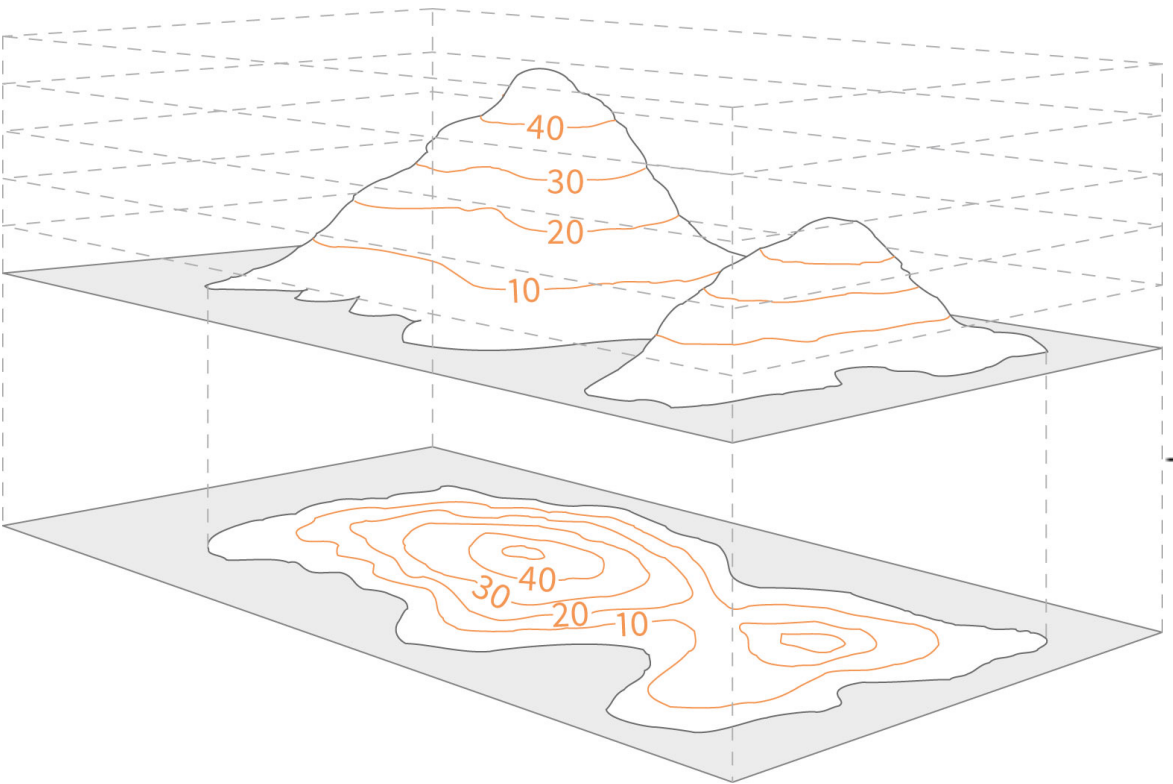
Multivariate Gaussian – Identity A

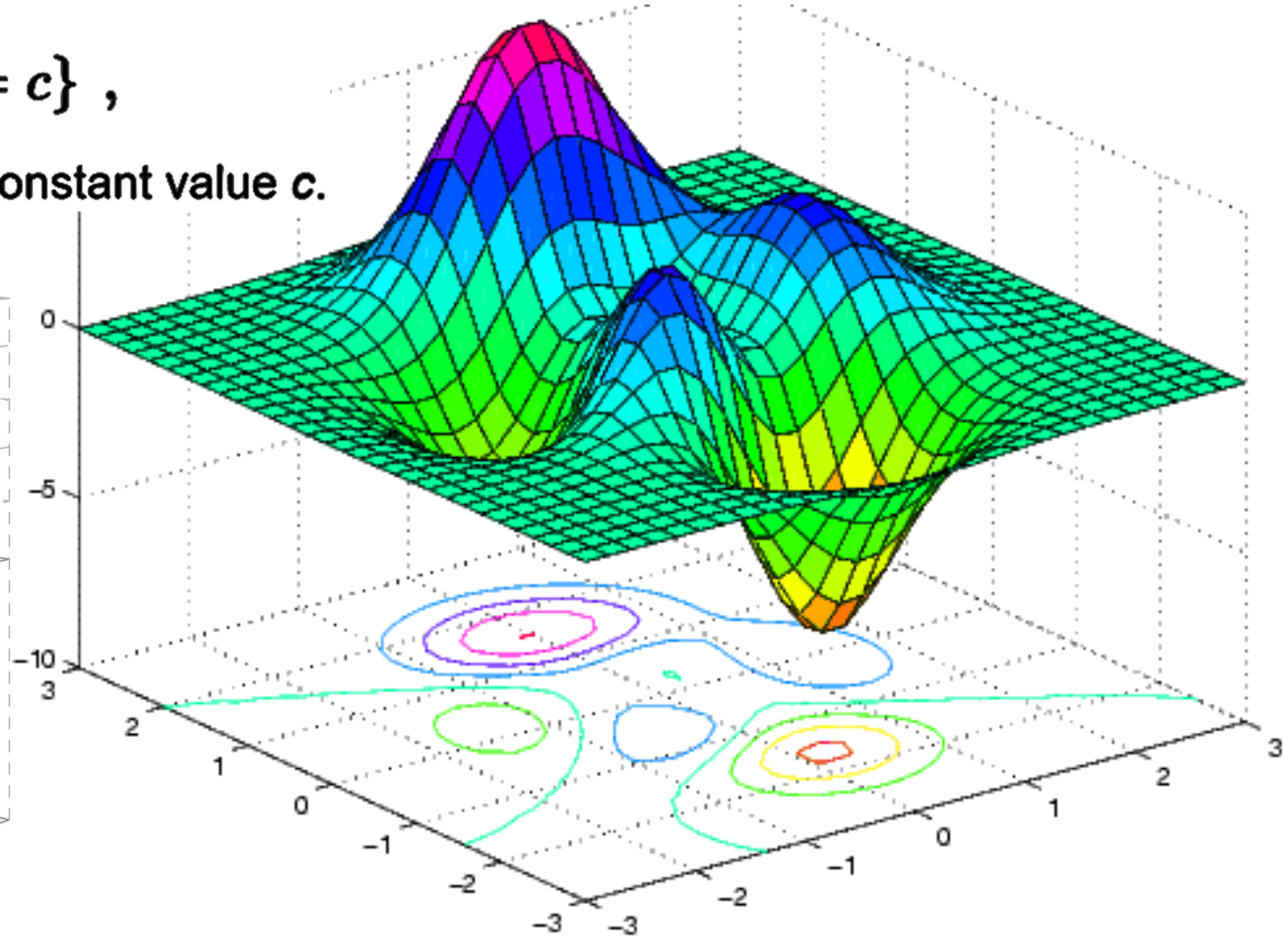
- What are the contours/level sets of the PDF ?

In mathematics, a **level set** of a real-valued function *f* of *n* real variables is a set of the form

$$L_c(f) = \{(x_1, \cdots, x_n) \mid f(x_1, \cdots, x_n) = c\} ,$$

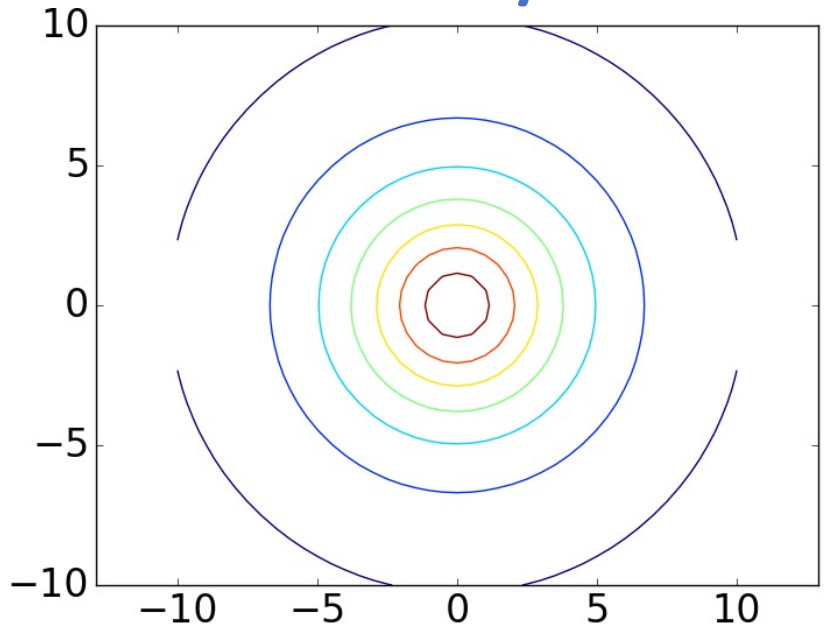
that is, a set where the function takes on a given constant value *c*.

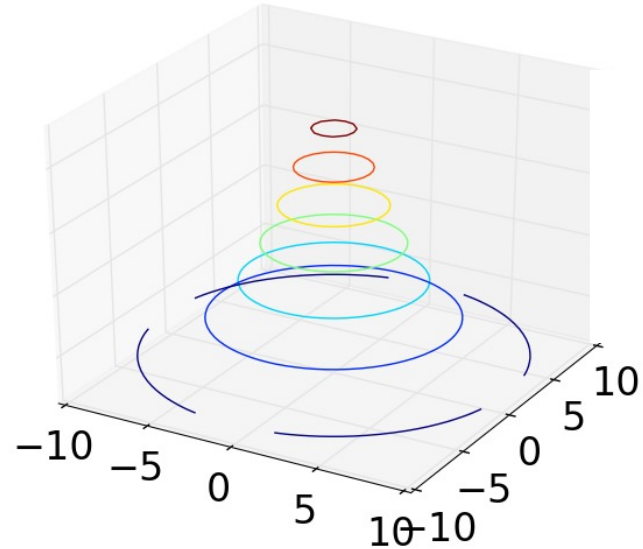




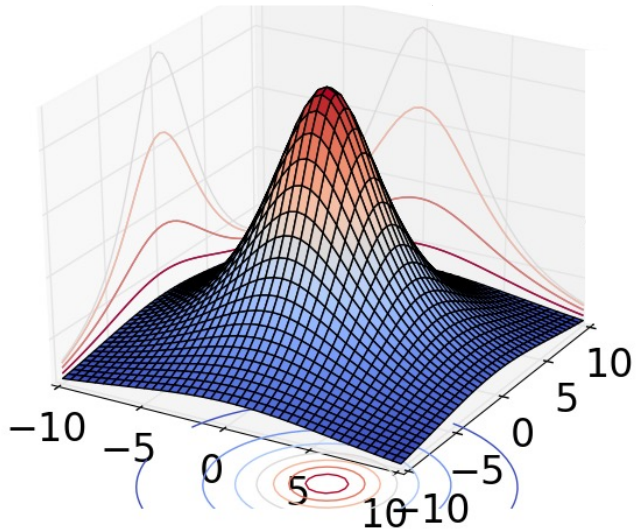
Multivariate Gaussian – Identity A

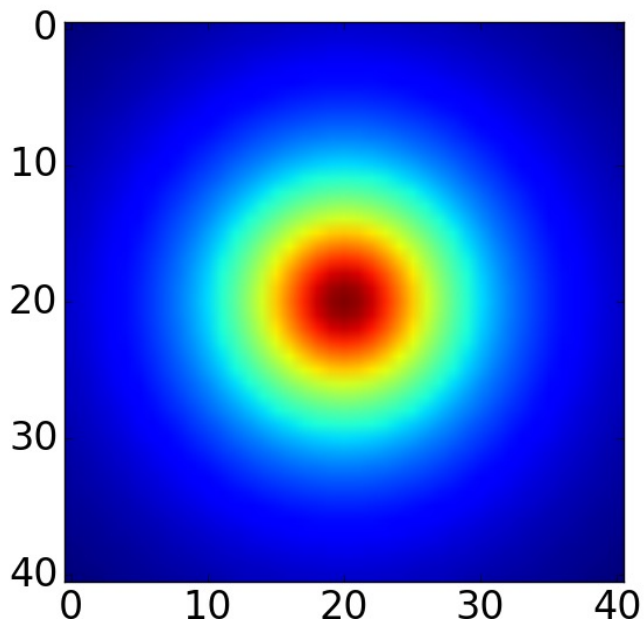
- Isotropic / spherical multivariate Gaussian
  - Contours / Level sets





$$p(w) = \prod_d p(w_d) = \frac{1}{(2\pi)^{D/2}} \exp(-0.5 w^\top w)$$

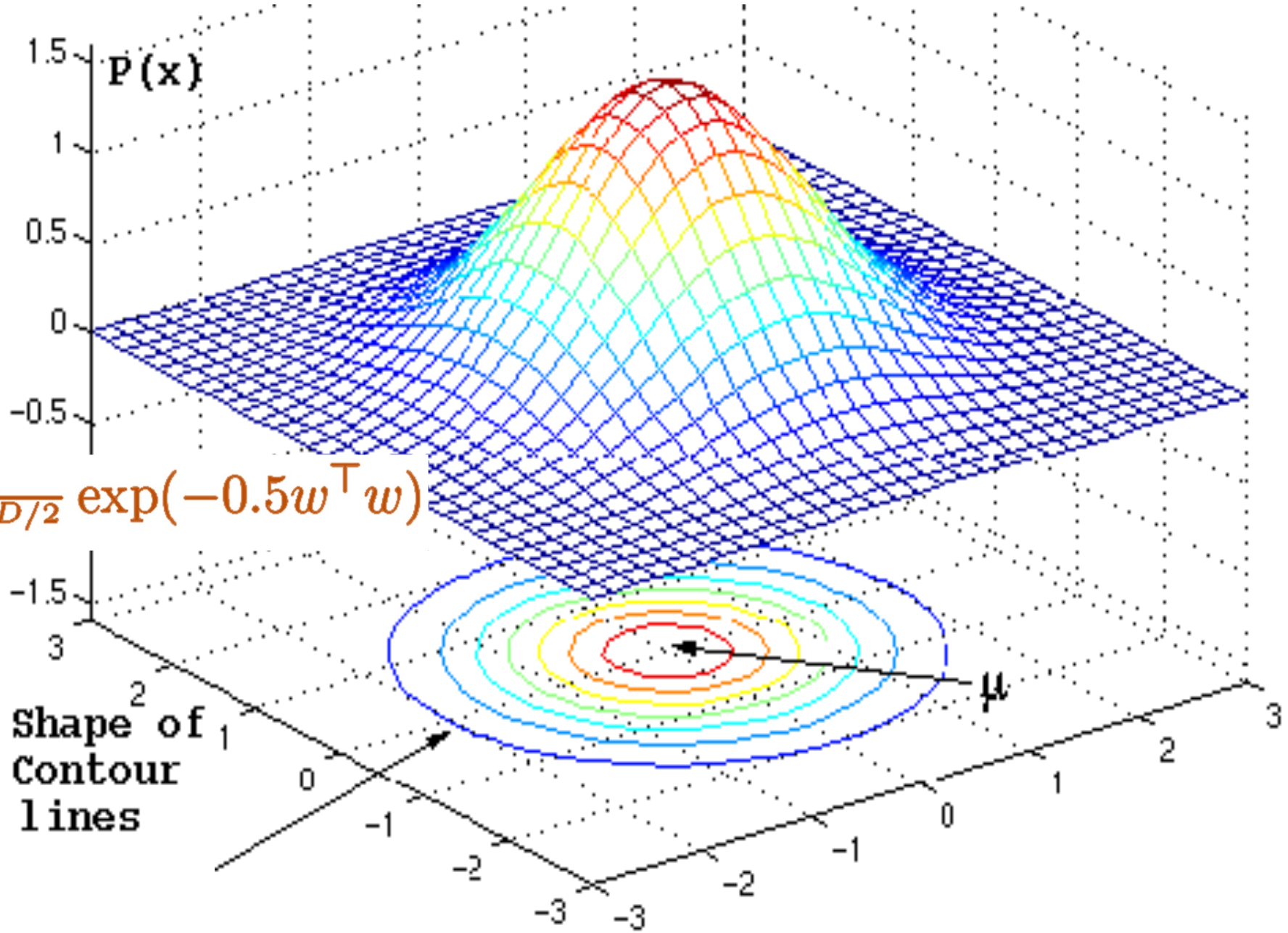




Multivariate Gaussian – Identity A

- Isotropic / spherical multivariate Gaussian
  - Contours / Level sets

$$p(w) = \prod_d p(w_d) = \frac{1}{(2\pi)^{D/2}} \exp(-0.5 w^\top w)$$



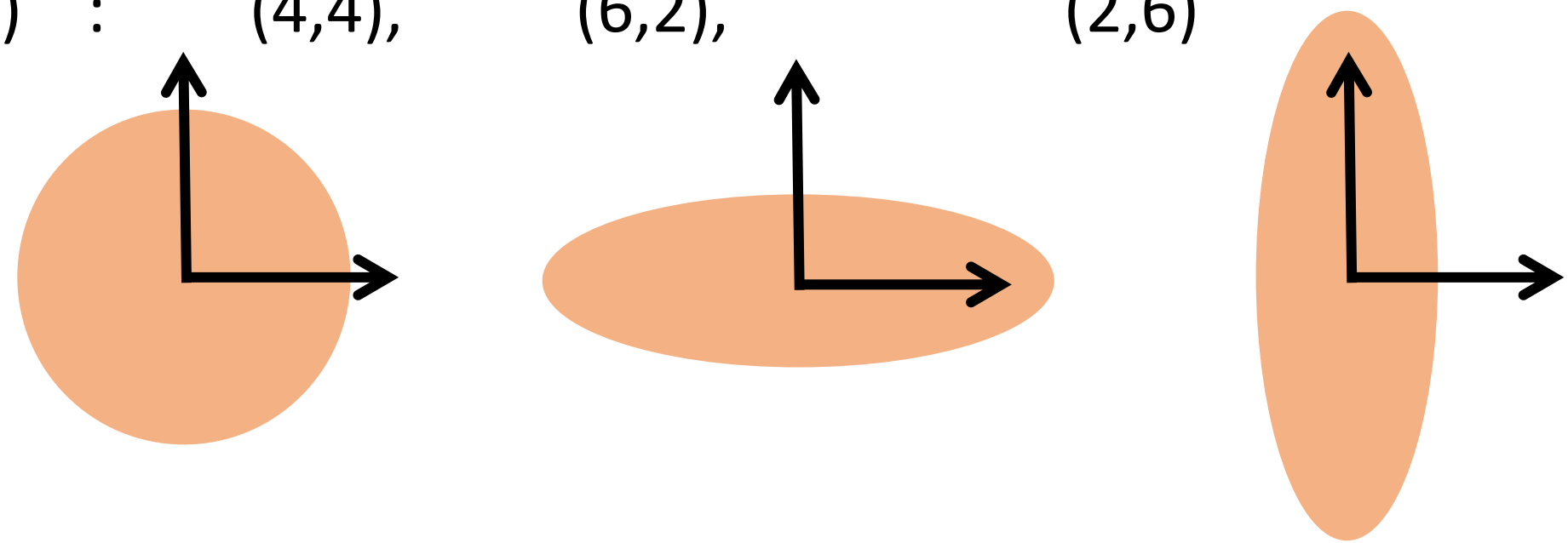
Multivariate Gaussian – Diagonal A

- $X := A\,W + \mu$
- What is PDF q(X) for non-singular square diagonal matrix A ?
- $X_1 = A_{11}\,W_1 + \mu_1$: Gaussian with mean μ_1, standard deviation $\sigma_1 = |A_{11}|$
- $X_2 = A_{22}\,W_2 + \mu_2$: Gaussian with mean μ_2, standard deviation $\sigma_2 = |A_{22}|$
- …
- $X_D = A_{DD}\,W_D + \mu_D$: Gaussian with mean μ_D, standard dev. $\sigma_D = |A_{DD}|$
- $P(X) := P(X_1, X_2, \ldots, X_D) = G(X_1; \mu_1, \sigma_1^2)\; G(X_2; \mu_2, \sigma_2^2)\; \ldots\; G(X_D; \mu_D, \sigma_D^2)$
- Any level set of PDF q(X) is a hyper-ellipsoid with:
  - Center at μ
  - Axes aligned with cardinal axes

Multivariate Gaussian – Diagonal A

- $X := A\,W + \mu$
- What is PDF q(X) for non-singular square diagonal matrix A ?
- $P(X) := P(X_1, X_2, \ldots, X_D) = G(X_1; \mu_1, \sigma_1^2)\, G(X_2; \mu_2, \sigma_2^2) \ldots G(X_D; \mu_D, \sigma_D^2)$
- Example 1-3 (left to right):

means	(μ_1, μ_2)	:	(0,0),	(0,0), (0,0)
variances	(σ_1^2, σ_2^2)	:	(4,4),	(6,2), (2,6)



Multivariate Gaussian – Non-Singular A

- $X := A\,W + \mu$
- What is PDF q(X) for non-singular square matrix A and $\mu = 0$?
- Recall: Transformation of Random Variables (multivariate case)
  - Transformation: $X := g(W) := A\,W$
  - Inverse transformation: $W = g^{-1}(X) = A^{-1}X$
  - Recall: Univariate case
    - $q_X(x) = p_W(g^{-1}(x)) \; |d/dx \; (g^{-1}(x))|$
    - We want local **scaling** in **lengths** caused by $g^{-1}(.)$
    - This is given by **magnitude of derivative** of $g^{-1}(.)$
  - Recall: Multivariate case
    - We want magnitude of local **scaling** in **volumes** caused by $g^{-1}(.)$
    - In our case, "derivative" generalizes to **Jacobian** of $g^{-1}(.)$, which is A^{-1}
    - In our case, "scale factor" is given by **magnitude of determinant** of A^{-1}

Multivariate Gaussian – Non-Singular A

- X := A W + μ
- What is PDF q(X) for non-singular square matrix A and μ = 0 ?
- Recall: Transformation of Random Variables (multivariate case)
  - Transformation: X := g(W) := A W
  - Inverse transformation: W = $g^{-1}(X) = A^{-1}X$
  - Multivariate case

$$q(X) = p(A^{-1}X)\frac{1}{|\det(A)|} = \frac{1}{(2\pi)^{D/2}|\det(A)|}\exp(-0.5X^\top(A^{-1})^\top A^{-1}X)$$

Let $C := AA^\top$. Then, $C^{-1} = (A^{-1})^\top A^{-1}$ and $\det(C) = \det(A)\det(A^\top) = (\det(A))^2$

$$q(X) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}}\exp(-0.5X^\top C^{-1}X)$$

Multivariate Gaussian – Non-Singular A

- But how do we interpret AA^T ?
- Within X := AW, consider random variables X_i and X_j
- $X_i := A_iW$ and $X_j := A_jW$, where A_i is i-th row of A
- $E[X_i] = 0$ for all i, because $E[W]=\mathbf{0}$
- Covariance $(X_i, X_j) = E[X_i X_j] = E[(\sum_d A_{id}W_d)(\sum_e A_{je}W_e)]$
- $= E[\sum_{d=e} A_{id}A_{je}W_dW_e + \sum_d \sum_{e\neq d} A_{id}A_{je}W_dW_e] = \sum_d A_{id}A_{jd}$
- = dot product of i-th row of A with j-th row of A
- = product of i-th row-vector of A with j-th column-vector of A^T
- = (i,j)-th element in AA^T
- Thus, $C := AA^T$ = covariance matrix

Multivariate Gaussian – Non-Singular A, Non-Zero μ

- If X := A W is a multivariate Gaussian,
  then Y := X + μ is a multivariate Gaussian with

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^\top C^{-1}(y-\mu))$$

- Proof:
  - Follows from transformation X := Y – μ := $g^{-1}(Y)$

Multivariate Gaussian – Special Cases

- Diagonal matrix
- Orthogonal matrix
  - Definition: **Real square matrix Q** whose **columns and rows** are **orthogonal unit** vectors (i.e., orthonormal vectors) $Q\,Q^T = Q^T\,Q$ = Identity matrix
  - Determinant det(Q) is either +1 or -1
  - "orthogonal" is an over-used term in mathematics
  - Sometimes also called orthonormal matrix
  - en.wikipedia.org/wiki/Orthogonal_matrix

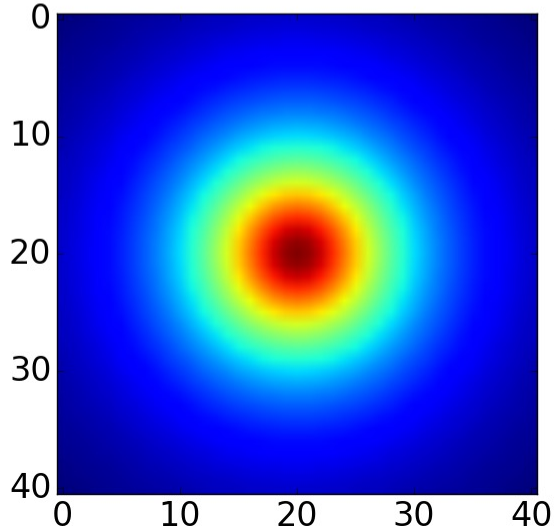
  - "**Rotation**" matrix
    - When det(Q) = +1, then Q is a (proper) **rotation** matrix
    - When det(Q) = –1, then Q models either "**reflection**" (called as an improper rotation) or a combination of (proper) **rotation** and "**reflection**"
    - "Rotation" is an over-used term (sometimes includes improper rotations) in math

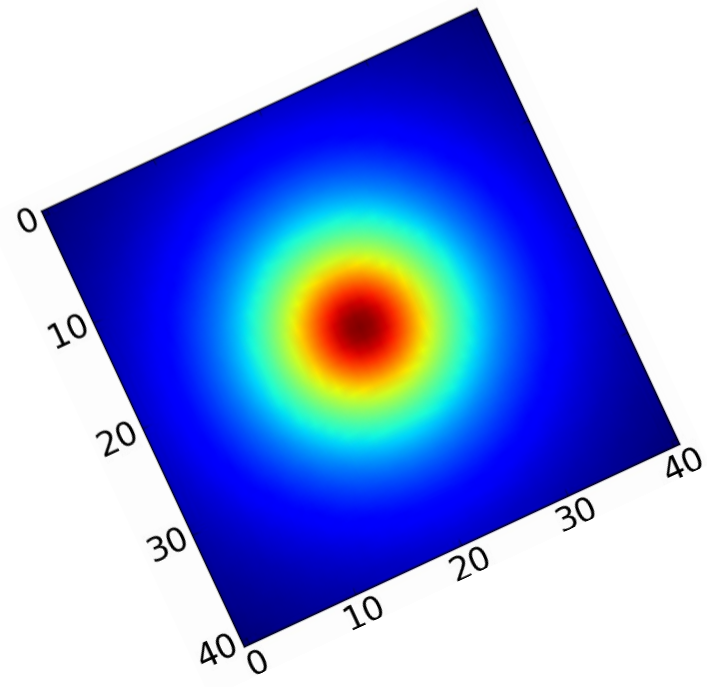
Multivariate Gaussian – Special Cases

- Property (Rotation and/or Reflection):
  If μ := 0; and A := R where R is orthogonal;
  then Y := RW has PDF:

$$P(y) = 1/(2\pi)^{D/2} \exp(-0.5 y^\top y)$$

- Proof:
  - Transformation of random vectors
    - |det(R)| = 1
    - Inverse transformation is
      $W = R^T Y$





So, $Q(y) = (1/(2\pi)^{D/2}) \exp(-0.5(R^\top y)^\top R^\top y) = (1/(2\pi)^{D/2}) \exp(-0.5 y^\top y)$

Thus, $Y := RW$ is also a zero-mean isotropic multivariate Gaussian, just like W

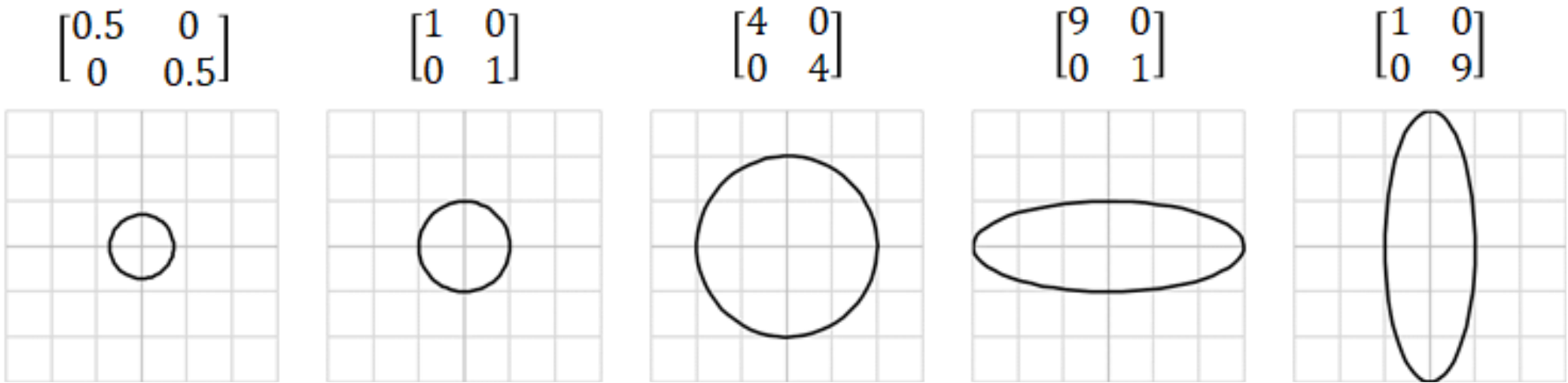
Multivariate Gaussian – Special Cases

- Property (Scaling):
  If μ := 0; and A := S square diagonal (with positive entries on diagonal);
  then Y := SW has PDF: $P(y) = (1/(2\pi)^{D/2})(1/\det(S))\exp(-0.5y^\top(S^2)^{-1}y)$

Proof: Transformation of random vectors. $|\det(S)| = \prod_d S_{dd}$
Inverse transformation is $W = S^{-1}Y$

So, $Q(y) = (1/(2\pi)^{D/2})(1/\det(S))\exp(-0.5(S^{-1}y)^\top S^{-1}y) = (1/(2\pi)^{D/2}(1/\det(S))\exp(-0.5y^\top(S^2)^{-1}y)$

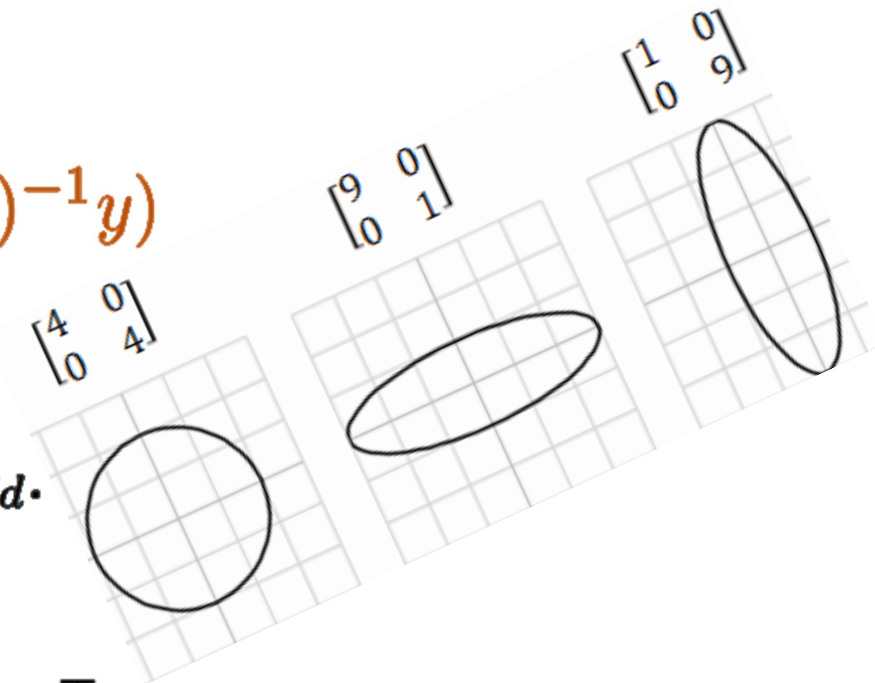
Thus, $Y := SW$ is a zero-mean anisotropic (axis-aligned) multivariate Gaussian.



Multivariate Gaussian – Special Cases

- Property (first Scaling, and then Rotation and/or Reflection):
  If μ = 0; square A = RS,
  then Y := RSW has the PDF:

$$P(y) = (1/(2\pi)^{D/2})(1/\det(S))\exp(-0.5y^{\top}(RS^2R^{\top})^{-1}y)$$



Proof: Transformation of random vectors. $|\det(RS)| = \prod_d S_{dd}$.
Inverse transformation is $W = (RS)^{-1}Y = S^{-1}R^{\top}y$

So, $Q(y) = (1/(2\pi)^{D/2})(1/\det(S))\exp(-0.5(S^{-1}R^{\top}y)^{\top}S^{-1}R^{\top}y)$
$= (1/(2\pi)^{D/2}(1/\det(S))\exp(-0.5y^{\top}(RS^2R^{\top})^{-1}y)$

Thus, $Y := RSW$ is zero-mean rotated+reflected anisotropic multivariate Gaussian with covariance $C = RS^2R^{\top}$.

Multivariate Gaussian – General Case

- A = RS is equivalent to most general case (for square A). Why ?
- Because, singular value decomposition of real-valued A' = USV^T where
  U & V are orthogonal & real-valued, and
  S is diagonal with non-negative real numbers on diagonal
- If X := A W is a multivariate Gaussian,
  then Y := X + μ is a multivariate Gaussian with

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^T C^{-1}(y-\mu))$$

- When $C = AA^T$, and A is invertible,
  then $C = USV^T VS^TU^T = U SS^T U^T$ where
  SS^T is square diagonal with positive real numbers on diagonal
- Thus, eigen decomposition of C is $U SS^T U^T$ where U is real-valued

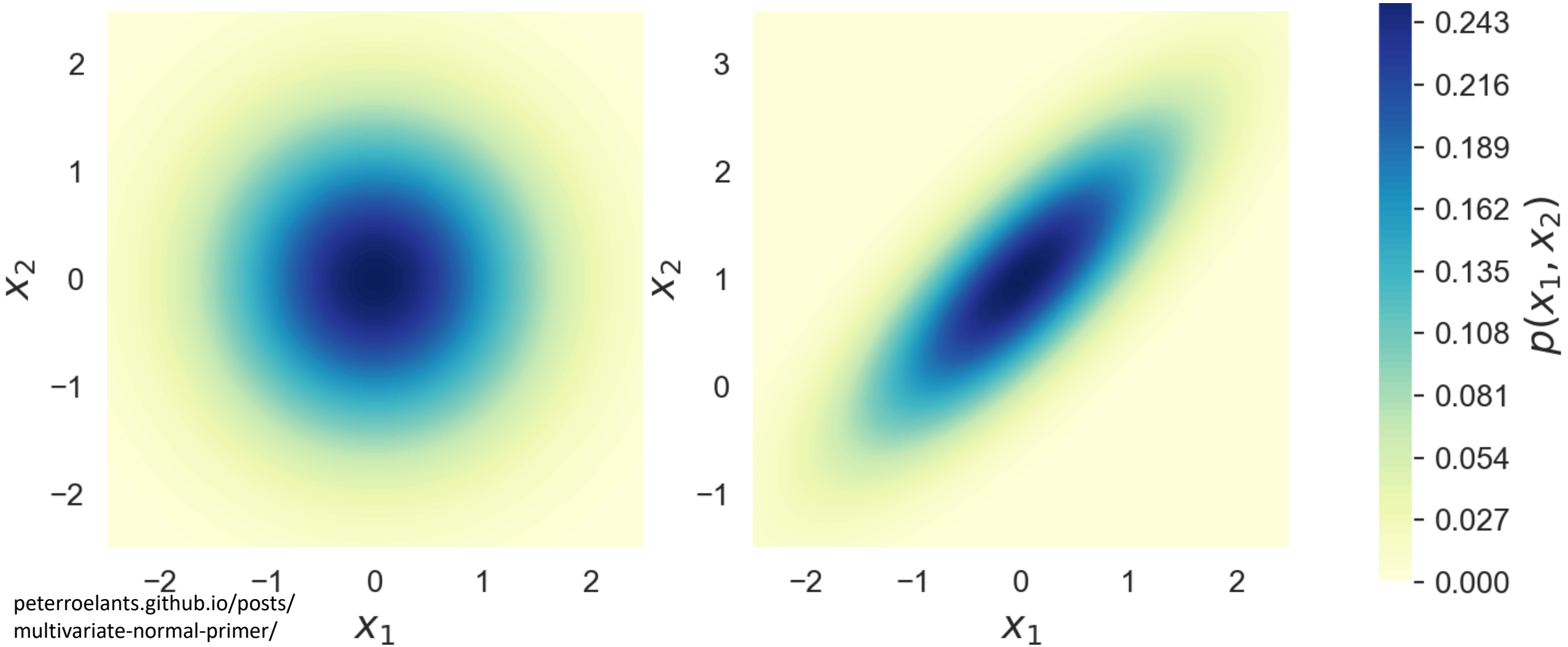
Multivariate Gaussian – Level Sets

- What are level sets of PDF: $p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^T C^{-1}(y-\mu))$
- Let $C = Q\ D\ Q^T$. Then, $C^{-1} = Q\ D^{-1}\ Q^T$ that is also SPD
- Each level set satisfies $(y - \mu)^T\ C^{-1}\ (y - \mu) = \alpha$, where $\alpha \geq 0$
  - Because C^{-1} is SPD; α becomes zero iff $y=\mu$
- So, $(y - \mu)^T\ Q\ D^{-1}\ Q^T\ (y - \mu) = \alpha$
- Change to roto-reflected coordinate system represented by orthogonal basis Q, where y maps to $y' = Q^T\ y$, and μ maps to $\mu' = Q^T\ \mu$
- Then, $(y' - \mu')^T\ D^{-1}\ (y' - \mu') = \alpha$, which is a hyper-ellipsoid:
  - in roto-reflected coordinate frame
    with center at μ' & axes along cardinal axes
    where half-lengths of axes are square root of diagonal elements in D^{-1}

Multivariate Gaussian – Level Sets

- Values of:
  - C=I, General C

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^\top C^{-1}(y-\mu))$$

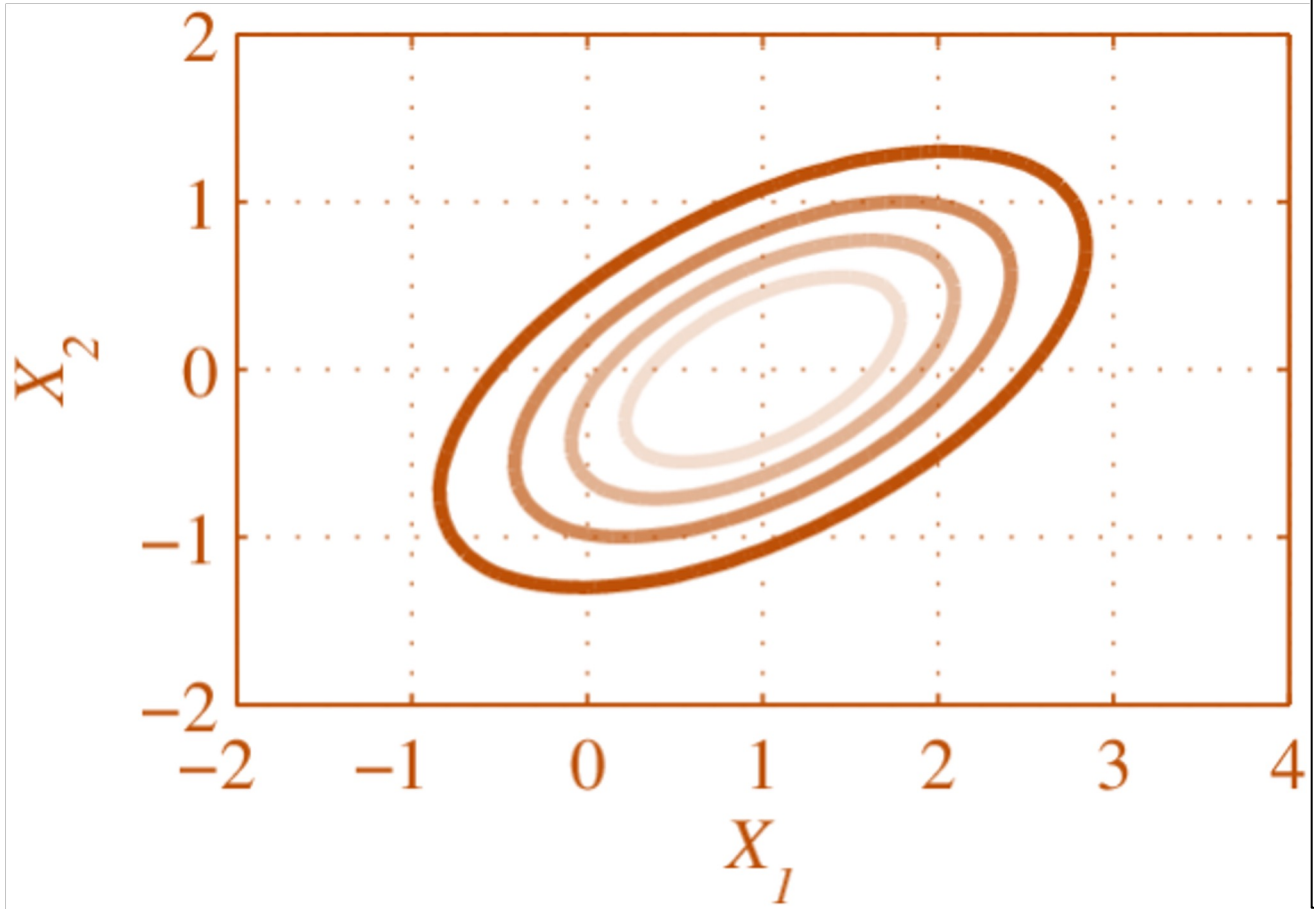


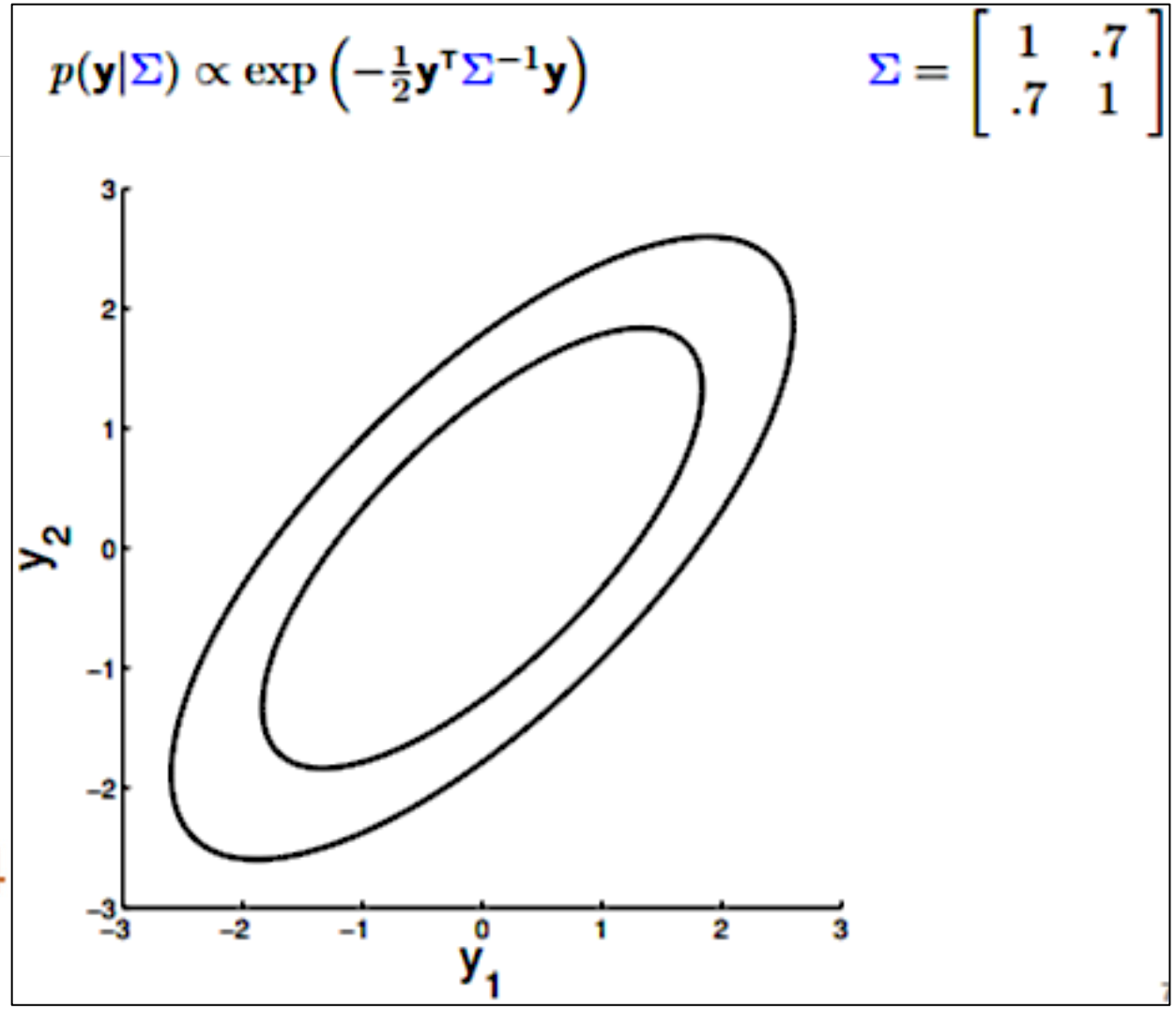
peterroelants.github.io/posts/multivariate-normal-primer/

Multivariate Gaussian – Level Sets

- Level sets of:

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^{\top} C^{-1}(y-\mu))$$





Multivariate Gaussian – ML Estimation

- Data: $\{y_1, \ldots, y_N\}$
- PDF: $$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^\top C^{-1}(y-\mu))$$
- ML estimate (MLE) for mean vector (= sample mean)
  - Take derivative of log-likelihood with respect to (w.r.t.) μ; assign to zero; solve
  - Quadratic form $a^\top Ba$
    $= \Sigma_i \Sigma_j\, a_i a_j B_{ij}$
  - Partial derivative w.r.t. a_k
    $= \Sigma_j\, a_j B_{kj} + \Sigma_i\, a_i B_{ik}$
    $= 2\, \Sigma_j\, B_{kj} a_j$ (because B is symmetric)
    = 2 (k-th row of B * column-vector a)
  - Scalar function, say f(a), of multiple scalar variables in column-vector 'a'
    - Jacobian df/da will be a row vector of same length as 'a'
      - Change in function value (df) = derivative (df/da) * change in variable (da)
    - Can be reshaped/rearranged into a column vector of same shape as 'a'

$$\frac{\partial}{\partial \mu}(\mu - x)^\top C^{-1}(\mu - x) = 2C^{-1}(\mu - x)$$

Multivariate Gaussian – ML Estimation

- Data: $\{y_1, \ldots, y_N\}$
- PDF: $$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^\top C^{-1}(y-\mu))$$
- MLE for covariance matrix (= sample covariance; uncorrected/biased)
  - Take derivative w.r.t. C; assign to zero; solve
  - Need partial derivatives w.r.t. C_{ij}
  - Scalar function, say f(C), of multiple scalar variables in C
    - Consider a (column)-vectorized form of C
    - Jacobian df/dC will be a row vector of the same length as (column)-vectorized C
    - Can be reshaped/rearranged into a matrix of the same shape as C

$$\frac{d}{dC}(x-\mu)^\top C^{-1}(x-\mu) = -C^{-\top}(x-\mu)(x-\mu)^\top C^{-\top}$$

$$\frac{d}{dC}\log(|C|) = \frac{1}{|C|}|C|C^{-\top} = C^{-\top}$$

Multivariate Gaussian – ML Estimation

- "Matrix Calculus"
  - http://www.ee.ic.ac.uk/hp/staff/dmb/matrix/calculus.html
  - https://en.wikipedia.org/wiki/Matrix_calculus
  - http://www.matrixcalculus.org/

Principal Component Analysis (PCA)

- We want to find principal mode of variation around mean
- Consider a general multivariate random variable X with PDF P(X)
  - We aren't assuming it to be a Gaussian

Consider observed multivariate data $\{x_i \in \mathbb{R}^D\}_{i=1}^N$
drawn from some PDF $P(X)$ with some mean μ and some covariance matrix C

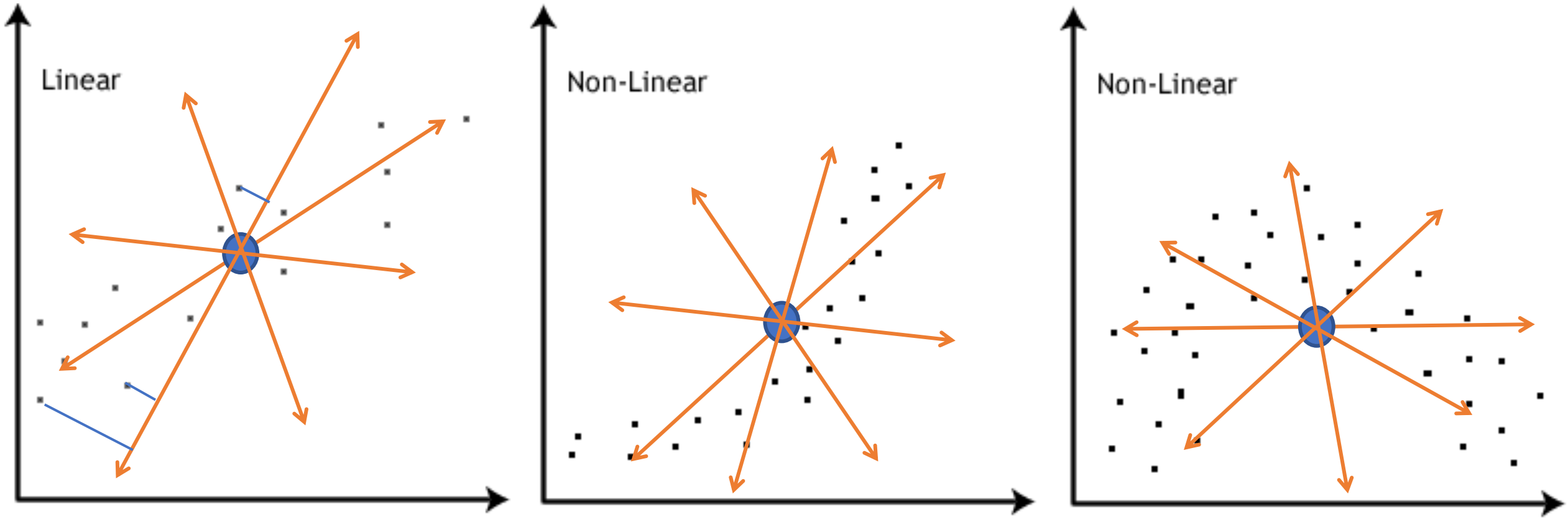
As $N \to \infty$ for the data, sample mean $\to \mu$, and sample covariance $\to C$

- We define "principal mode of variation around mean":=
  direction, through mean, such that
  when data projected on line with that direction passing through mean,
  then projected data has maximum variance

Principal Component Analysis (PCA)

- Direction (through mean) of maximal variance

Find the "direction" v (i.e., $\| v \|_2 = 1$) such that the data projected on the 1D space indicated by (i) mean μ and (ii) direction v that passes through μ has the maximal variance



Principal Component Analysis (PCA)

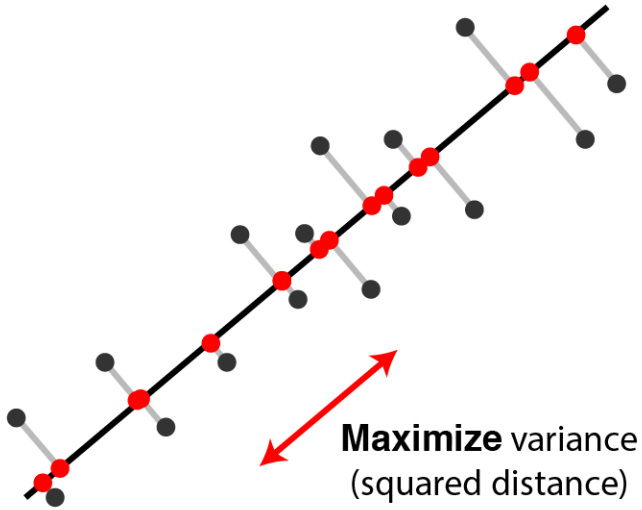
- Direction (through mean) of maximal variance

For finite N, we perform the analysis in a shifted coordinate frame where the sample mean $\sum_{i=1}^{N} x_i/N = 0$

Assume the sample mean for $\{x_i \in \mathbb{R}^D\}_{i=1}^{N}$ is at the origin

Projected data are $\langle x_i, v\rangle v$

Sample mean of projected data, in the 1D space, is $\sum_i \langle x_i, v\rangle v = 0$

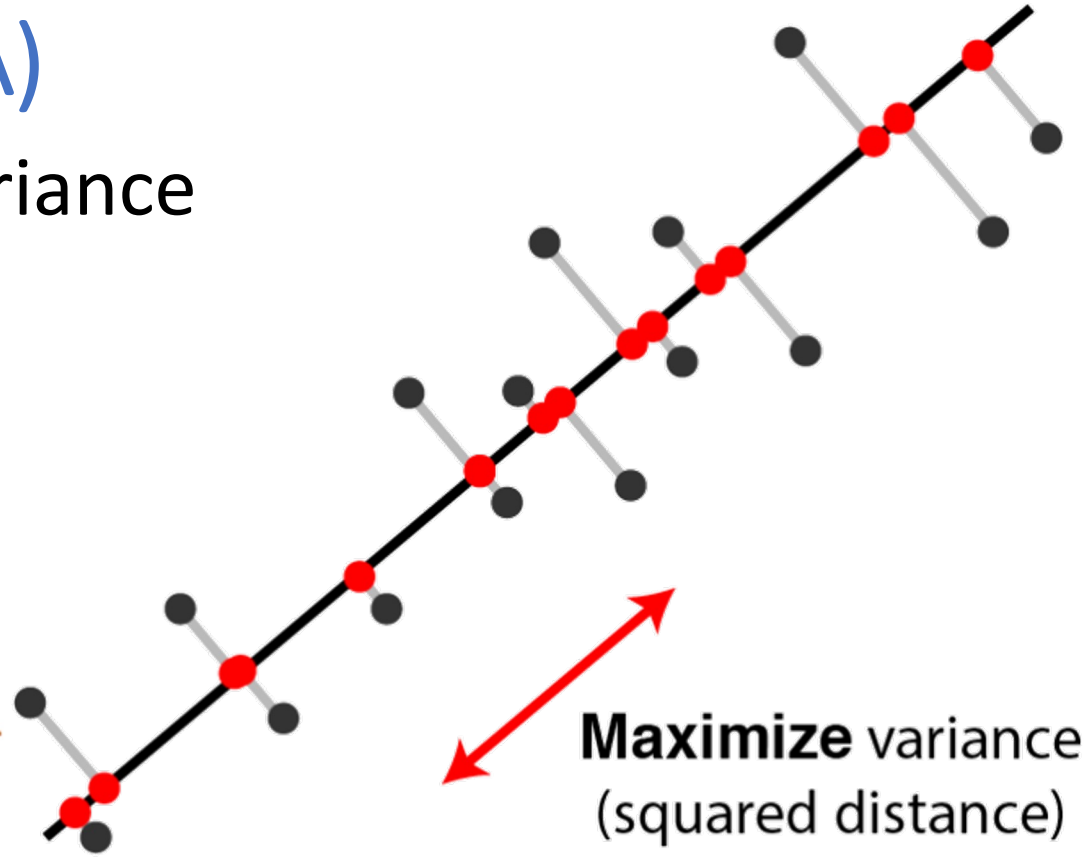


Distance of projected data from the sample mean (i.e., origin) is $\|\langle x_i, v\rangle v\|_2 = |\langle x_i, v\rangle|$

Sample variance of projected data, in the 1D space, is $\sum_i \langle x_i, v\rangle^2/N$

Principal Component Analysis (PCA)

- Direction (through mean) of maximal variance

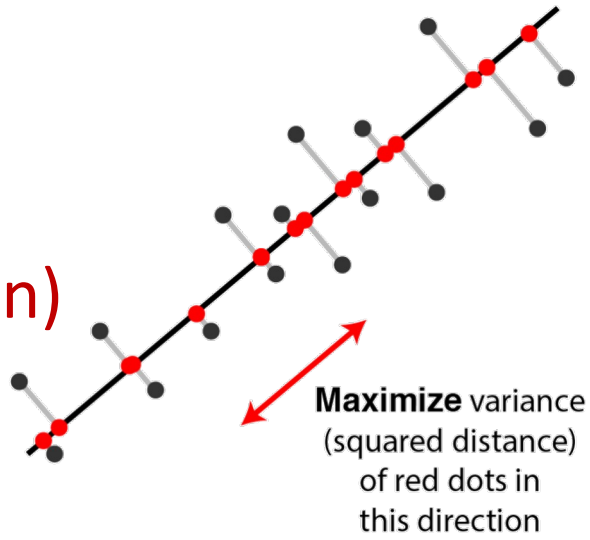


Optimal direction

$$= \arg\max_{v:\|v\|_2=1} \sum_i \langle x_i, v\rangle^2/N$$
$$= \arg\max_{v:\|v\|_2=1} \sum_i (x_i^\top v)^2/N$$
$$= \arg\max_{v:\|v\|_2=1} \sum_i (x_i^\top v)^\top (x_i^\top v)/N$$
$$= \arg\max_{v:\|v\|_2=1} \sum_i v^\top x_i x_i^\top v/N$$
$$= \arg\max_{v:\|v\|_2=1} v^\top (\sum_i x_i x_i^\top/N) v$$
$$= \arg\max_{v:\|v\|_2=1} v^\top C v \text{ (where } C \text{ is the sample covariance matrix)}$$

Principal Component Analysis (PCA)

- Direction (through mean) of maximal variance
  - When covariance matrix C is diagonal (sample mean at origin)
  - Let d-th element on diagonal of C be C_{dd}
  - Let d-th element in vector v be v^d



Without loss of generality, let the diagonal elements be sorted in descending order,

The optimal direction is $\arg\max_{v:\|v\|_2=1} \sum_d C_{dd}(v^d)^2$

The unit-norm constraint on v is: $\sum_d (v^d)^2 = 1$

- Objective function is a convex combination of { C_{dd} } with weights { $(v^d)^2$ }

The "objective function" $\sum_d C_{dd}(v^d)^2$ is maximized when $v^1 = 1$, and $v^d = 0$ for $d = 2, \cdots, D$

The maximal variance then is C_{11} that is the eigenvalue corresponding to the principal eigenvector $[1, 0, \cdots, 0]^\top$

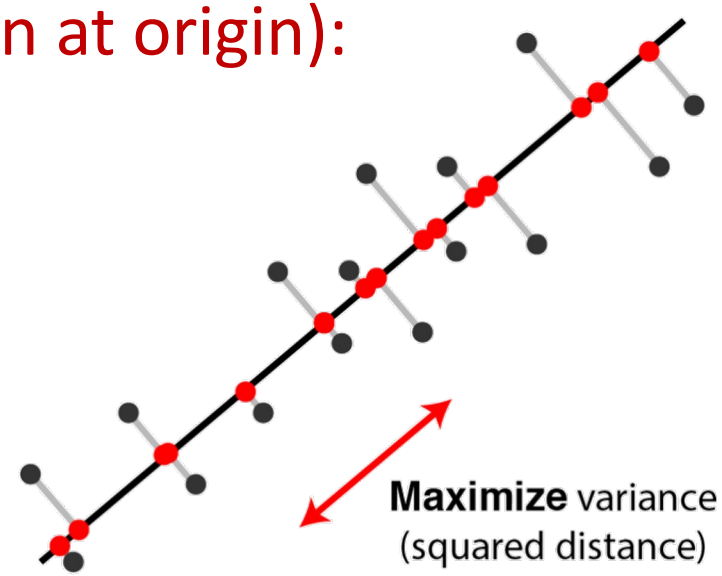
Principal Component Analysis (PCA)

- Directions (through mean) of maximal variance
  - When covariance matrix C is diagonal (and sample mean at origin):

Now find the 2nd direction u that is:
(i) orthogonal to v and
(ii) maximizes the variance of the data projected onto it

Optimal direction
$= \arg\max_{u:\|u\|_2=1, u\perp v} \sum_i \langle x_i, u\rangle^2$
$= \arg\max_{u:\|u\|_2=1, u\perp v} \sum_d C_{dd}(u^d)^2$, where we know $C_{dd} \geq 0$

This is maximized when $u^d = 1$ for $d = 2$, and $u^d = 0$ for all other d



- **2nd mode of variation** is **2nd cardinal axis (another eigenvector of C). Variance along that mode** = **second-largest eigenvalue of C** = C_{22}
- Similar arguments hold for 3rd, 4th, ... directions

Principal Component Analysis (PCA)

- Directions (through mean) of maximal variance
  - For a general SPD covariance matrix C (and sample mean at origin):

Then, $C = Q\Lambda Q^\top$, where the diagonal of Λ has sorted values (high to low)

Then, $\max_v v^\top C v = \max_v v^\top Q\Lambda Q^\top v = \max_{u:=Q^\top v} u^\top \Lambda u$

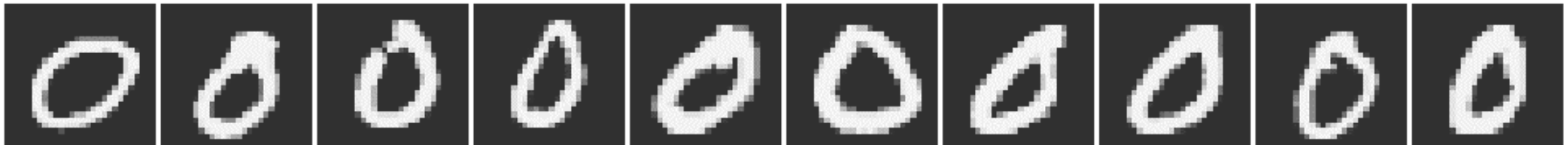
Thus, the principal mode of variation is given by $u = [1, 0, \cdots, 0]^\top$ (in the Q basis)

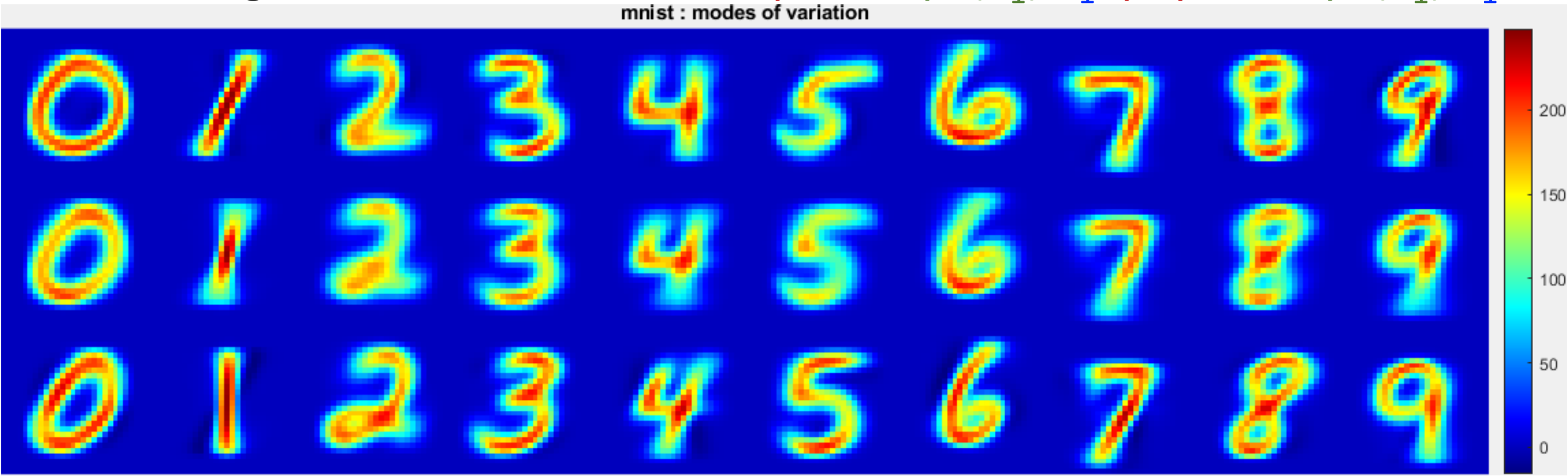
that is equivalent to $v = Qu =$ first column of Q (in the original basis)

The maximal variance then is Λ_{11} that is the eigenvalue corresponding to the principal eigenvector

Simlarly, the remaining modes of variation will be the other columns of Q

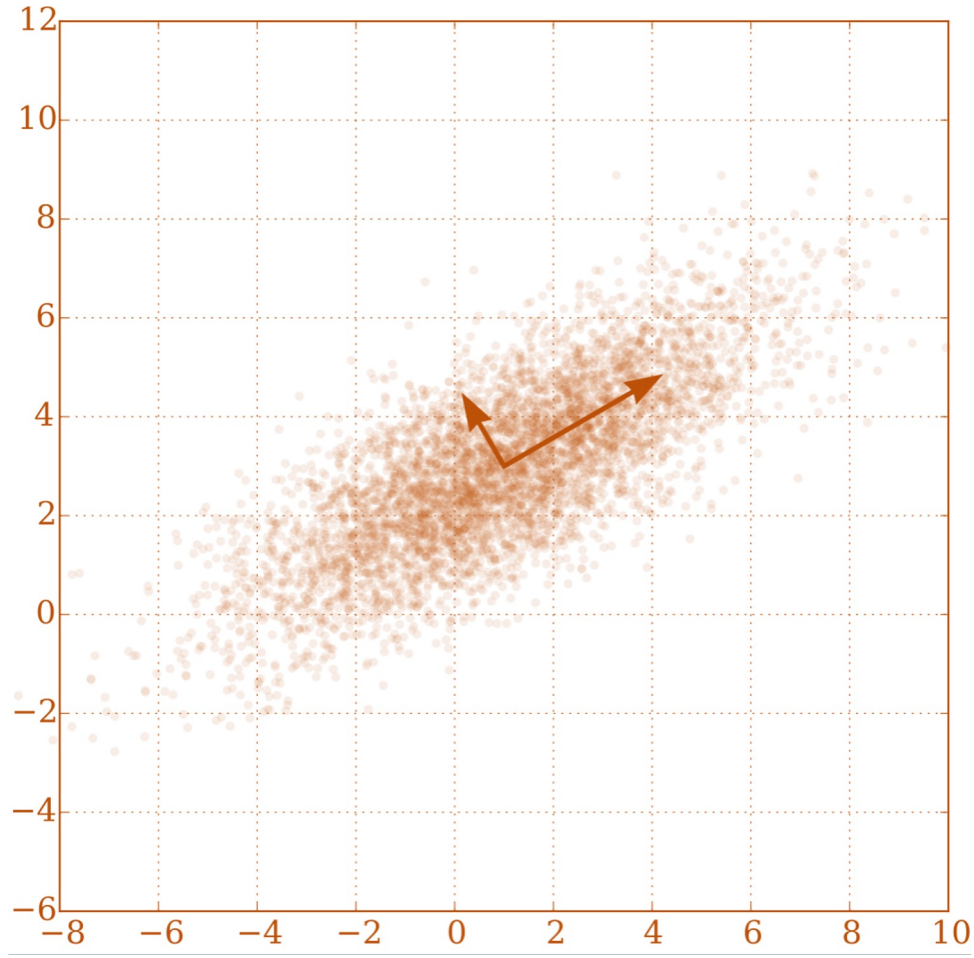
Principal Component Analysis (PCA)

- Directions (through mean) of maximal variance
- Data 
- Mean vector/"image" μ. Eigen vector/"image" v_i. Eigenvalue λ_i.
- Visualizing 1st mode of variation: [$\mu - k\ \mathrm{sqrt}(\lambda_1)\ v_1$, μ, $\mu + k\ \mathrm{sqrt}(\lambda_1)\ v_1$]



Principal Component Analysis (PCA)

- PCA applied to data from a multivariate Gaussian distribution
  - Consider X is multivariate Gaussian
  - If X := AW + b, then:
    - Principal modes of variation are directions given by eigenvectors of covariance matrix $C := AA^T$
    - Principal modes of variation are along axes of hyper-ellipsoids that are level sets of P(X)
    - Variances along principal modes of variation are eigenvalues of C
  - If X := RSW + b, then:
    - Principal modes of variation are column vectors of orthogonal matrix R i.e., eigenvectors of $C = RS^2R^T$
    - Variances along principal modes of variation are eigenvalues of C, i.e., diagonal elements in S^2

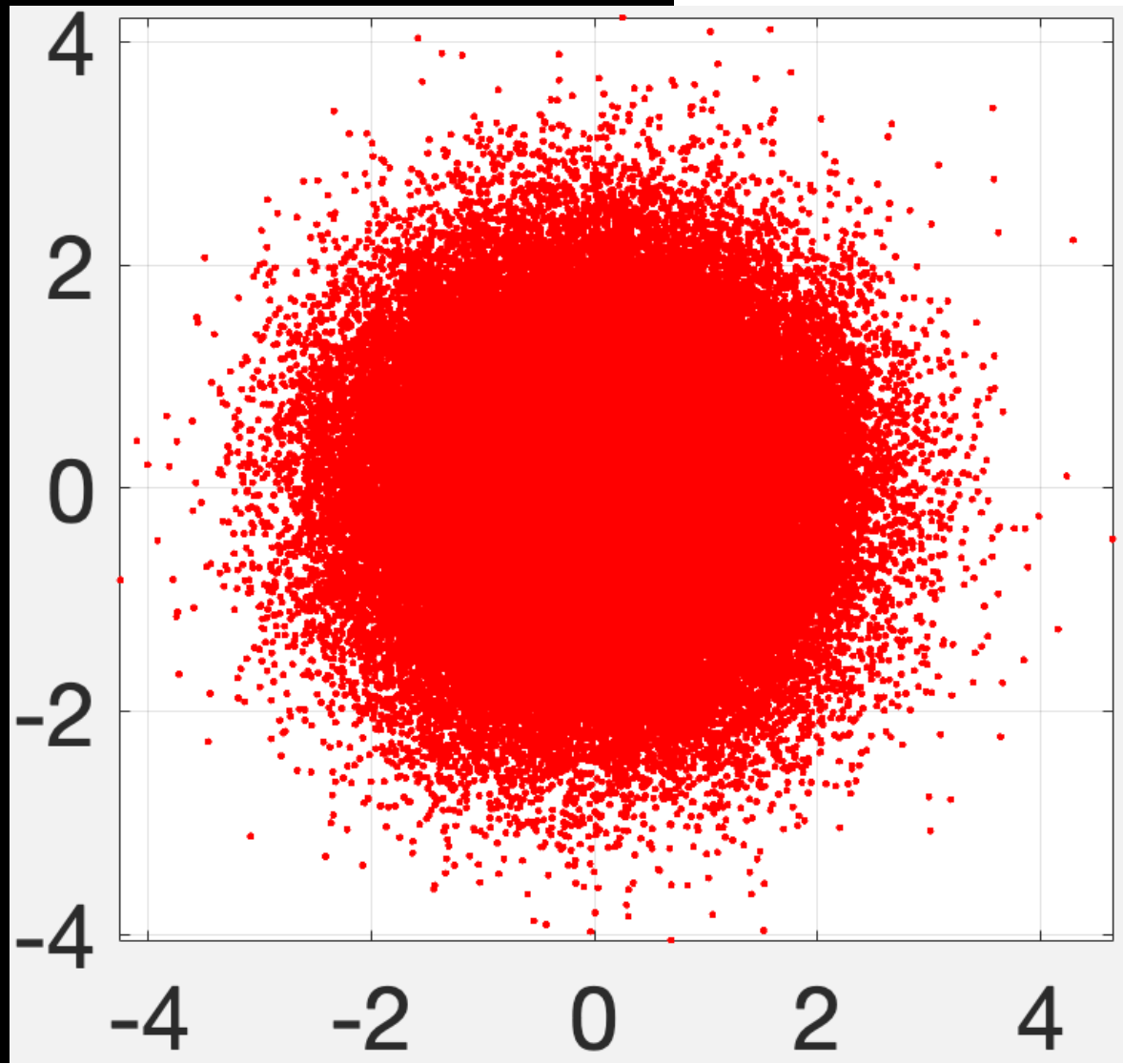


Principal Component Analysis (PCA): Tricky Cases

- Example 1: rng(0); N = 1e5; data = randn (2,N)

```
>> rng(0); N=1e5; data=(randn(2,N)); C=cov(data'), [V,D]=eig(C)

C =

    1.0007   -0.0023
   -0.0023    0.9966


V =

   -0.4112   -0.9115
   -0.9115    0.4112


D =

    0.9956         0
         0    1.0018
```

C=I ← independence

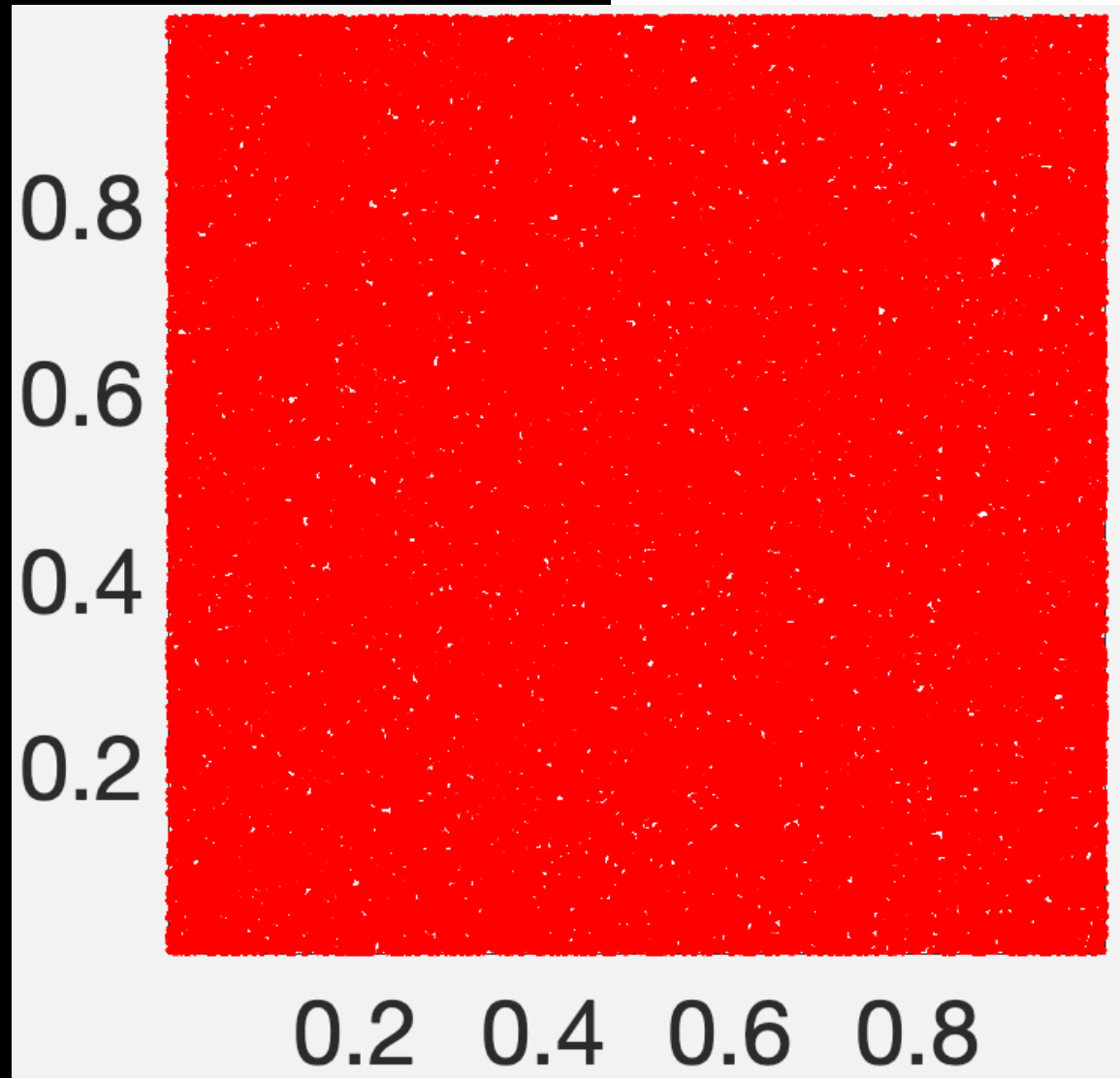


Principal Component Analysis (PCA): Tricky Cases

- Example 2: rng(0); N = 1e5; data = rand (2,N)

```
>> rng(0); N=1e5; data=(rand(2,N)); C=cov(data'), [V,D]=eig(C)

C =

    0.0835   -0.0001
   -0.0001    0.0831


V =

   -0.1710   -0.9853
   -0.9853    0.1710


D =

    0.0831         0
         0    0.0835
```

$C \propto I$ ← independence

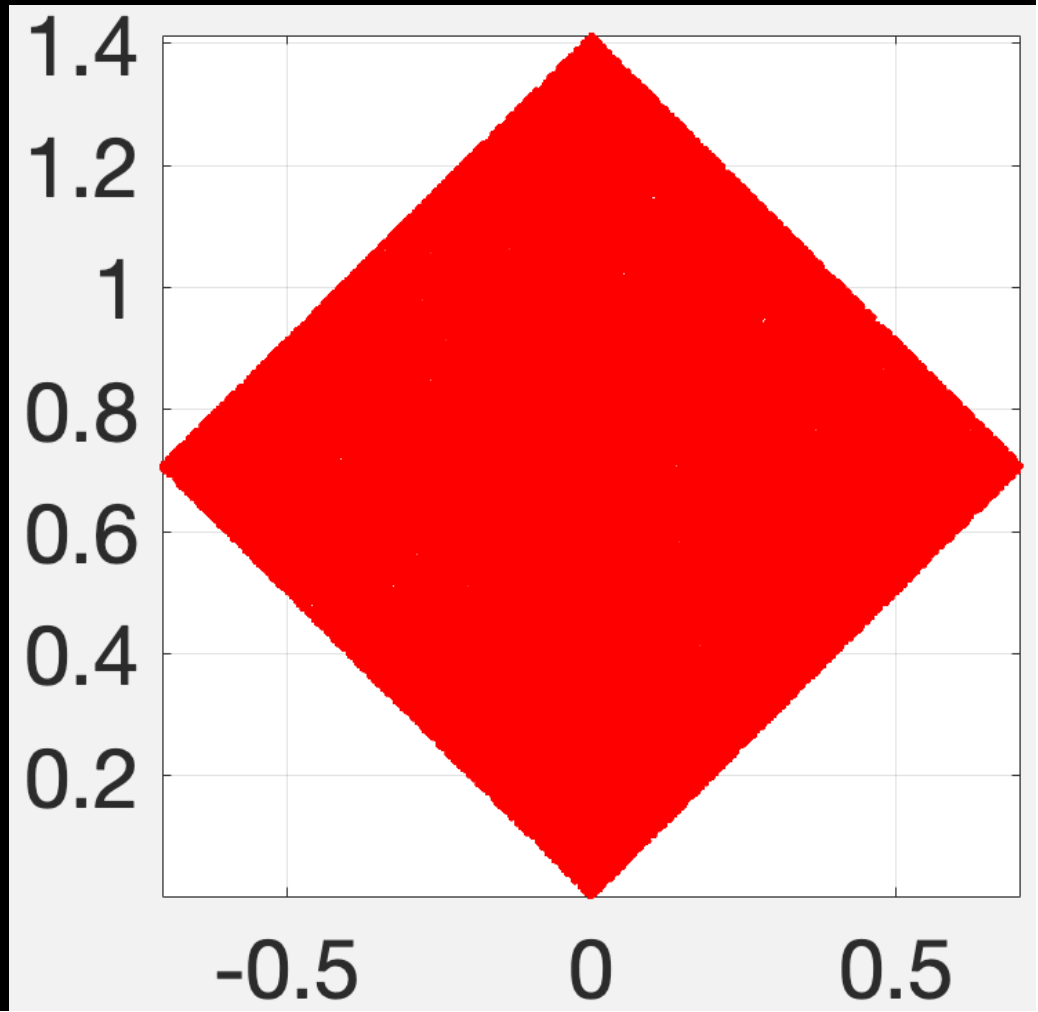


Principal Component Analysis (PCA): Tricky Cases

- Example 3: rng(0); N = 1e5; data = [[1 -1];[1 1]]/sqrt(2) * (rand (2,N))

```
>> rng(0); N=1e5; data=[[1 -1];[1 1]]/sqrt(2)*(rand(2,N)); C=cov(data'), [V,D]=eig(C)

C =

    0.0834    0.0002
    0.0002    0.0832


V =

    0.5758   -0.8176
   -0.8176   -0.5758


D =

    0.0831         0
         0    0.0835
```

$C \propto I$ ← why ?



Principal Component Analysis (PCA)

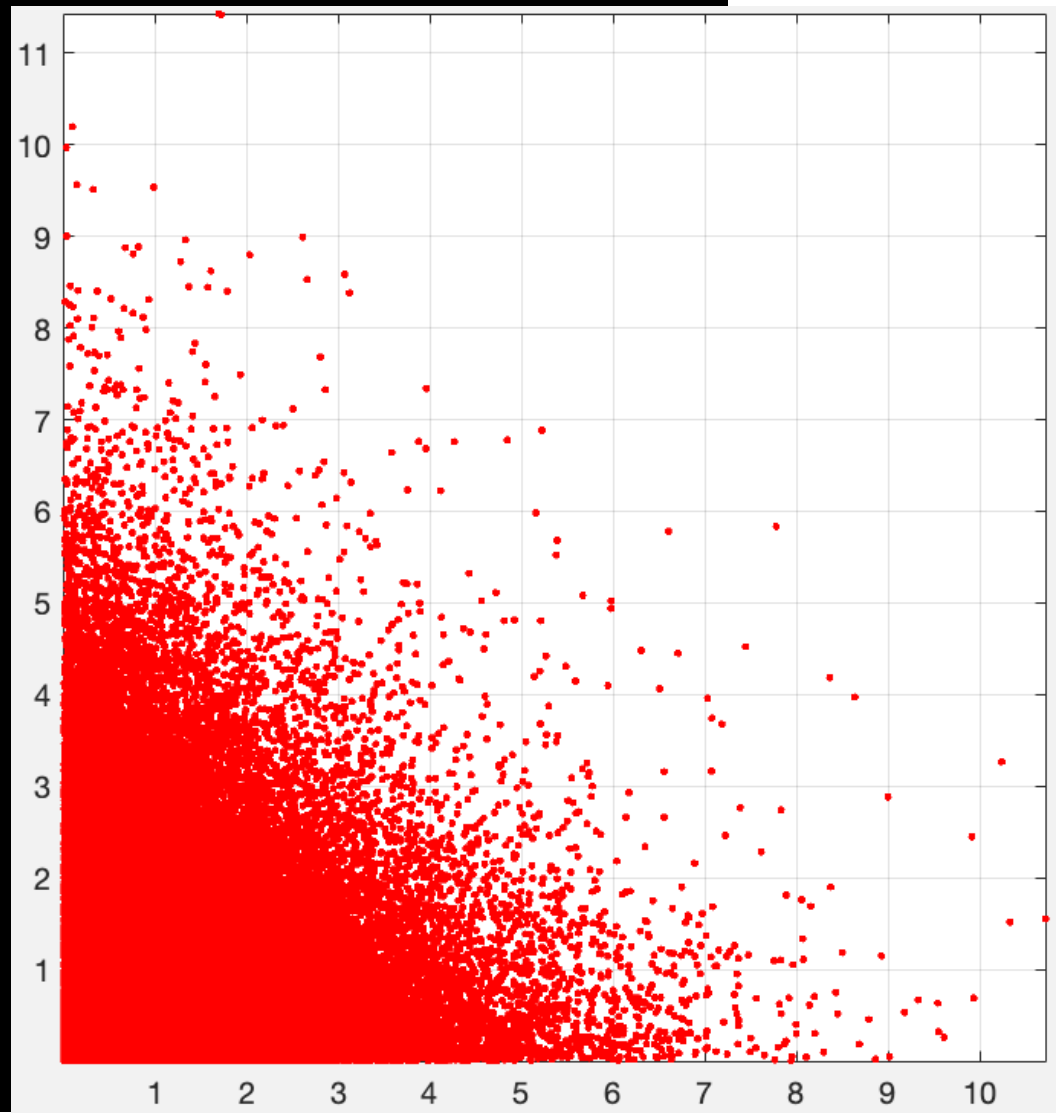
- What happens to covariance matrix when we rotate data ?
- In general
  - Let $C_X := E[(X - \mu_X)(X - \mu_X)^\top] = E[XX^\top] - \mu_X \mu_X^\top$
  - Let $Y := RX$ (for any invertible R)
  - Then, $\mu_Y := E[Y] = E[RX] = RE[X] = R\mu_X$
  - Then, $C_Y := E[(Y - \mu_Y)(Y - \mu_Y)^\top]$
  - $= E[YY^\top] - \mu_Y \mu_Y^\top$
  - $= E[RXX^\top R^\top] - R\mu_X \mu_X^\top R^\top$
  - $= R(E[XX^\top] - \mu_X \mu_X^\top) R^\top$
  - $= R\mathrm{C_X}\mathrm{R}^\top$
- If $C_X = a\mathrm{I}$ and R is orthogonal, then $C_Y = RaI\mathrm{R}^\top = a\mathrm{I} = C_X$
  - So, eigenvalues don't change; eigenvector set can be any orthogonal basis
  - So, directions of maximal variance aren't unique

Principal Component Analysis (PCA): Tricky Cases

- Example 4 (C=I): rng(0); N = 1e5; data = exprnd (1,[2,N])

```
>> rng(0); N=1e5; data=(exprnd(1,[2,N])); C=cov(data'), [V,D]=eig(C)

C =

    1.0030    0.0001
    0.0001    0.9991


V =

    0.0147   -0.9999
   -0.9999   -0.0147


D =

    0.9991         0
         0    1.0030
```

C=I ← independence



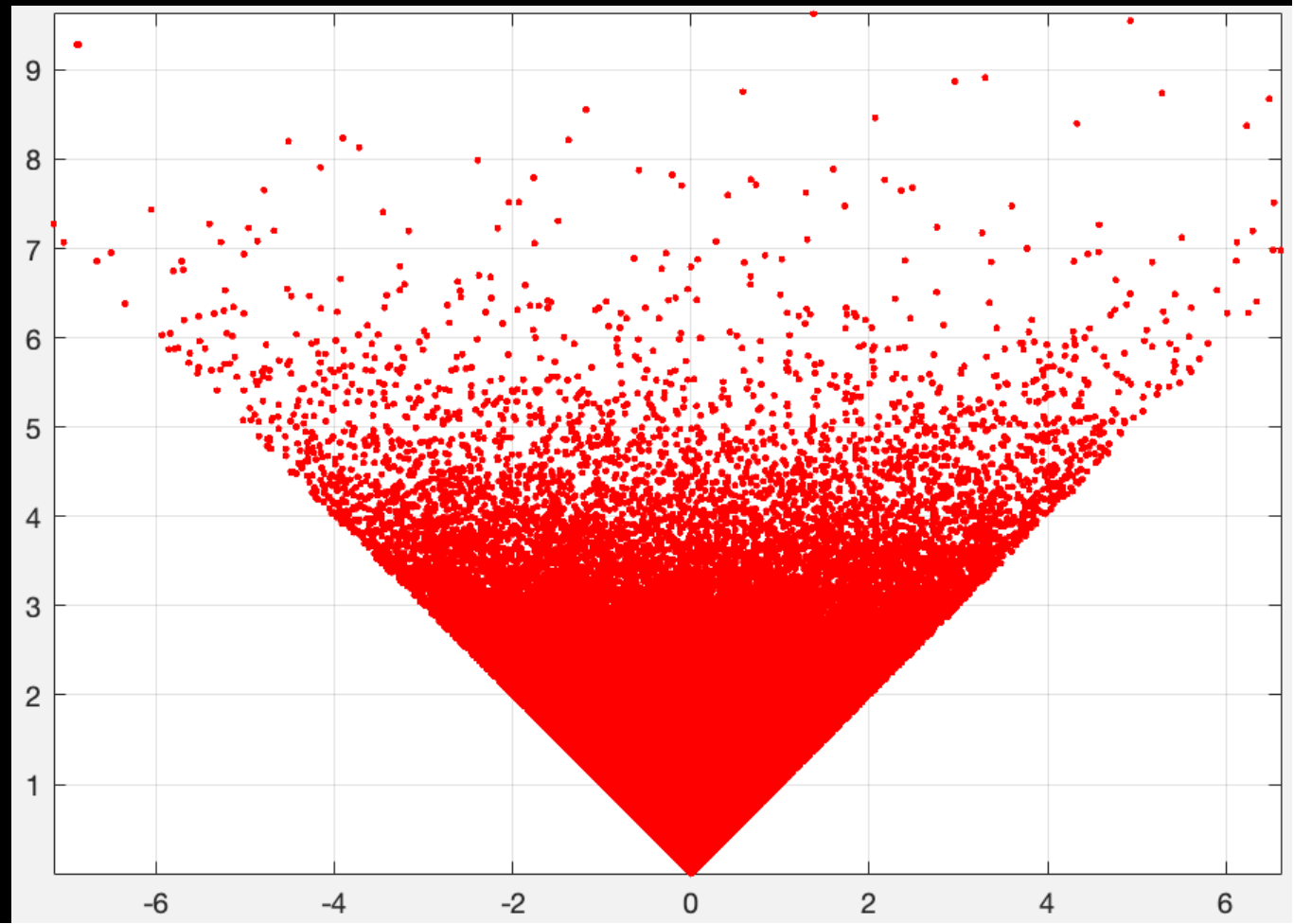
Principal Component Analysis (PCA): Tricky Cases

- Example 5: rng(0); N=1e5; data = [[1 -1];[1 1]]/sqrt(2) * exprnd (1,[2,N])

```
>> rng(0); N=1e5; data= [[1 -1];[1 1]]/sqrt(2) * (exprnd(1,[2,N])); C=cov(data'), [V,D]=eig(C)
close
```

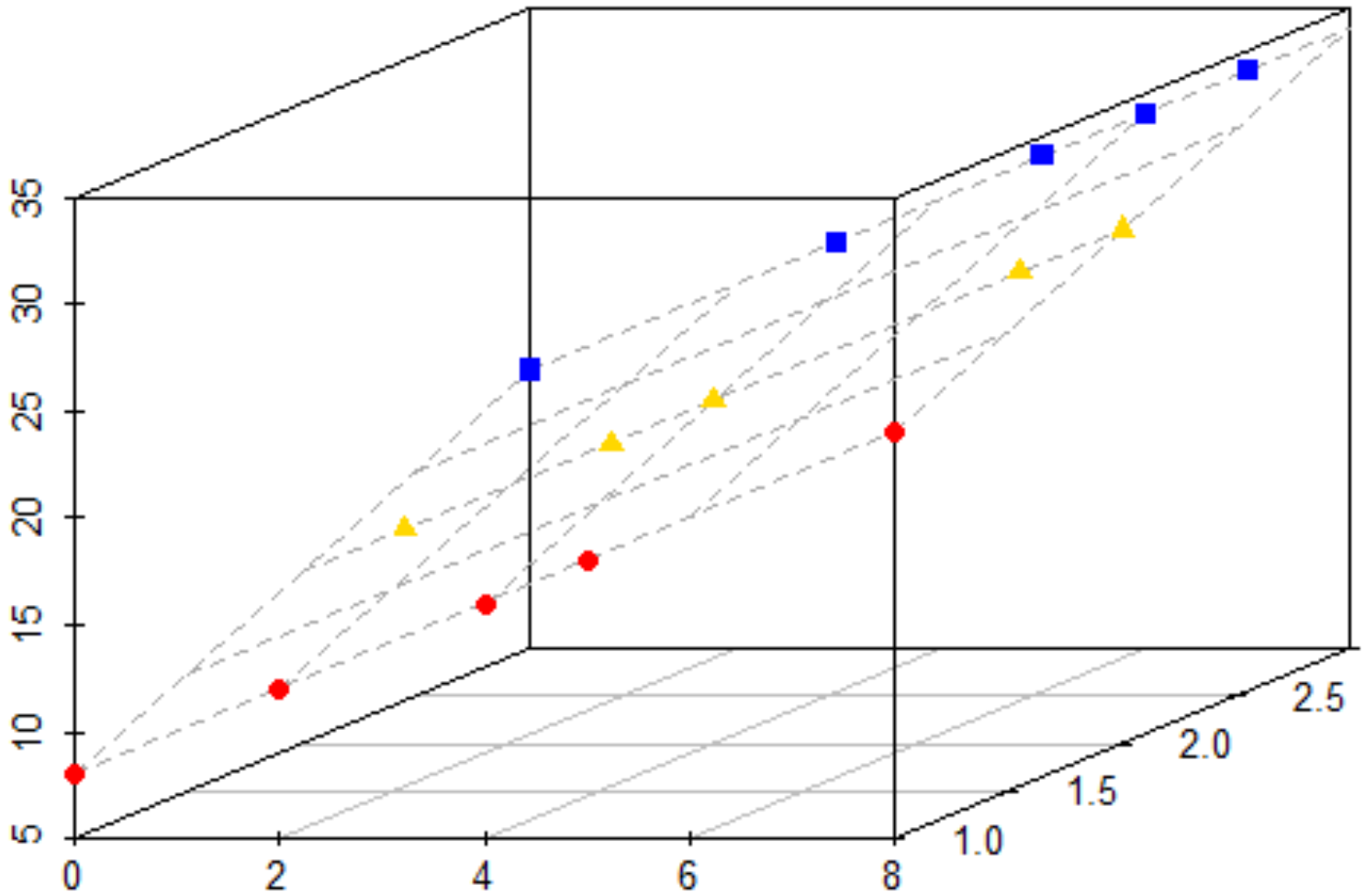
C=I ← same argument as before

```
C =

    1.0010    0.0020
    0.0020    1.0011


V =

   -0.7174    0.6966
    0.6966    0.7174


D =

    0.9991         0
         0    1.0030
```



Principal Component Analysis (PCA)

- Subspaces of maximal variance
- How to find multi-D lower-dimensional spaces that maximize total dispersion/variance = empirical average of squared distance from mean



Principal Component Analysis (PCA)

- Subspaces of maximal variance:

  When covariance matrix **C is diagonal** (and sample mean is at origin)

Let a 2D space be defined by (i) mean μ and (ii) orthogonal directions v_1 and v_2 through μ

Then the goal is to find $\arg\max_{v_1,v_2} v_1^\top C v_1 + v_2^\top C v_2$

under the constraints $\|v_1\| = 1$, $\|v_2\| = 1$, and $\langle v_1, v_2 \rangle = 0$

The objective function is

$$
\begin{aligned}
& v_1^\top C v_1 + v_2^\top C v_2 \\
&= \sum_{d=1}^{D} C_{dd} (v_1^d)^2 + \sum_{d=1}^{D} C_{dd} (v_2^d)^2 \\
&= \sum_{d=1}^{D} C_{dd} [(v_1^d)^2 + (v_2^d)^2]
\end{aligned}
$$

Optimal direction

$$
\begin{aligned}
&= \arg\max_{v:\|v\|_2=1} \textstyle\sum_i \langle x_i, v \rangle^2 / N \\
&= \arg\max_{v:\|v\|_2=1} \textstyle\sum_i (x_i^\top v)^2 / N \\
&= \arg\max_{v:\|v\|_2=1} \textstyle\sum_i (x_i^\top v)^\top (x_i^\top v) / N \\
&= \arg\max_{v:\|v\|_2=1} \textstyle\sum_i v^\top x_i x_i^\top v / N \\
&= \arg\max_{v:\|v\|_2=1} v^\top (\textstyle\sum_i x_i x_i^\top / N) v \\
&= \arg\max_{v:\|v\|_2=1} v^\top C v \text{ (where } C \text{ is the}
\end{aligned}
$$

Principal Component Analysis (PCA)

- Subspaces of maximal variance:
  When covariance matrix C is diagonal (and sample mean is at origin)

Now, consider v_1 and v_2 as the first two columns vectors of a basis (of D vectors) for the complete D-dimensional Euclidean space

Then, for any dimension d, we have $(v_1^d)^2 + (v_2^d)^2 + \cdots + (v_D^d)^2 = 1$ (each row in the basis has unit norm)

Thus, for any d, we have the constraint $(v_1^d)^2 + (v_2^d)^2 \leq 1$

Another constraint is: $\sum_{d=1}^{D}[(v_1^d)^2 + (v_2^d)^2] = 1 + 1 = 2$ (each column, i.e., v_1 and v_2, has unit norm)

Principal Component Analysis (PCA)

- Subspaces of maximal variance:

  When covariance matrix C is diagonal (and sample mean is at origin)

Let us define a vector a such that its d-th component $a^d := (v_1^d)^2 + (v_2^d)^2$

Then, the new objective function is $\sum_{d=1}^{D} C_{dd} a^d$

and the new constraints are: (i) $\forall d, 0 \leq a^d \leq 1$ and (ii) $\sum_d a^d = 2$ (two)

- Now this problem is similar to what we had solved before
- Like before, we have a "convex" combination of $\{C_{dd}\}$ with weights $\{0 \leq a^d \leq 1\}$
- So, increase a^1 to its limit (i.e., 1) and then increase a^2 to its limit (i.e., 1)

The solution for this problem is $a^1 = a^2 = 1$ and all other $a^d = 0$ (because $C_{11} \geq C_{22} \geq \cdots$)

Principal Component Analysis (PCA)

- Subspaces of maximal variance:

  When covariance matrix C is diagonal (and sample mean is at origin)

The solution for this problem is $a^1 = a^2 = 1$ and all other $a^d = 0$ (because $C_{11} \geq C_{22} \geq \cdots$)

That implies:

(i) $v_1^d = v_2^d = 0$ for all $d > 2$; so that optimal 2D space is the one spanned by the first two cardinal axes, and

(ii) to maximize $\sum_{d=1}^{D} C_{dd}[(v_1^d)^2 + (v_2^d)^2]$ (where $C_{11} \geq C_{22}$), we choose $v_1^1 = 1, v_2^1 = 0$ and $v_1^2 = 0, v_2^2 = 1$

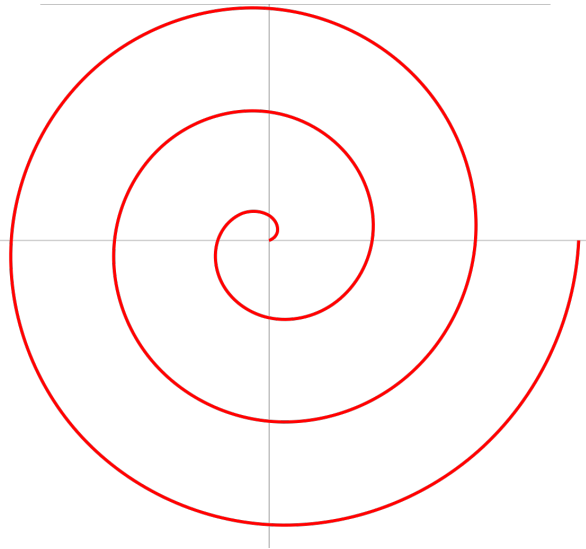
Thus, total variance in the (optimal) 2D space is $\sum_{d=1}^{D} C_{dd}[(v_1^d)^2 + (v_2^d)^2] = C_{11} + C_{22}$ that is the sum of the top two eigenvalues

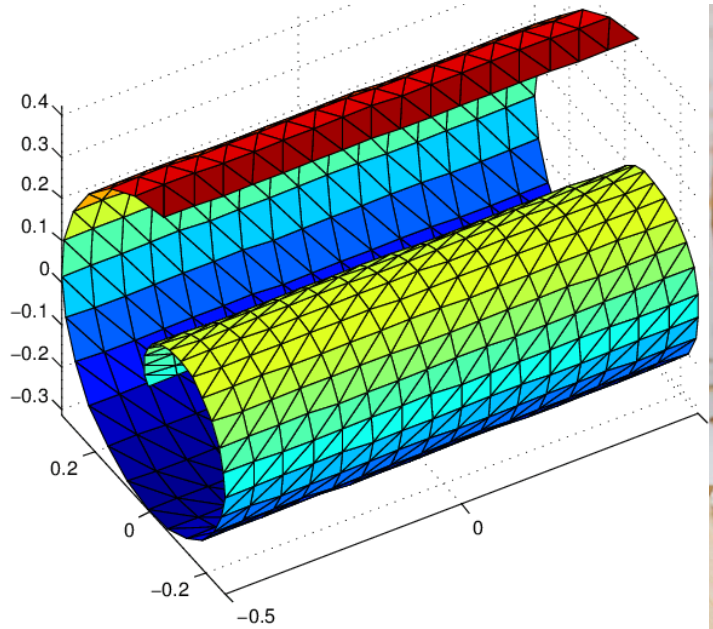
- Solution = any orthogonal basis within space spanned by first 2 cardinal axes
- **Similar arguments hold for lower-dimensional spaces of dimension 3, 4, …**
- **Similar arguments hold for a general SPD covariance matrix C**

Principal Component Analysis (PCA)

- Applications: **Dimensionality reduction**
  - **Intrinsic dimension**: Minimum number of variables (degrees of freedom) required to represent signal
    - Consider a multivariate random vector X of N scalar variables: $x = (x_1, \ldots, x_N)$
    - Consider a function g(.), and M<N scalar variables $a_1, \ldots, a_M$ such that **every $x^i \sim P(X)$** can be written as $x^i = \mathbf{g}(a^i_1, \ldots, a^i_M)$ for some $a^i_1, \ldots, a^i_M$, then signal X needs only M variables for representation
    - Here, "intrinsic dimension" of X is M, instead of the "representation dimension" = N

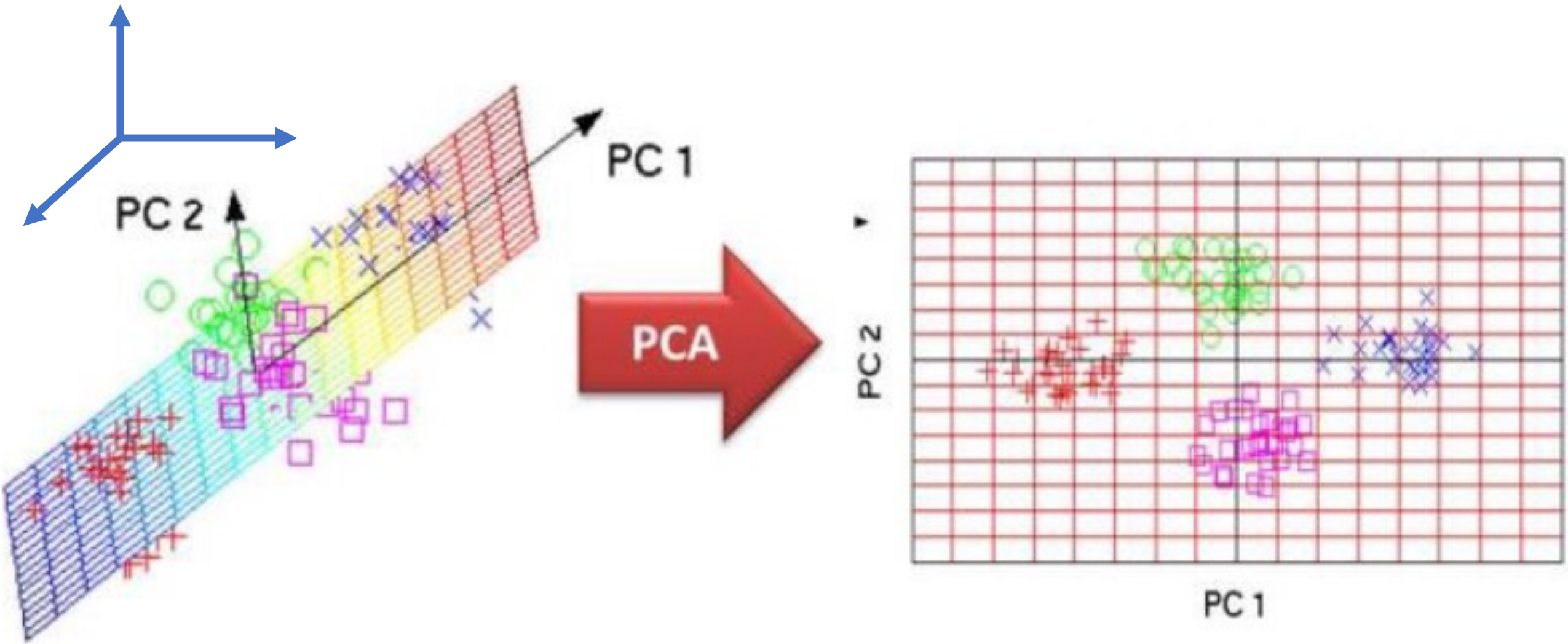
$$(b\theta\cos\theta, b\theta\sin\theta)$$





Principal Component Analysis (PCA)

- Applications: Dimensionality reduction



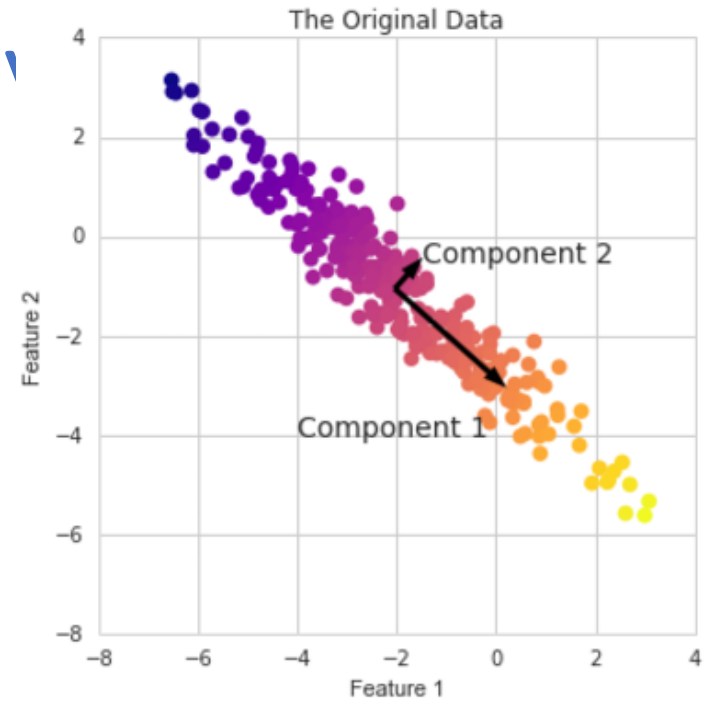
https://medium.com/analytics-vidhya/dimensionality-reduction-using-principal-component-analysis-pca-41e364615766

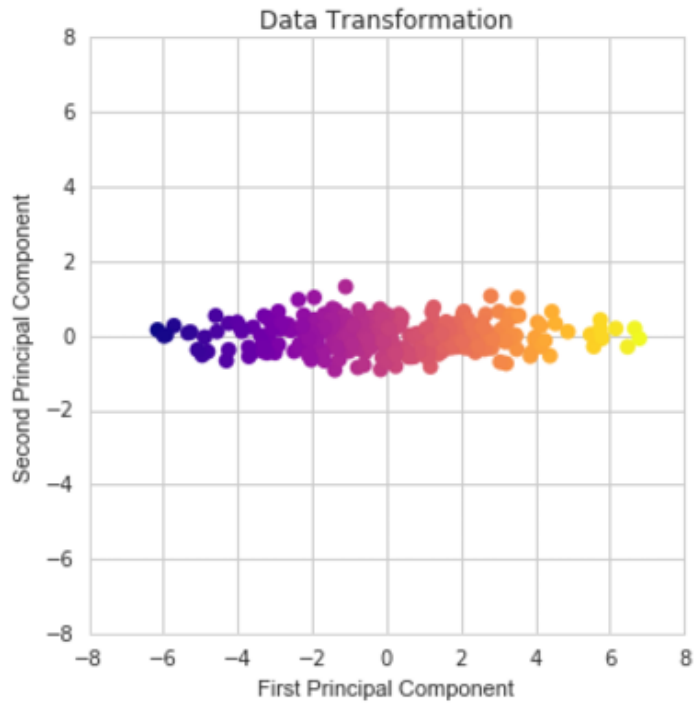
Principal Component Analysis (PCA)

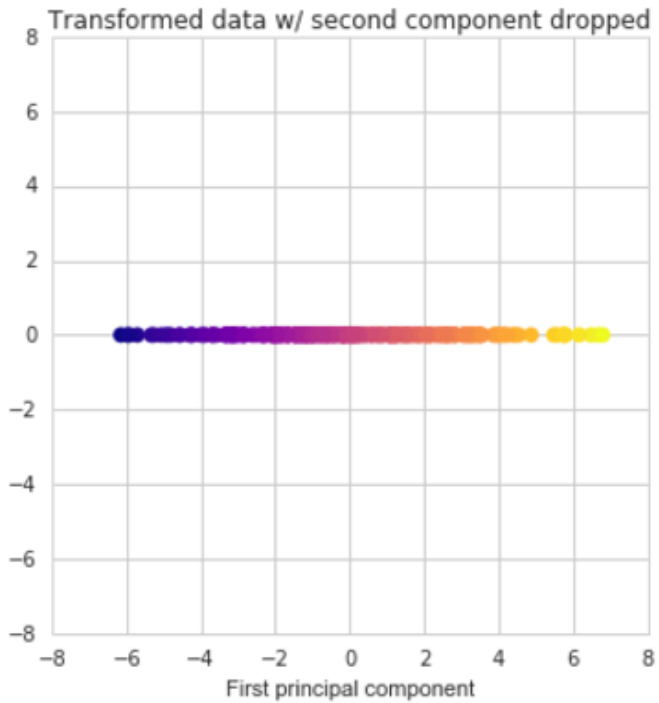
- Applications: Dimensionality reduction
- Acquired data is corrupted with errors
  - e.g., measurement errors
  - Such errors make signal representation seem to be of dimension higher than intrinsic dimension
- Dimensionality reduction:
  Transformation of data
  from a higher-dimensional space into a lower-dimensional space
  so that
  lower-D representation (ideally close to its intrinsic dimension)
  retains some meaningful properties of original data
- PCA can perform **linear (transformation)** dimensionality reduction

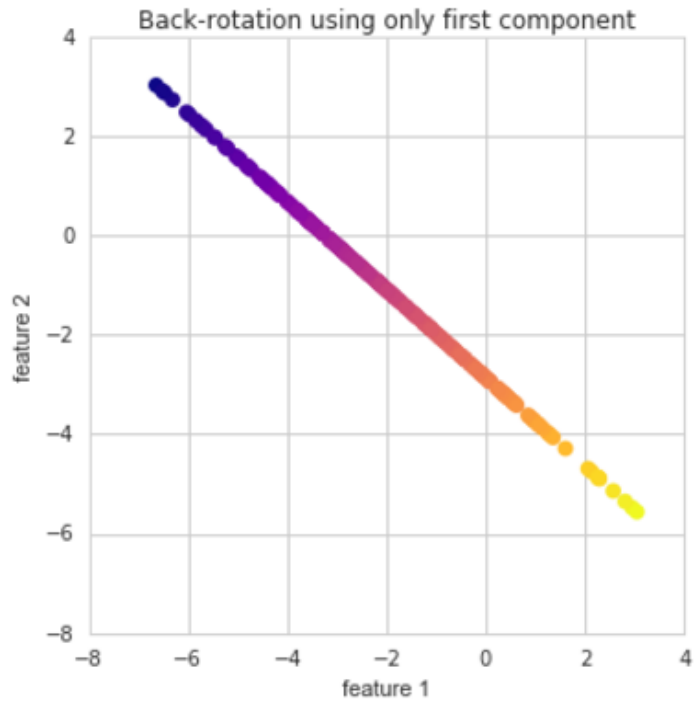
Principal Component Anal

- Applications: Dimensionality reduction
- Using PCA
- X may be N dimensional
- PCA finds an M-dimensional space that captures most of variability/variance in data





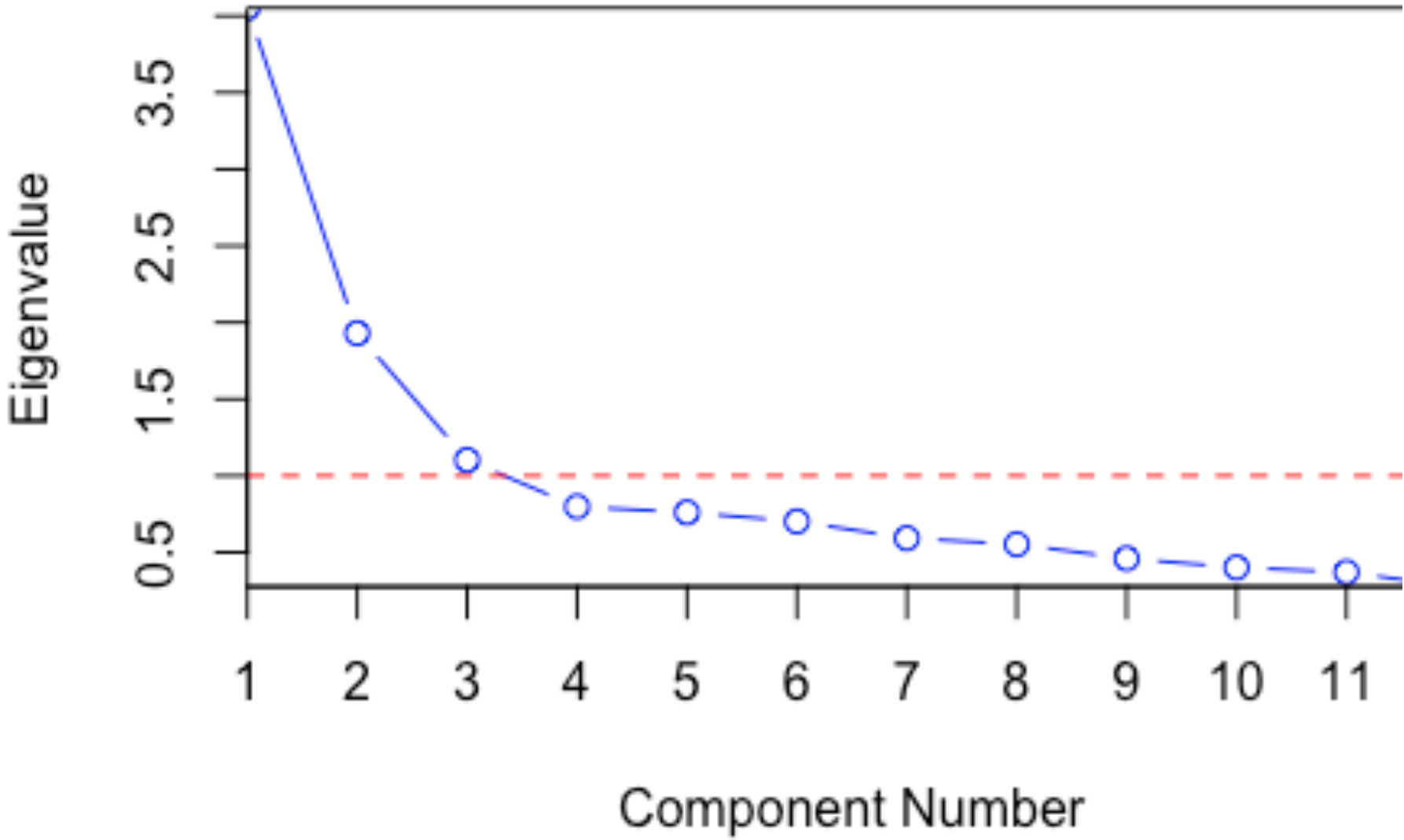




bennymachinelearning.blogspot.com/2017/08/machine-learning-principal-component.html

Principal Component Analysis (PCA)

- Applications: Dimensionality reduction
- Using PCA
- X may be N dimensional
- PCA can find an M-dimensional space (often $M \ll N$) that captures most of variability/variance in data



Multivariate Gaussian – Mahalanobis Distance

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^{\top} C^{-1}(y-\mu))$$

- Term $(y-\mu)^{\top} C^{-1} (y-\mu)$ appearing in exponent
  = squared Mahalanobis distance
  of point y from mean μ
- $d(y,\mu; C)^2$:= $(y-\mu)^{\top} C^{-1} (y-\mu)$ (where C is SPD)
- Prasanta Chandra Mahalanobis
  founded
  Indian Statistical Institute (ISI) in Kolkata

Multivariate Gaussian – Mahalanobis

Multivariate Gaussian – Mahalanobis Distance

$$p(y) = \frac{1}{(2\pi)^{D/2}|C|^{0.5}} \exp(-0.5(y-\mu)^{\top} C^{-1}(y-\mu))$$

- $d(y,\mu; C)^2 := (y-\mu)^T C^{-1} (y-\mu)$ (where C is SPD)
- Generalizes Euclidean distance in a multidimensional space
- When C is Identity:
  - Mahalanobis distance = Euclidean distance
- When C is diagonal:
  - Mahalanobis distance rescales “units” along each dimension based on standard deviation of marginal along that dimension

Multivariate Gaussian – Mahalanobis Distance

- $d(y,\mu; C)^2 := (y-\mu)^T C^{-1} (y-\mu)$ (where C is SPD)
- Property: Mahalanobis distance is a "distance metric"
- Proof:
  - A distance metric is a function $d(.,.) \rightarrow R$ that needs to satisfy 3 properties:

  1) identity of indiscernibles: $d(x,y) = 0$ iff $x = y$
  2) symmetry: $d(x,y) = d(y,x)$
  3) triangle inequality: $d(x,y) \leq d(x,z) + d(z,y)$

  - These imply non-negativity (i.e., $d(x,z) \geq 0$, for all x,z):
    $0 = d(x,x) \leq d(x,z) + d(z,x) = 2\, d(x,z)$
  - In our case of SPD matrix C:

  1) C being S**PD** implies: $d(x,y; C) = 0$ iff $x=y$
  2) Because of specific quadratic form of $d(.,.; C)^2$: $d(x,y; C) = d(y,x; C)$

  - C being S**PD** implies: $d(x,y; C) \geq 0$ for all x,y

Multivariate Gaussian – Mahalanobis Distance

- Mahalanobis distance is a distance metric (for **diagonal C**):

3) Triangular inequality: $d(x,y) \leq d(x,z) + d(z,y), \forall x, y, z$

Let $u := x - z$ and $v := z - y$

Then $u + v = x - y$

$$\text{LHS} = \sqrt{(u+v)^\top C^{-1}(u+v)}$$

$$\text{RHS} = \sqrt{u^\top C^{-1} u} + \sqrt{v^\top C^{-1} v}$$

$$\text{LHS}^2 = (u+v)^\top C^{-1}(u+v)$$

$$= \sum_d (u_d + v_d)^2 / \sigma_d^2 \text{ (assuming } C \text{ is diagonal)}$$

$$= \sum_d u_d^2/\sigma_d^2 + \sum_d v_d^2/\sigma_d^2 + 2\sum_d u_d v_d/\sigma_d^2$$

- 1st 2 terms same
- Cauchy-Shwartz for 3rd terms

$$\text{RHS}^2 = u^\top C^{-1} u + v^\top C^{-1} v + 2\sqrt{u^\top C^{-1} u}\sqrt{v^\top C^{-1} v}$$

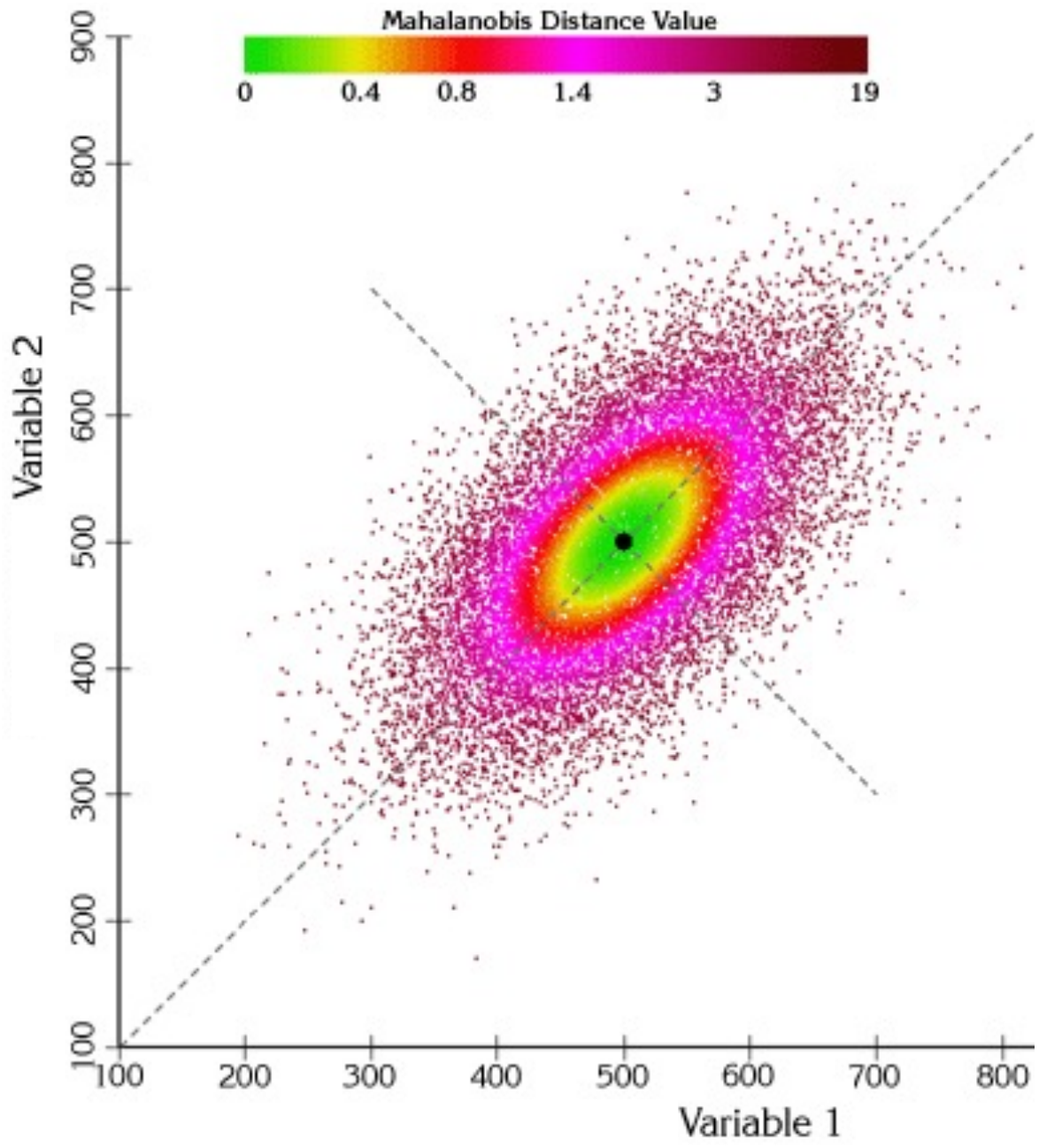
$$= \sum_d u_d^2/\sigma_d^2 + \sum_d v_d^2/\sigma_d^2 + 2\sqrt{\sum_d u_d^2/\sigma_d^2}\sqrt{\sum_d v_d^2/\sigma_d^2}$$

Multivariate Gaussian – Mahalanobis Distance

- Mahalanobis distance is a distance metric (for **general SPD C**):
- Use eigen-decomposition of $C = VDV^T$
- Define $a := V^T (x - z)$ and $b := V^T (z - y)$
- Process with same analysis as before

- **A level set of a multivariate Gaussian PDF is locus of points with equal Mahalanobis distance from mean**
  - See next slide …

Multivariate Gaussian – Mahalanobis Distance

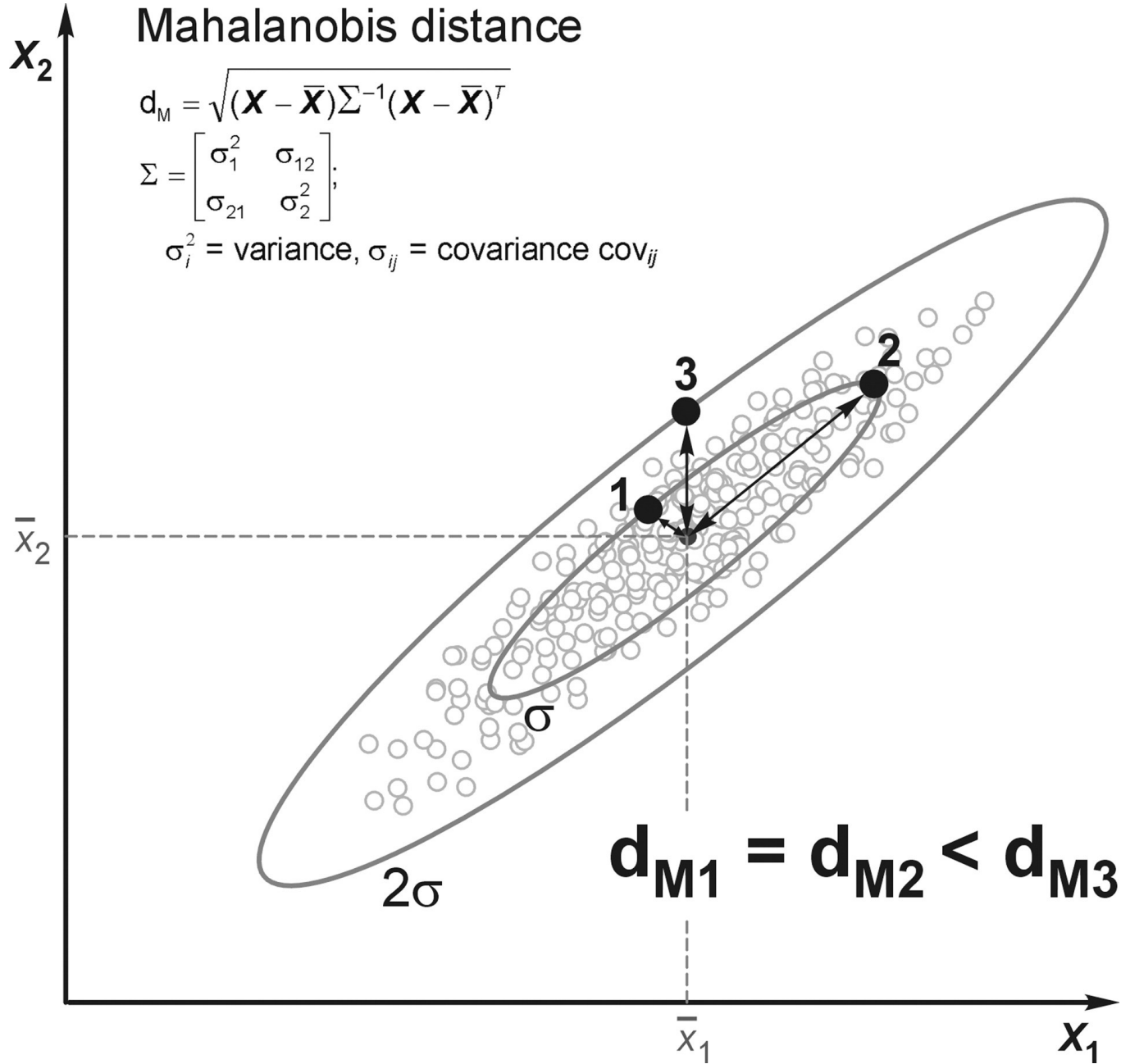


Mahalanobis distance

$$d_M = \sqrt{(\boldsymbol{X} - \bar{\boldsymbol{X}})\Sigma^{-1}(\boldsymbol{X} - \bar{\boldsymbol{X}})^T}$$

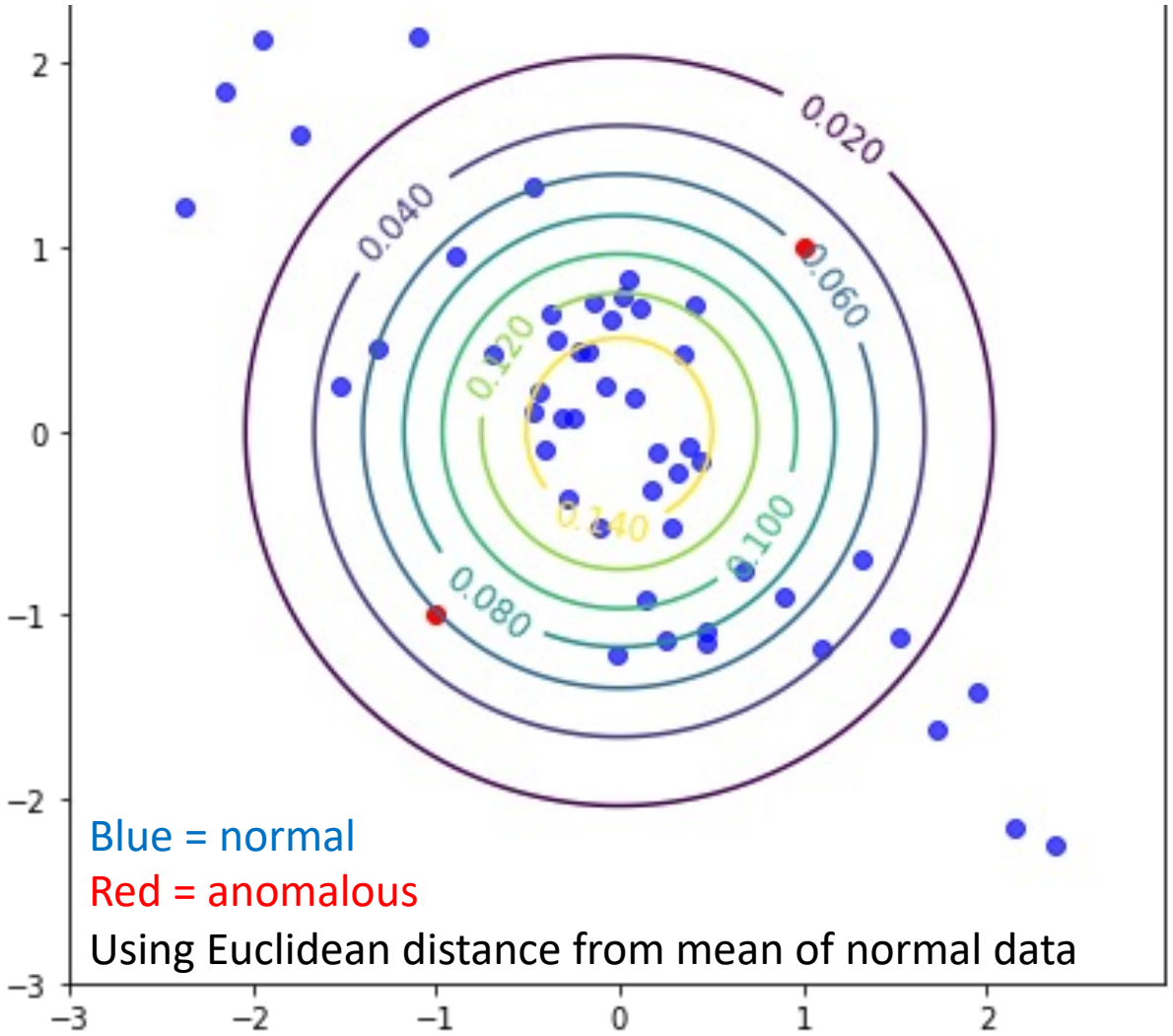
$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix};$$

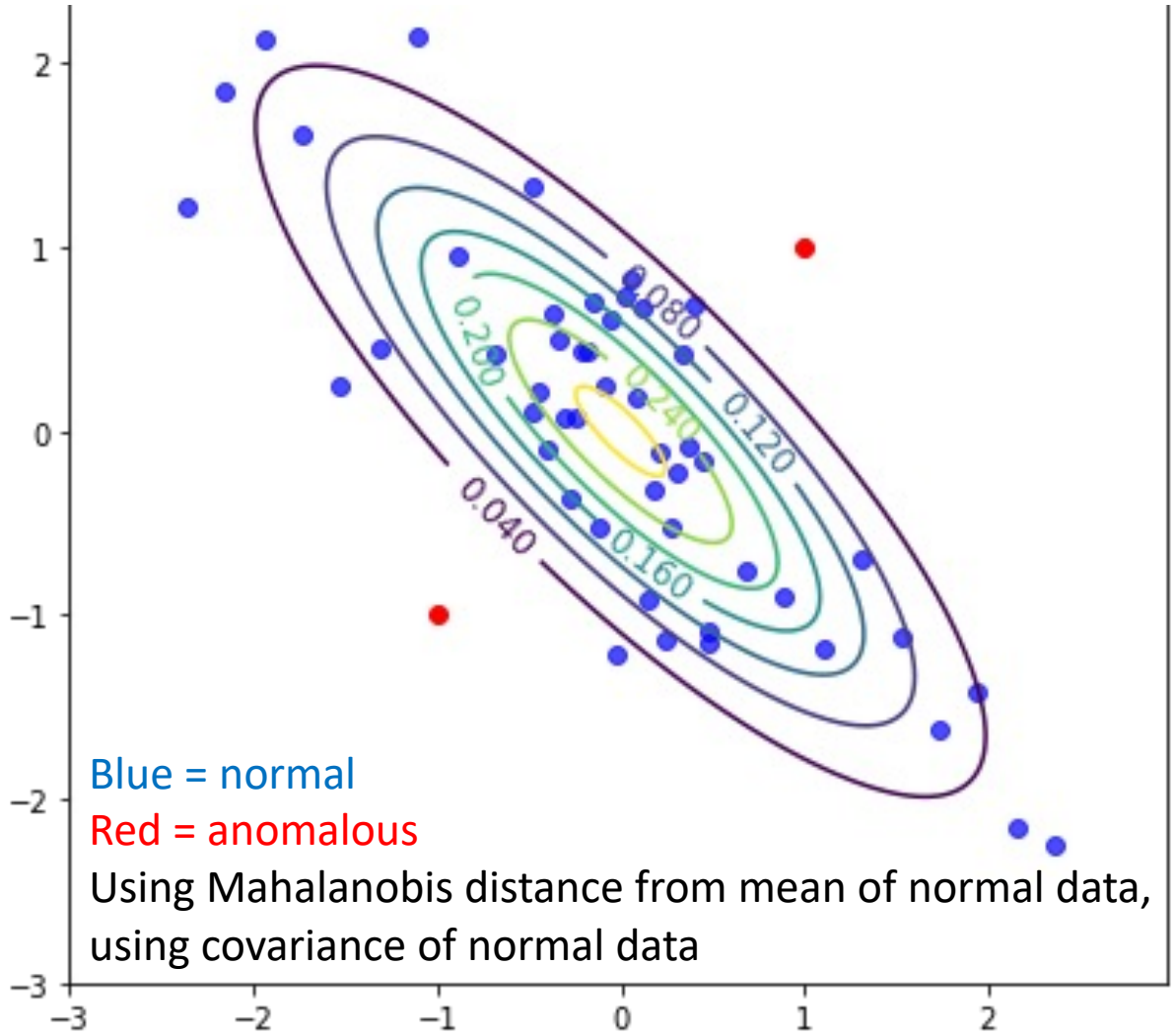
σ_i^2 = variance, σ_{ij} = covariance cov_{ij}



Multivariate Gaussian – Applications

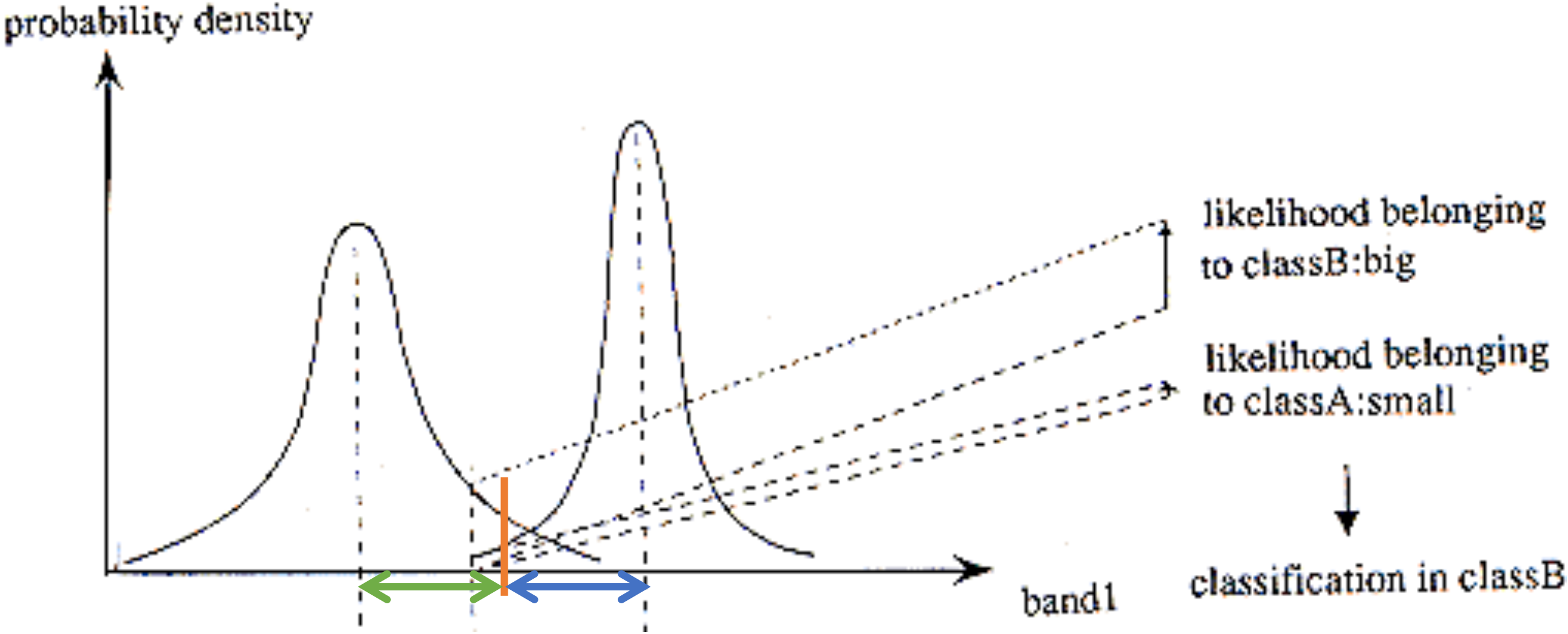
- Multivariate Gaussian (Mahalanobis distance) for anomaly detection





Multivariate Gaussian – Applications

- Multivariate Gaussian for maximum-likelihood classification (in 1D)
  - Effect of Euclidean distance vs. Mahalanobis distance



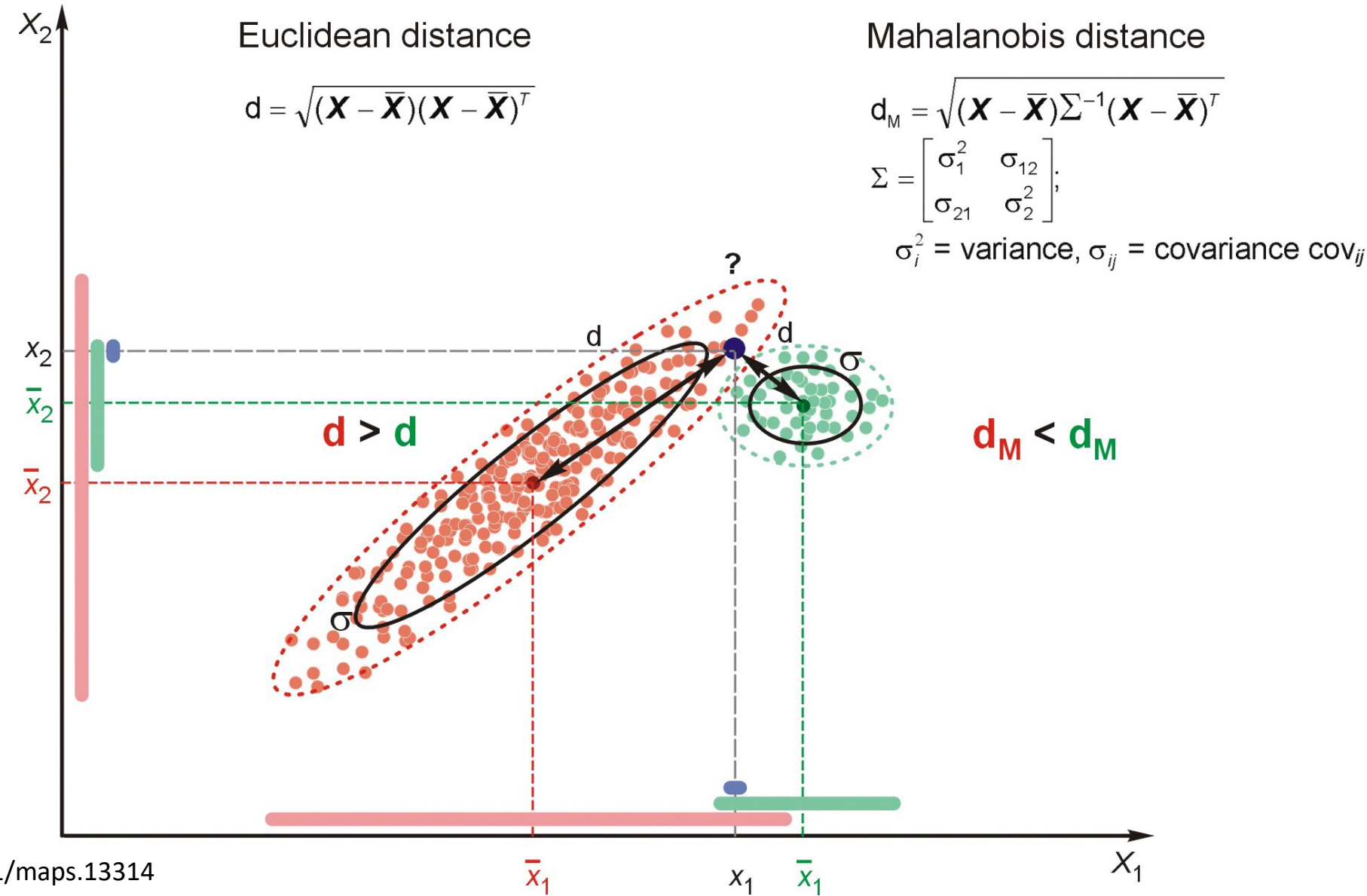
http://sar.kangwon.ac.kr/etc/rs_note/rsnote/cp11/cp11-7.htm

Multivariate Gaussian – Applications

- Multivariate Gaussian for maximum-likelihood classification
- How do decision boundaries look like ?
- $P(x|Class_1) = G(x; m_1, C_1)$
- $P(x|Class_2) = G(x; m_2, C_2)$
- Decision boundary comprises all points x at which likelihoods are equal
  - $\{ x : P(x|Class_1) = P(x|Class_2) \}$
  - $\{ x : 0 = \log (P(x|Class_1) / P(x|Class_2)) \}$
- At any point in the domain 'x', the log likelihood-ratio =
  $\log (P(x|Class_1) / P(x|Class_2)) =$
  $-0.5 (x - m_1)^T C_1^{-1} (x - m_1) - 0.5 \log (\det (C_1))$
  $+ 0.5 (x - m_2)^T C_2^{-1} (x - m_2) + 0.5 \log (\det (C_2))$
- In general, decision surface is a hyper-quadric

Multivariate Gaussian – Applications

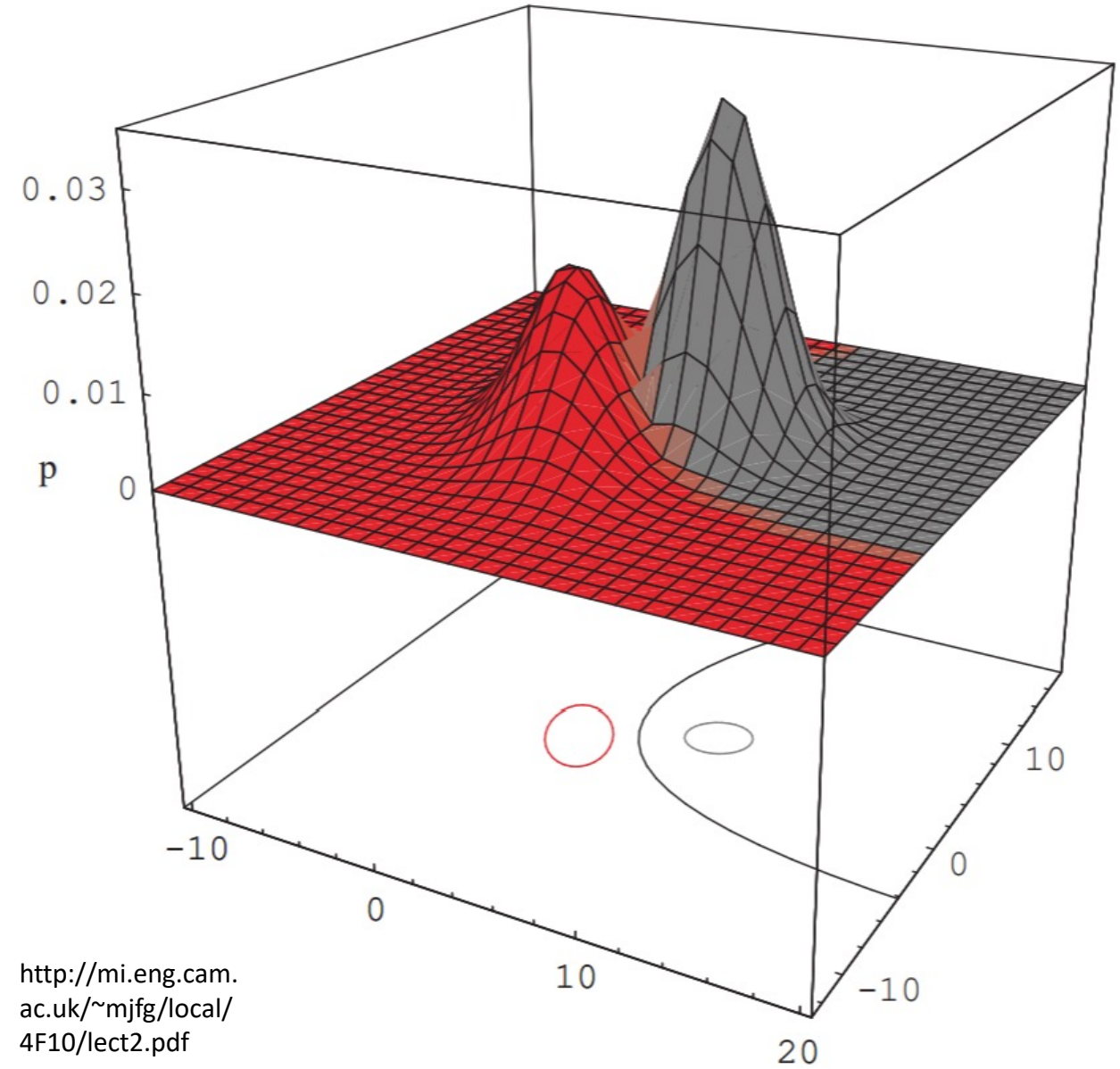
- Multivariate Gaussian for maximum-likelihood **classification (in 2D)**
- If $\det(C_1)=\det(C_2)$, then likelihood-based classification is equivalent to Mahalanobis-distance based classification

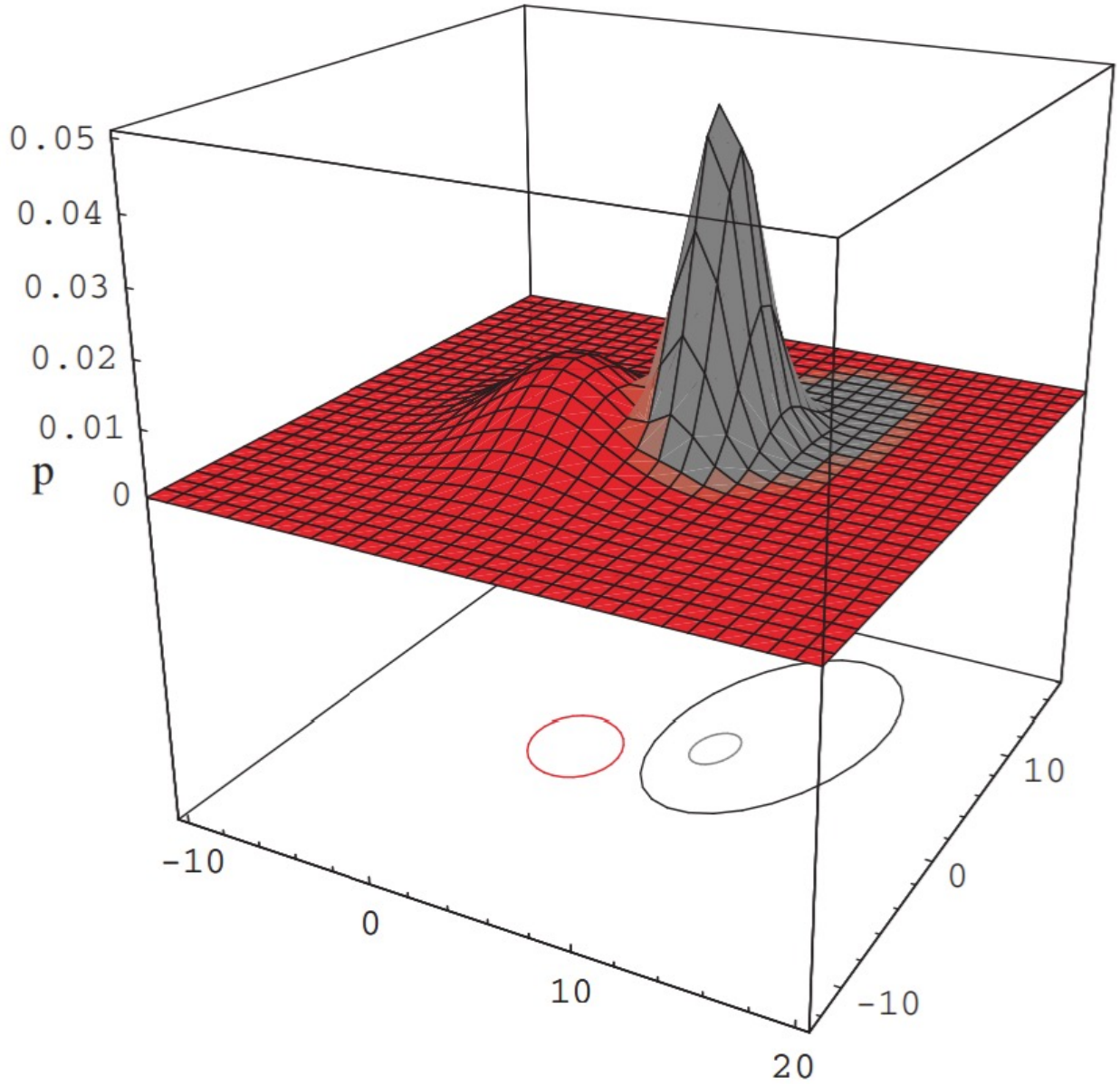


https://onlinelibrary.wiley.com/doi/full/10.1111/maps.13314

Multivariate Gaussian – Applications

- Multivariate Gaussian for ML classification: decision boundaries

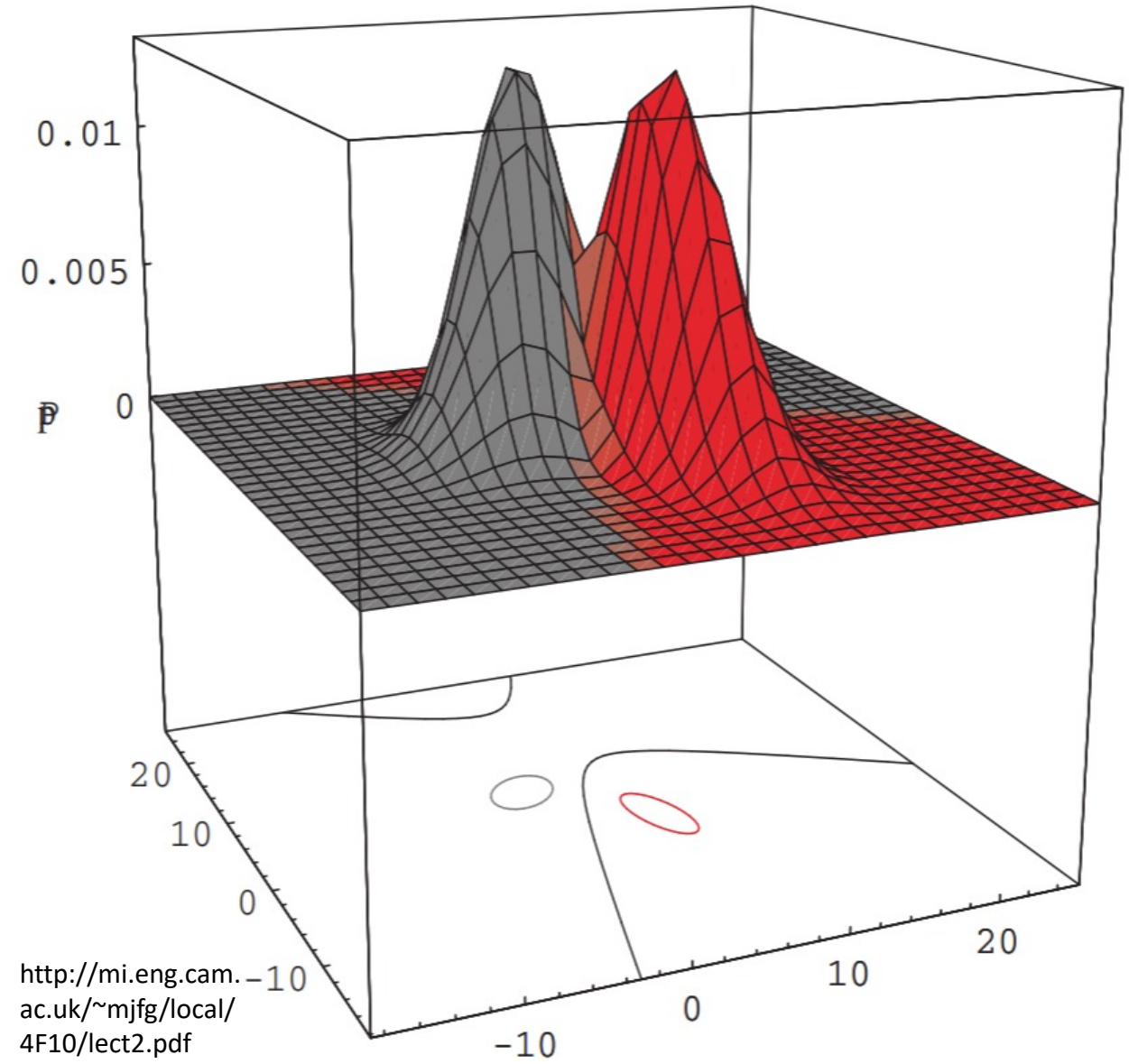


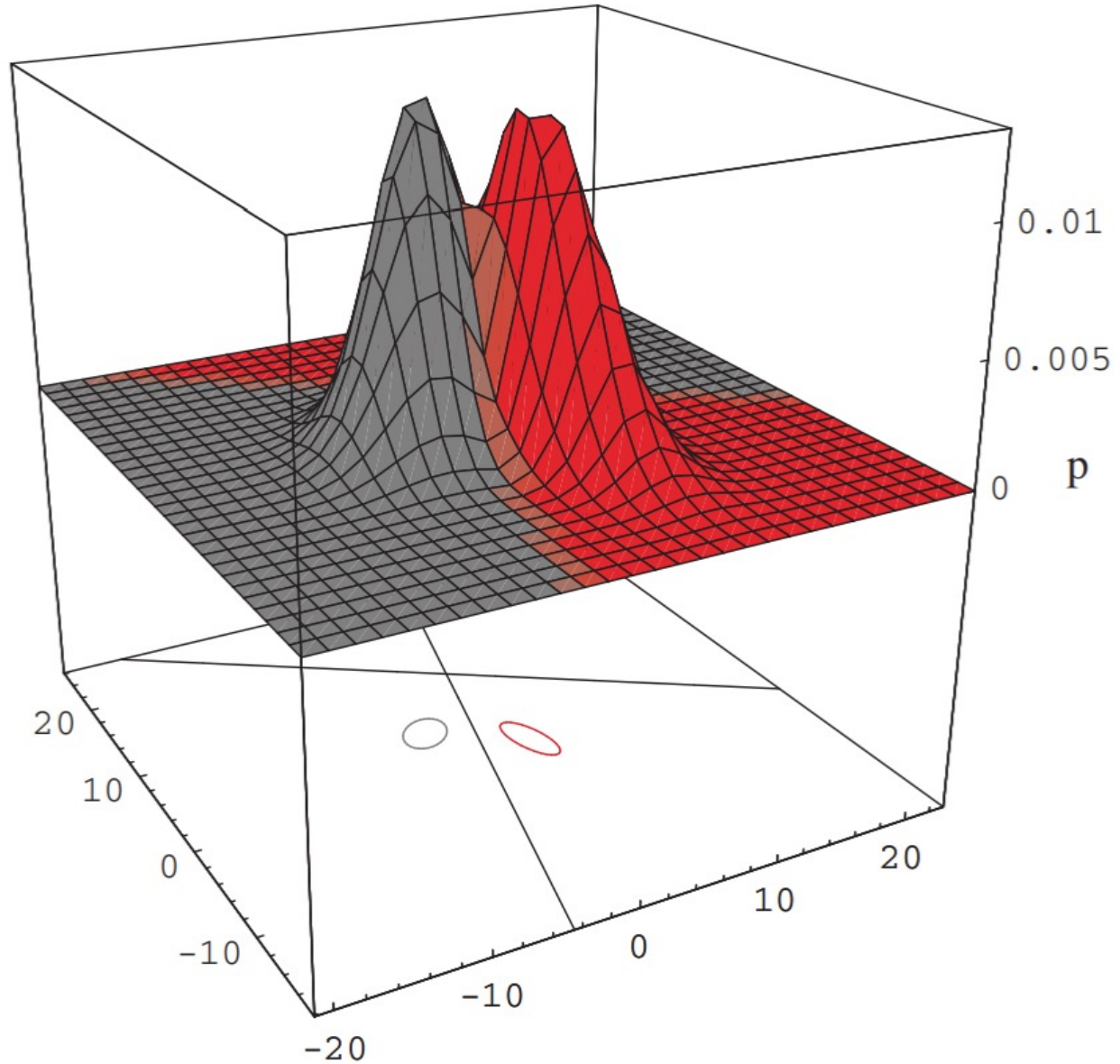


http://mi.eng.cam.ac.uk/~mjfg/local/4F10/lect2.pdf

Multivariate Gaussian – Applications

- Multivariate Gaussian for ML classification: decision boundaries

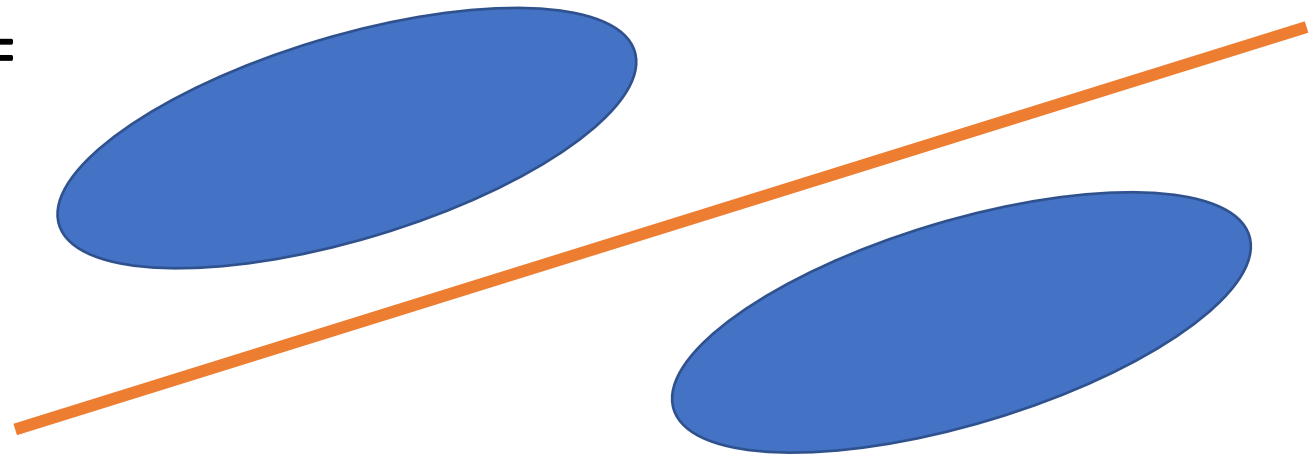




http://mi.eng.cam.ac.uk/~mjfg/local/4F10/lect2.pdf

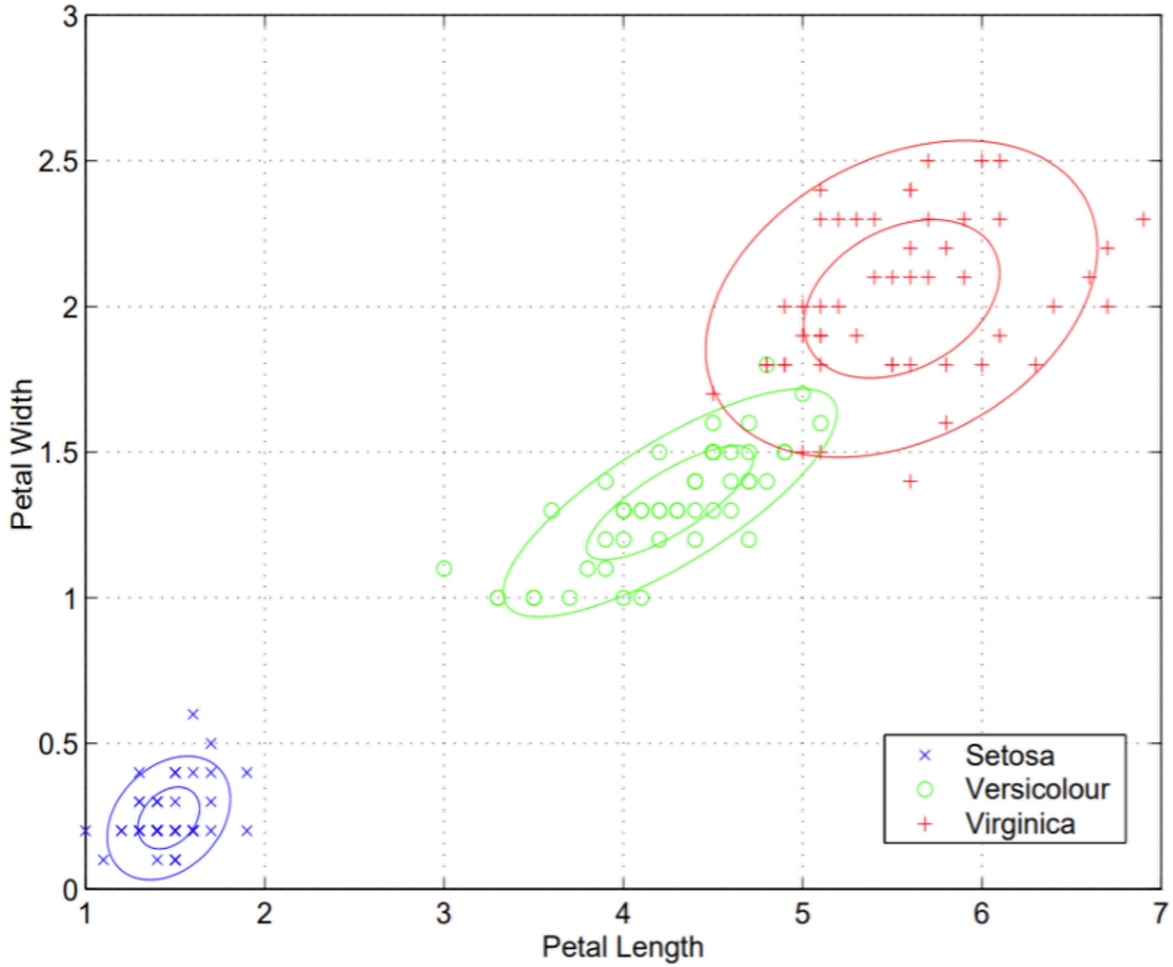
Multivariate Gaussian – Applications

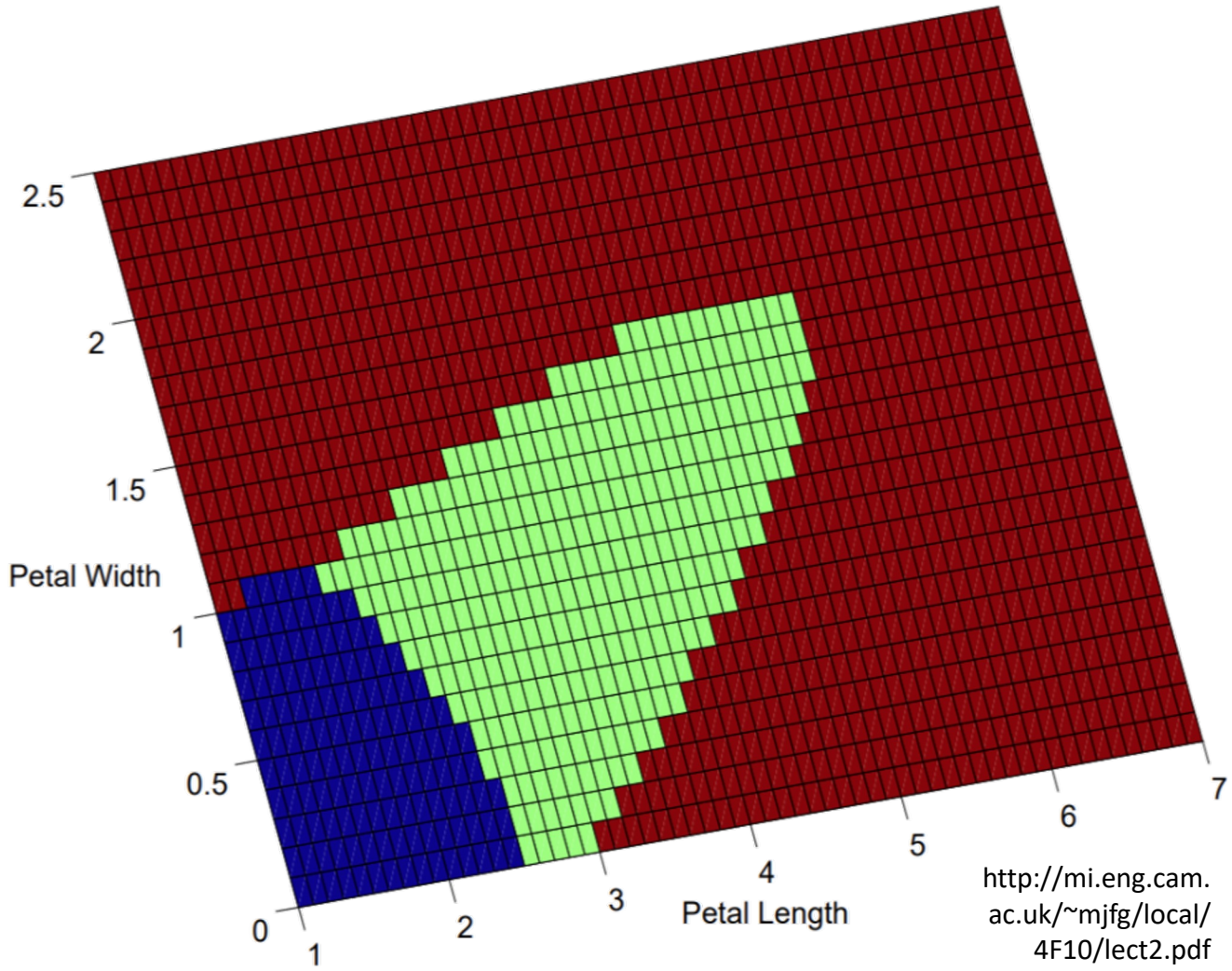
- Multivariate Gaussian for maximum-likelihood **classification**
- Decision boundaries
- When $C_1 = C_2 = C$, then decision boundary is:
- $0 = \log (P(x|Class_1) / P(x|Class_2)) =$
  $-0.5\,(x - m_1)^T C^{-1} (x - m_1)$
  $+0.5\,(x - m_2)^T C^{-1} (x - m_2)$
- 0
  $=$
  $+ (m_2 - m_1)^T C^{-1} x$
  $+0.5\, m_1^T C^{-1} m_1$
  $-0.5\, m_2^T C^{-1} m_2$
- Decision surface is a hyper-plane



Multivariate Gaussian – Applications

- Multivariate Gaussian for maximum-likelihood **classification**
- archive.ics.uci.edu/ml/datasets/Iris





http://mi.eng.cam.ac.uk/~mjfg/local/4F10/lect2.pdf

PCA for Face Recognition

Reprinted from Journal of the Optical Society of America A, Vol. 4, page 519, March 1987
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Low-dimensional procedure for the characterization of human faces

L. Sirovich and M. Kirby

Division of Applied Mathemetics, Brown University, Providence, Rhode Island 02912

Received August 25, 1986; accepted November 10, 1986

A method is presented for the representation of (pictures of) faces. Within a specified framework the representation is ideal. This results in the characterization of a face, to within an error bound, by a relatively low-dimensional vector. The method is illustrated in detail by the use of an ensemble of pictures taken for this purpose.

PCA for F

- IEEE CVPR 1991

Face Recognition Using Eigenfaces

Matthew A. Turk and Alex P. Pentland

Vision and Modeling Group, The Media Laboratory
Massachusetts Institute of Technology

Abstract

We present an approach to the detection and identification of human faces and describe a working, near-real-time face recognition system which tracks a subject's head and then recognizes the person by comparing characteristics of the face to those of known individuals. Our approach treats face recognition as a two-dimensional recognition problem, taking advantage of the fact that faces are are normally upright and thus may be described by a small set of 2-D characteristic views. Face images are projected onto a feature space ("face space") that best encodes the variation among known face images. The face space is defined by the "eigenfaces", which are the eigenvectors of the set of faces; they do not necessarily correspond to isolated features such as eyes, ears, and noses. The framework provides the ability to learn to recognize new faces in an unsupervised manner.

another. The approach transforms face images into a small set of characteristic feature images, called "eigenfaces", which are the principal components of the initial training set of face images. Recognition is performed by projecting a new image into the subspace spanned by the eigenfaces ("face space") and then classifying the face by comparing its position in face space with the positions of known individuals.

Automatically learning and later recognizing new faces is practical within this framework. Recognition under reasonably varying conditions is achieved by training on a limited number of characteristic views (e.g., a "straight on" view, a 45° view, and a profile view). The approach has advantages over other face recognition schemes in its speed and simplicity, learning capacity, and relative insensitivity to small or gradual changes in the face image.

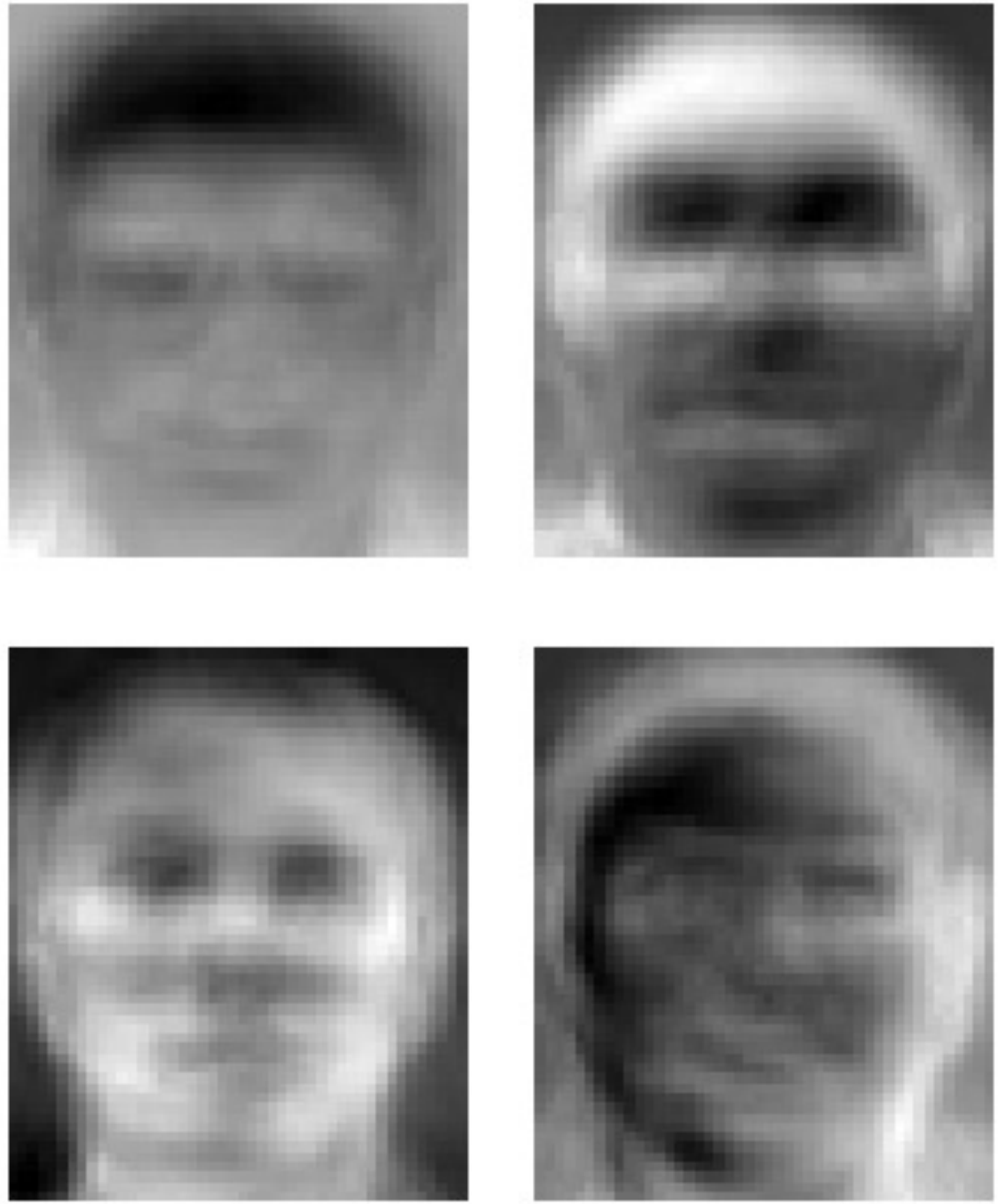
1.1 Background and related work

PCA for Face Recognition

- AT&T Face Dataset
- git-disl.github.io/GTDLBench/datasets/att_face_dataset/
- 40 persons
- 10 images per person

PCA for Face Recognition

- en.wikipedia.org/wiki/Eigenface



PCA for Face Recogni

- Turk and Pentland 1991 CVPR
- Algorithm for face recognition: Summary
  - "Face Space" determined by a set of principal eigenvectors

1. **Initialization: Acquire the training set of face images and calculate the eigenfaces, which define the *face space*.**
2. **When a new face image is encountered, calculate a set of weights based on the input image and the M eigenfaces by projecting the input image onto each of the eigenfaces.**
3. **Determine if the image is a face at all (whether known or unknown) by checking to see if the image is sufficiently close to "face space."**
4. **If it is a face, classify the weight pattern as either a known person or as unknown.**
5. **(Optional) If the same unknown face is seen several times, calculate its characteristic weight pattern and incorporate into the known faces (i.e., learn to recognize it).**

PCA for Face Recognition

- Turk and Pentland 1991 CVPR
- Algorithm for face recognition
- Case 1: face of person recognized + identified
  - May use kNN classifier
- Case 2: face of unknown person
- Case 3: not a face
- Case 4: not a face

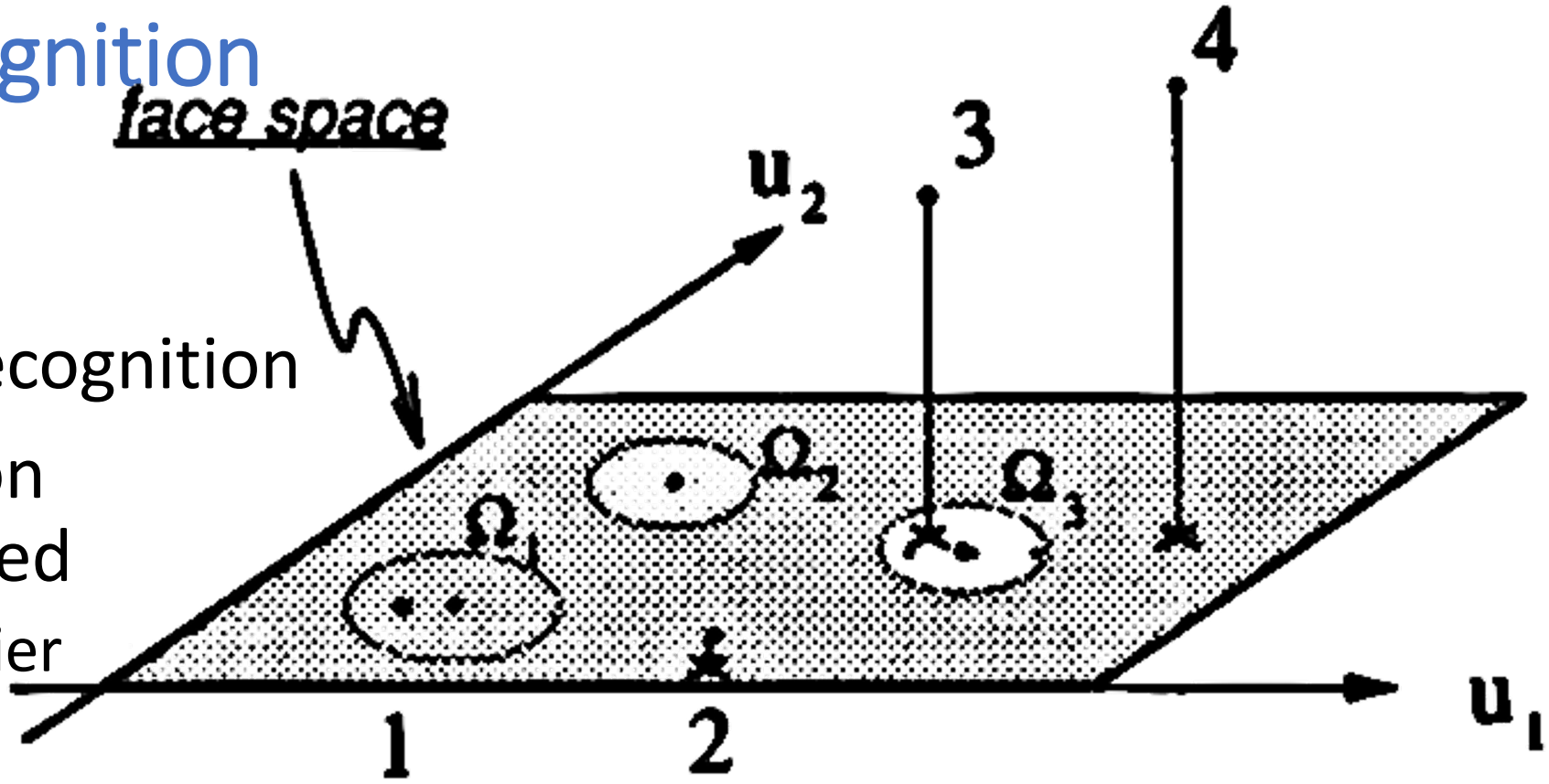


Figure 5: A simplified version of face space to illustrate the four results of projecting an image into face space. In this case, there are two eigenfaces (u_1 and u_2) and three known individuals (Ω_1, Ω_2, and Ω_3).

Modern Face Recognition

- viso.ai/computer-vision/deepface/

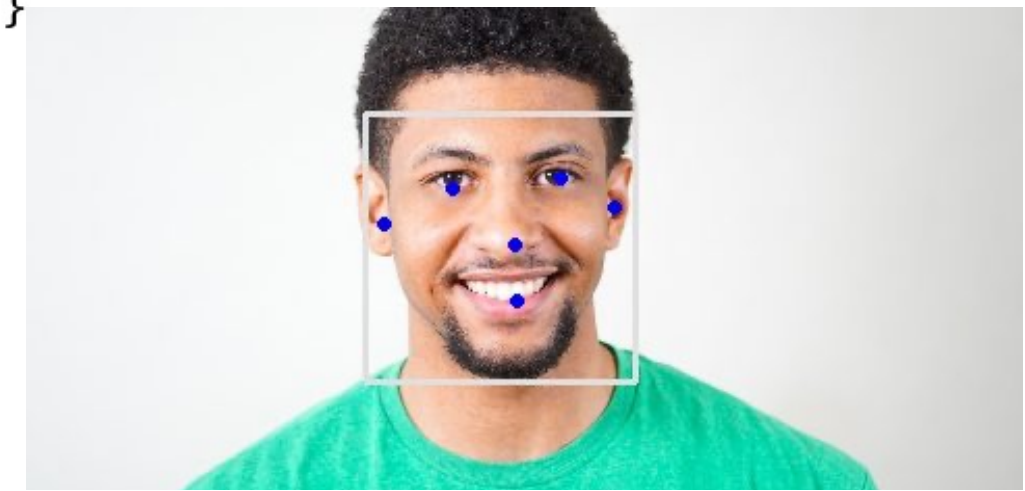
DeepFace: A Popular Open Source Facial Recognition Library

How to use the DeepFace Library to apply Deep Face Recognition and Facial Attribute Analysis with the most powerful Face Recognition models.

Nico Klingler

January 29th, 2023


```
{
  "verified": True,
  "distance": 0.32450073146188274,
  "max_threshold_to_verify": 0.4,
  "model": "VGG-Face",
  "similarity_metric": "cosine"
}
```

Modern Face Recognition

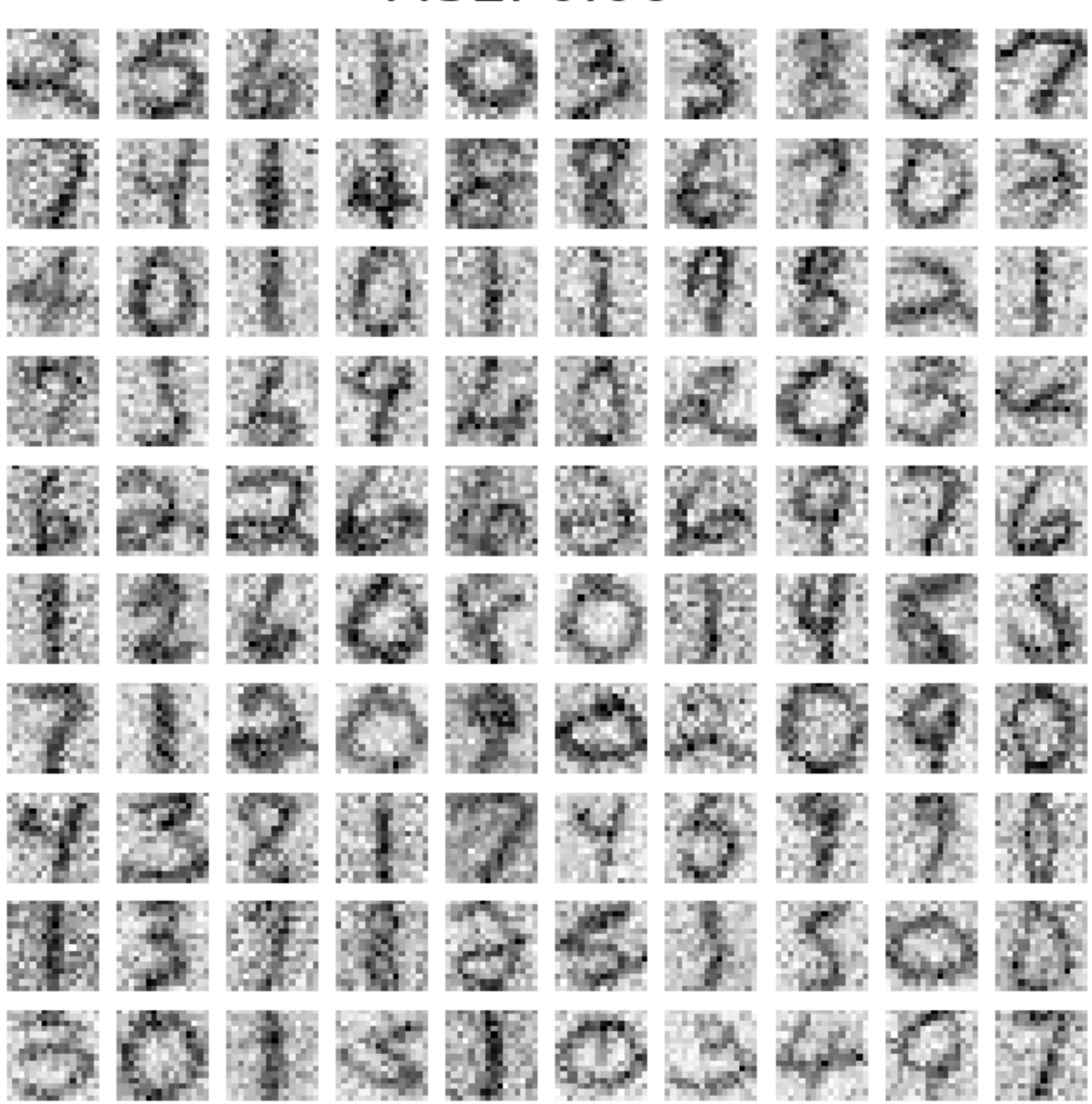
```
{
   "emotion":{
      "angry":7.603101671639384e-14,
      "disgust":2.7474185705216866e-21,
      "fear":1.688688161735822e-14,
      "happy":100.0,
      "sad":4.205067717644173e-10,
      "surprise":7.103817571484745e-13,
      "neutral":4.4851553027136504e-08
   },
   "dominant_emotion":"happy",
   "age":31,
   "gender":"Woman",
   "race":{
      "asian":0.9087088517844677,
      "indian":1.1444833129644394,
      "black":0.09399998234584928,
      "white":66.56872034072876,
      "middle eastern":16.655877232551575,
      "latino hispanic":14.628209173679352
   },
   "dominant_race":"white"
}
```

PCA for Image Denoising

- USPS dataset
- en.wikipedia.org/wiki/MNIST_database#USPS_database
- 16×16 grayscale images digitized from handwritten zip codes that appeared on U.S. mail passing through Buffalo, New York post office
- Uncorrupted images →

PCA for Image Denoising

- USPS dataset
- Noisy images
- Mean square error (averaged across set) = 0.06



PCA for Image Denoising

- Noisy USPS dataset

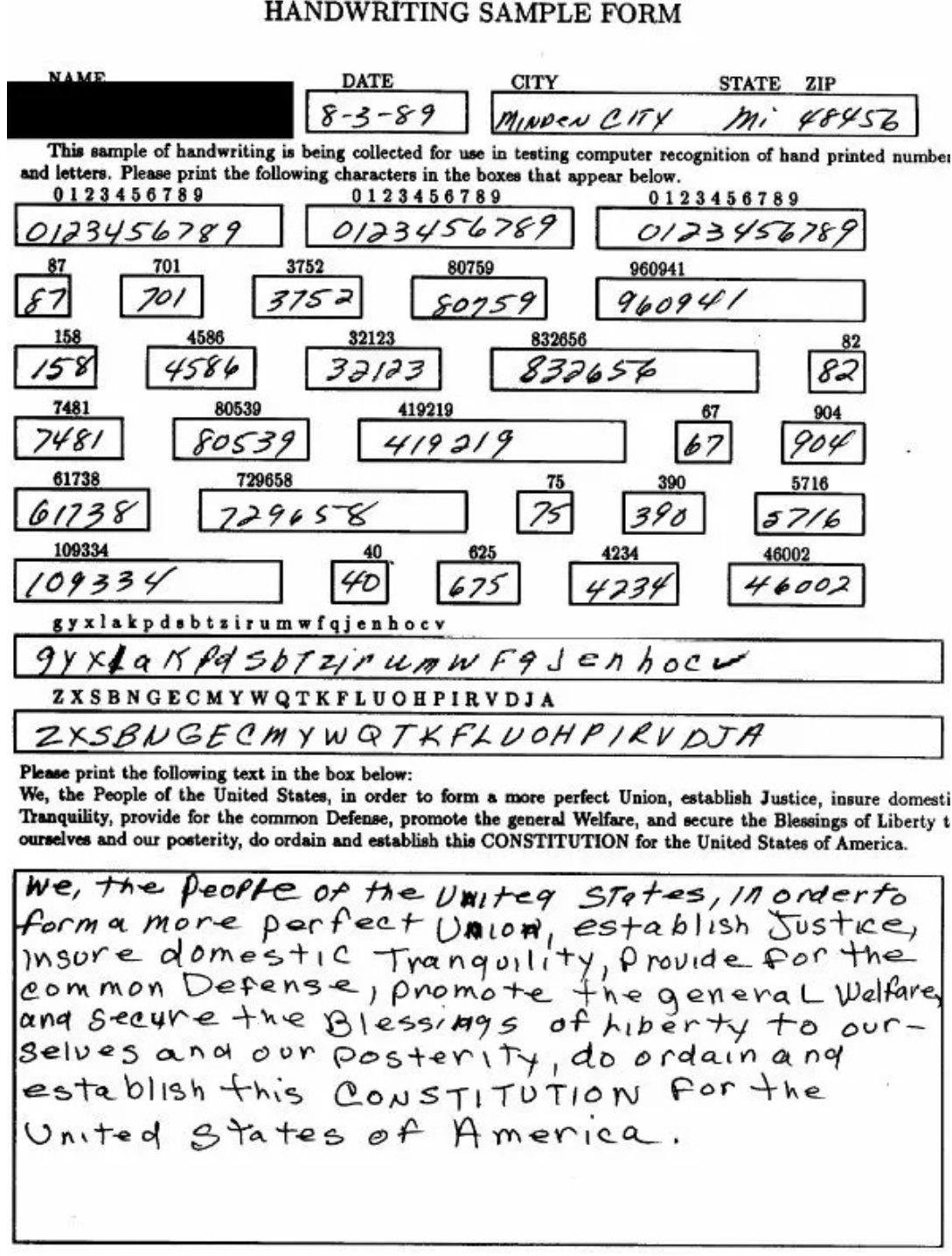
1) Learn principal subspace
   using PCA
   on noisy images
2) Project onto
   principal subspace

- Mean square error
  (averaged across set)
  = 0.01

scikit-learn.org/stable/auto_examples/applications/plot_digits_denoising.html

NIST Special Database 19

- www.nist.gov/srd/nist-special-database-19
- Contains NIST's entire corpus of training materials for handprinted document and character recognition
- Handprinted Sample Forms from 3600 writers,
  810,000 character images
  isolated from their forms, ground truth classifications for those images, reference forms for further data collection, and software utilities for image management and handling



HANDWRITING SAMPLE FORM

NAME [redacted] DATE 8-3-89 CITY MINDEN CITY STATE MI ZIP 48456

This sample of handwriting is being collected for use in testing computer recognition of hand printed numbe and letters. Please print the following characters in the boxes that appear below.

0 1 2 3 4 5 6 7 8 9 | 0 1 2 3 4 5 6 7 8 9 | 0 1 2 3 4 5 6 7 8 9

0123456789 | 0123456789 | 0123456789

87 | 701 | 3752 | 80759 | 960941

158 | 4586 | 32123 | 832656 | 82

7481 | 80539 | 419219 | 67 | 904

61738 | 729658 | 75 | 390 | 5716

109334 | 40 | 625 | 4234 | 46002

g y x l a k p d s b t z i r u m w f q j e n h o c v

gyxlakpdsbtzirumwfqjenhocv

Z X S B N G E C M Y W Q T K F L U O H P I R V D J A

ZXSBNGECMYWQTKFLUOHPIRVDJA

Please print the following text in the box below:
We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domesti Tranquility, provide for the common Defense, promote the general Welfare, and secure the Blessings of Liberty t ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domestic Tranquility, provide for the common Defense, promote the general Welfare, and secure the Blessings of liberty to ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

Extended MNIST (EMNIST)

- www.nist.gov/itl/products-and-services/emnist-dataset
- en.wikipedia.org/wiki/MNIST_database#Further_versions
- "newer dataset developed and released by NIST to be the (final) successor to MNIST, released in 2017"
- arxiv.org/pdf/1702.05373v1

EMNIST: an extension of MNIST to handwritten letters

Gregory Cohen, Saeed Afshar, Jonathan Tapson, and André van Schaik
The MARCS Institute for Brain, Behaviour and Development
Western Sydney University
Penrith, Australia 2751

Extended MNIST (EMNIST)

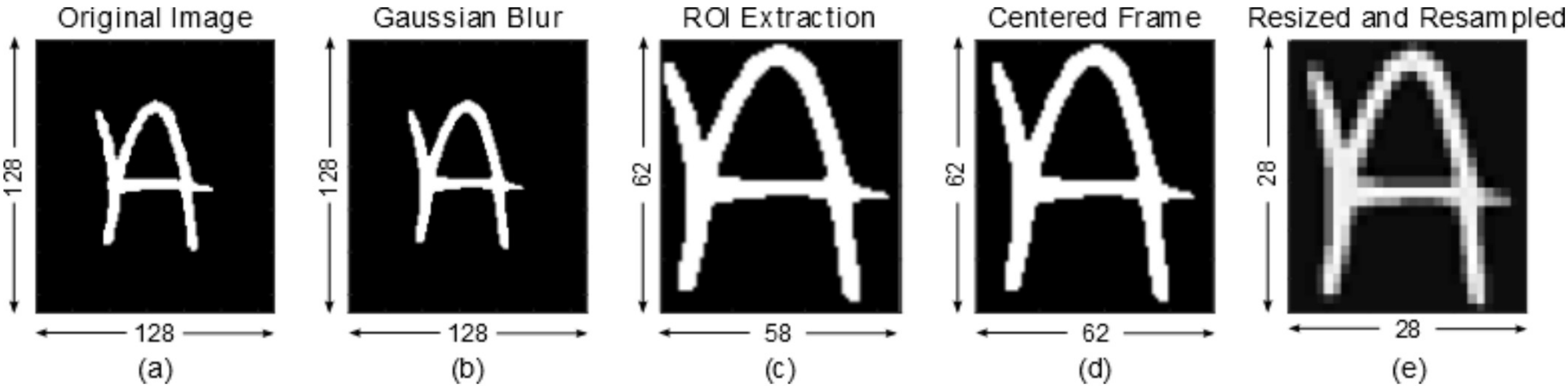


Fig. 1. **Diagram of the conversion process used to convert the NIST dataset.** The original images are stored as 128×128 pixel binary images as shown in (a). A Gaussian filter with $\sigma = 1$ is applied to the image to soften the edges as shown in (b). As the characters do not fill the entire image, the region around the actual digit is extracted (c). The digit is then placed and centered into a square image (d) with the aspect ratio preserved. The region of interest is padded with a 2 pixel border when placed into the square image, matching the clear border around all the digits in the MNIST dataset. Finally, the image is down-sampled to 28×28 pixels using bi-cubic interpolation. The range of intensity values are then scaled to $[0, 255]$, resulting in the 28×28 pixel gray-scale images shown in (e).

Street View House Numbers (SVHN) Dataset (In The Wild)

- 630,420 digits with bounding boxes in house numbers captured in Google Street View
- ufldl.stanford.edu/housenumbers
- Much more challenging to recognize than MNIST or USPS

More on PCA

- arxiv.org/pdf/1404.1100

A Tutorial on Principal Component Analysis

Jonathon Shlens*

Google Research
Mountain View, CA 94043

(Dated: April 7, 2014; Version 3.02)

Principal component analysis (PCA) is a mainstay of modern data analysis - a black box that is widely used but (sometimes) poorly understood. The goal of this paper is to dispel the magic behind this black box. This manuscript focuses on building a solid intuition for how and why principal component analysis works. This manuscript crystallizes this knowledge by deriving from simple intuitions, the mathematics behind PCA. This tutorial does not shy away from explaining the ideas informally, nor does it shy away from the mathematics. The hope is that by addressing both aspects, readers of all levels will be able to gain a better understanding of PCA as well as the when, the how and the why of applying this technique.

More on PCA

- arxiv.org/abs/1804.02502

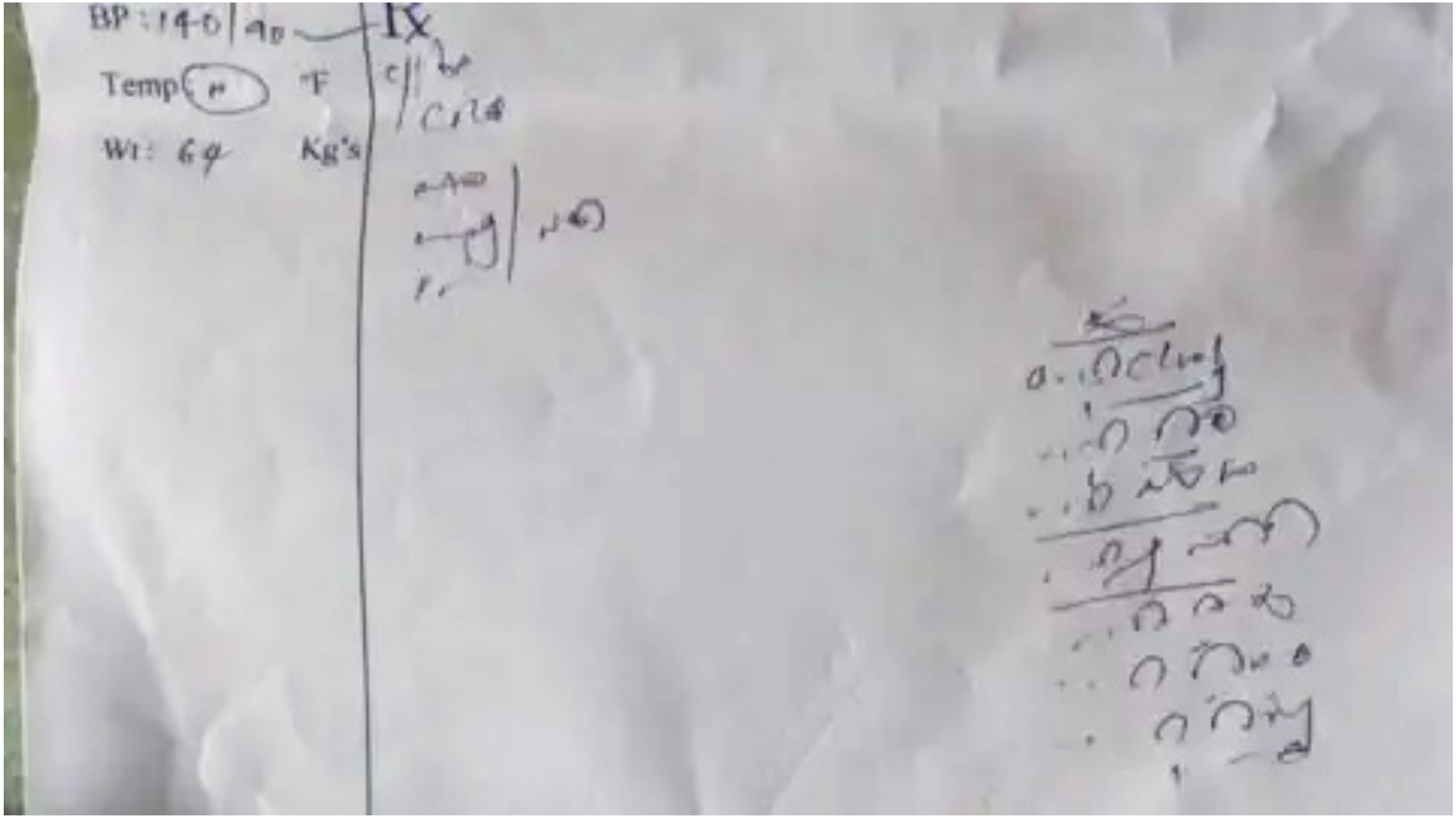
Principal Component Analysis: A Natural Approach to Data Exploration

Authors: Felipe L. Gewers, Gustavo R. Ferreira, Henrique F. De Arruda, Filipi N. Silva, Cesar H. Comin, Diego R. Amancio, Luciano Da F. Costa | Authors Info & Claims

ACM Computing Surveys (CSUR), Volume 54, Issue 4 • Article No.: 70, Pages 1 - 34
https://doi.org/10.1145/3447755

- en.wikipedia.org/wiki/Principal_component_analysis

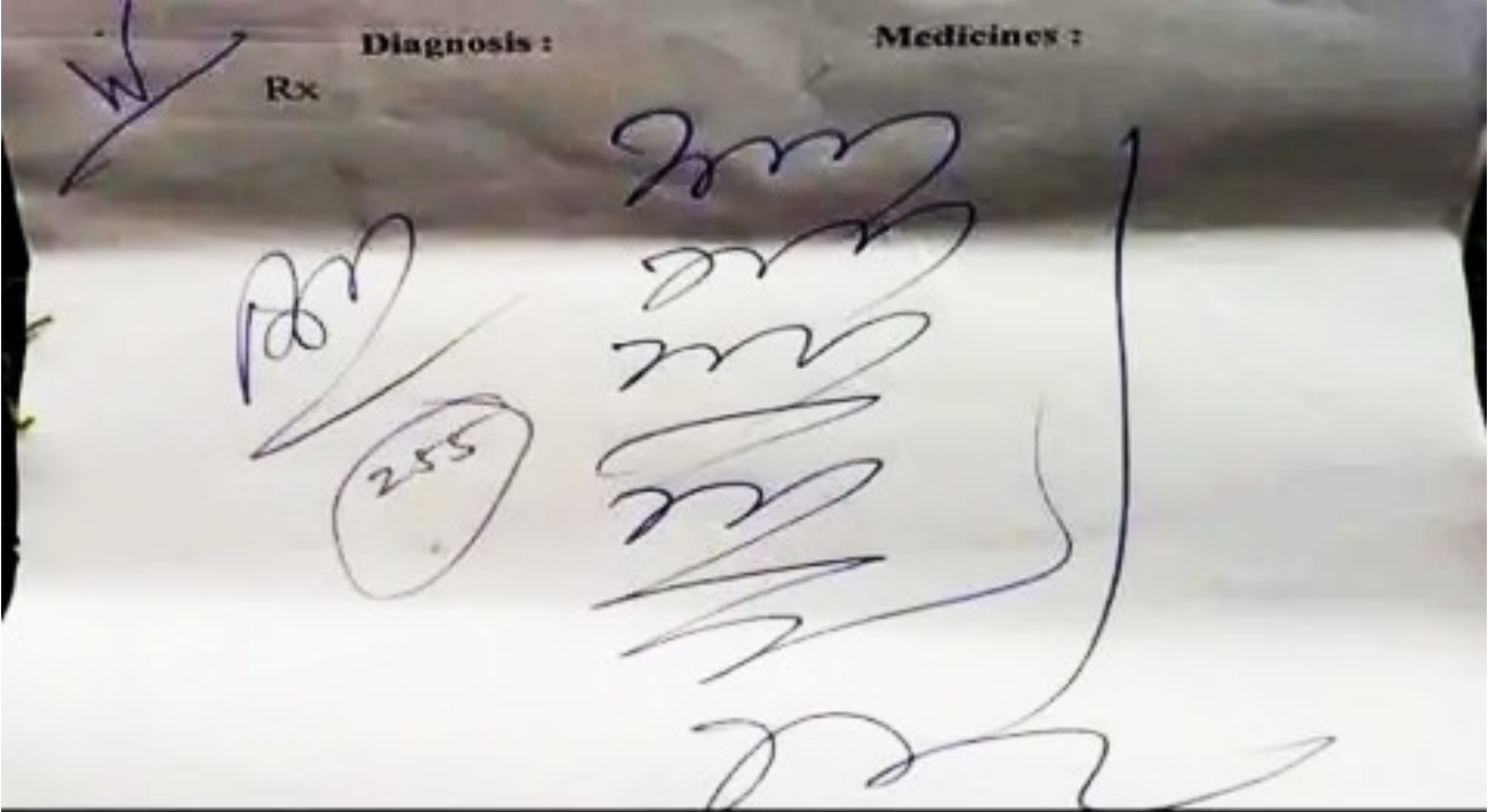
Beyond the Wild (Out of the World)



Diagnosis :

Medicines :

Rx



Beyond the Wild (Out of the World)

- Punjab and Haryana High Court has ruled:
  "legible medical prescription is a fundamental right"

BBC

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No more scribbling: Indian court tells doctors to fix their handwriting

7 days ago

Share

Geeta Pandey BBC News, Delhi

Beyond the Wild (Out of the World)

- “Bad handwriting by doctors is responsible for 7000 deaths every year.”
  - Worldwide

2018:31:1;47-48
doi: 10.4103/0970-258X.243417
PMID: 30348927

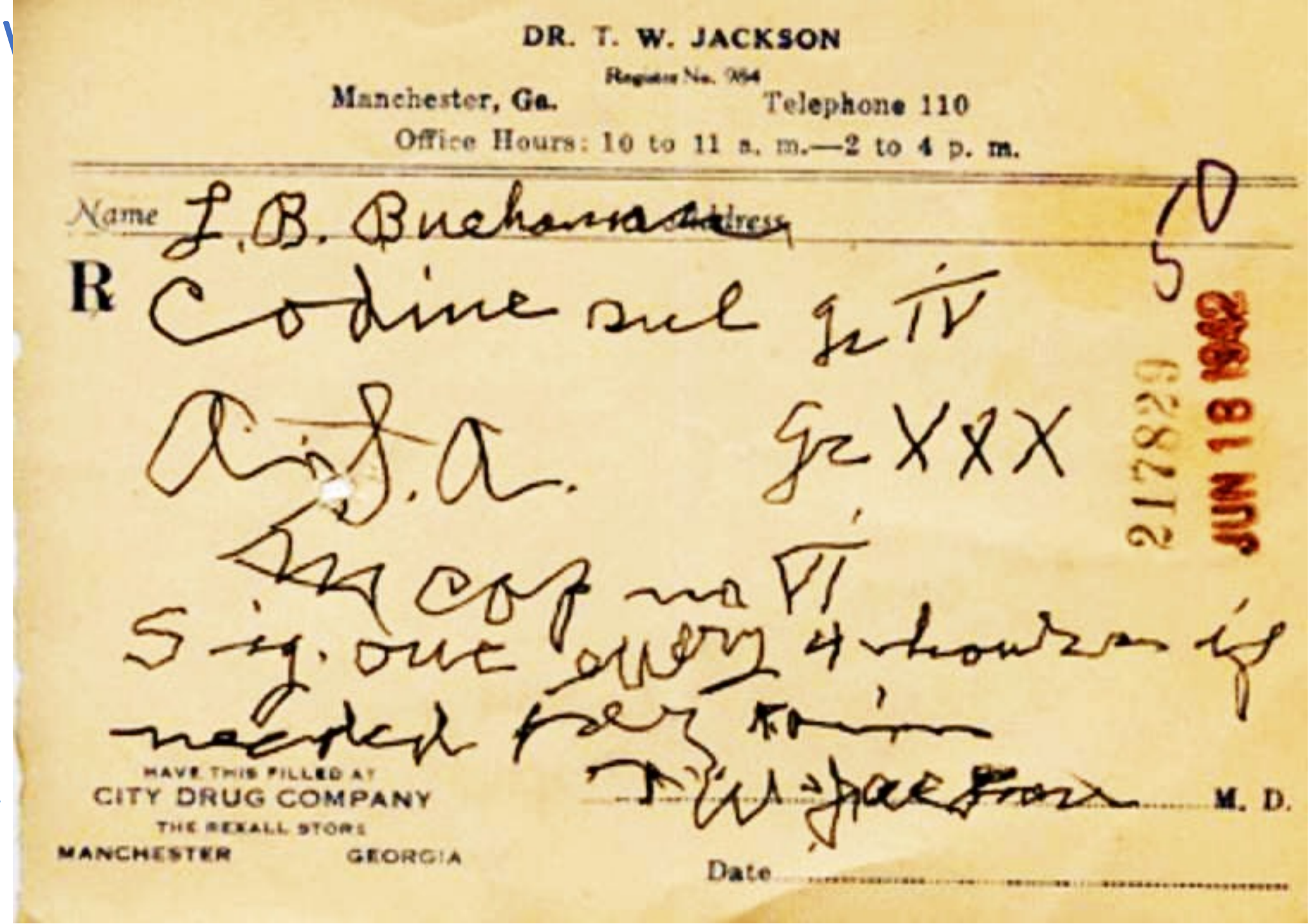
The National Medical Journal of India
on the frontline of Indian Medicine

What’s wrong with doctors’ handwriting?

Santosh K Chaturvedi
Dean, Behavioual Sciences and Department of Psychiatry, National Institute of Mental Health and Neurosciences, Hosur Road, Bengaluru 560029, Karnataka, India

Beyond the W

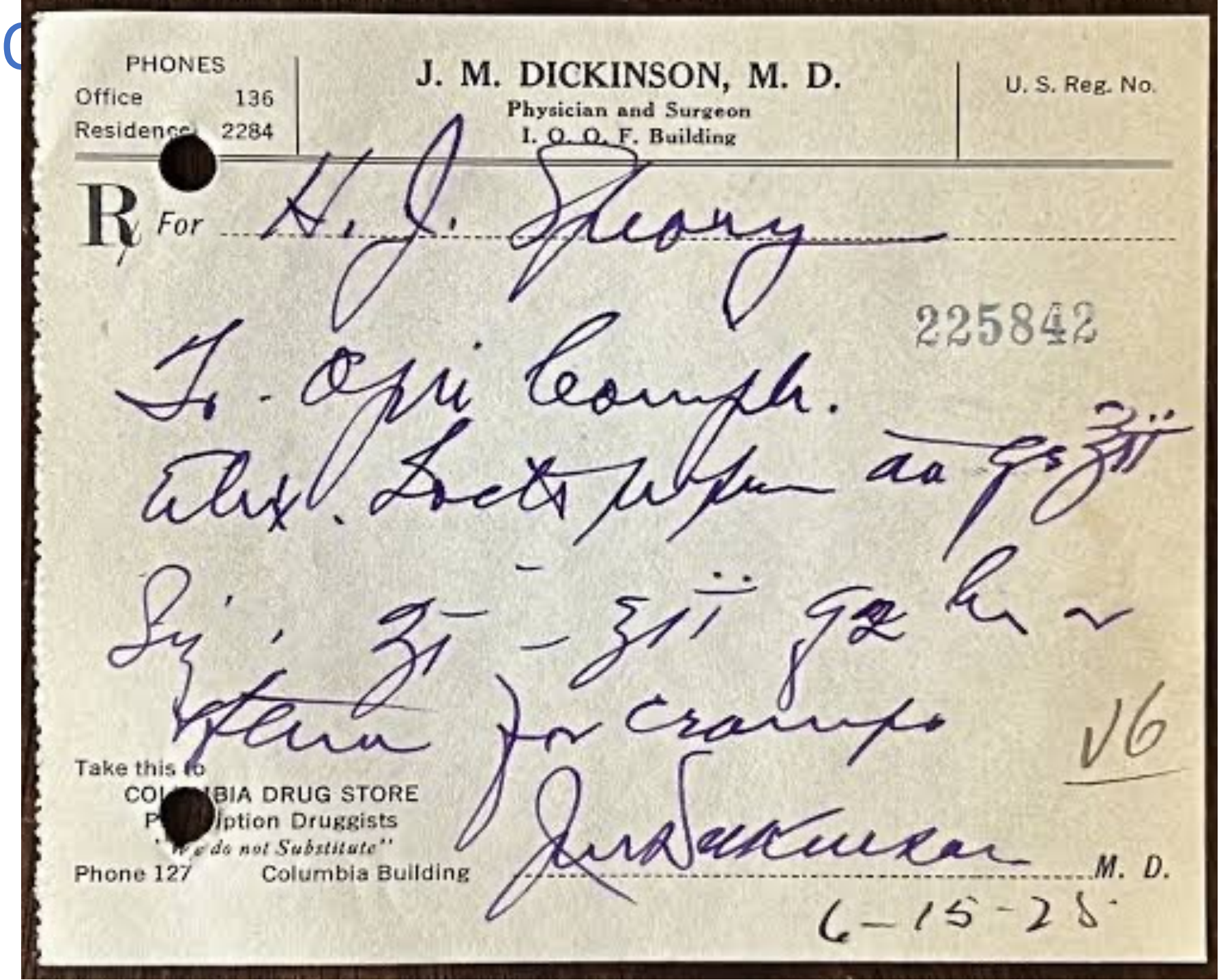
- USA
- 1942



boingboing.net/2022/12/22/google-tries-to-decipher-doctors-bad-handwriting.html

Beyond the Wild (C

- USA
- 2025


PHONES
Office 136
Residence 2284

J. M. DICKINSON, M. D.
Physician and Surgeon
I. O. O. F. Building

U. S. Reg. No.

℞ For H. J. Storey

225842

T. - opii camph.
Aut. Lact. pulv. aa gr. ii
Sig. ii - iii q2h p.r.n.
for cramps

V6

Take this to
COLUMBIA DRUG STORE
Prescription Druggists
"We do not Substitute"
Phone 127 Columbia Building

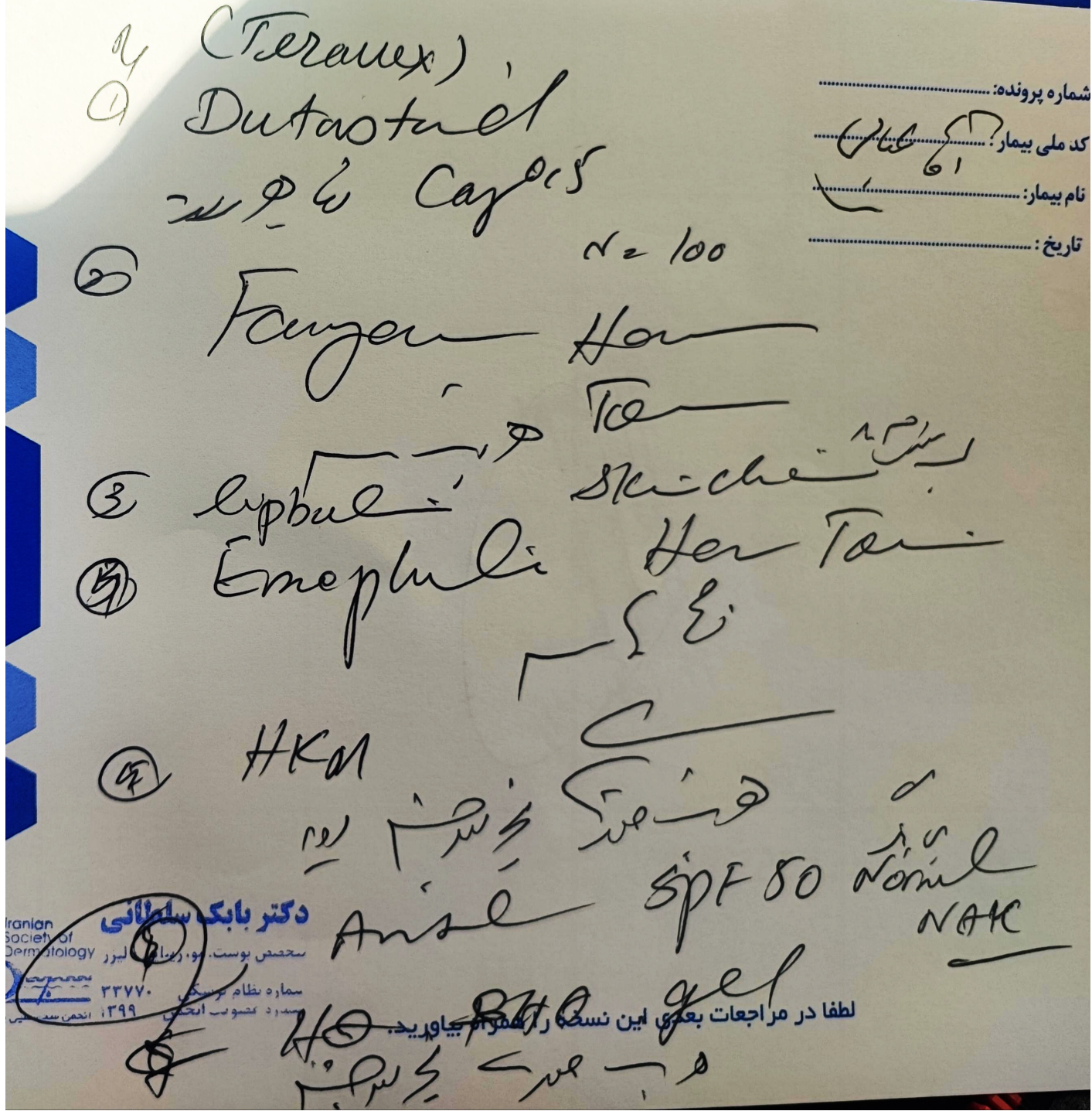
J. M. Dickinson M. D.

6-15-28

www.medboundhub.com/t/google-ai-to-help-interpret-doctors-handwriting/5464

Beyond the Wild (Out of th

- Iran



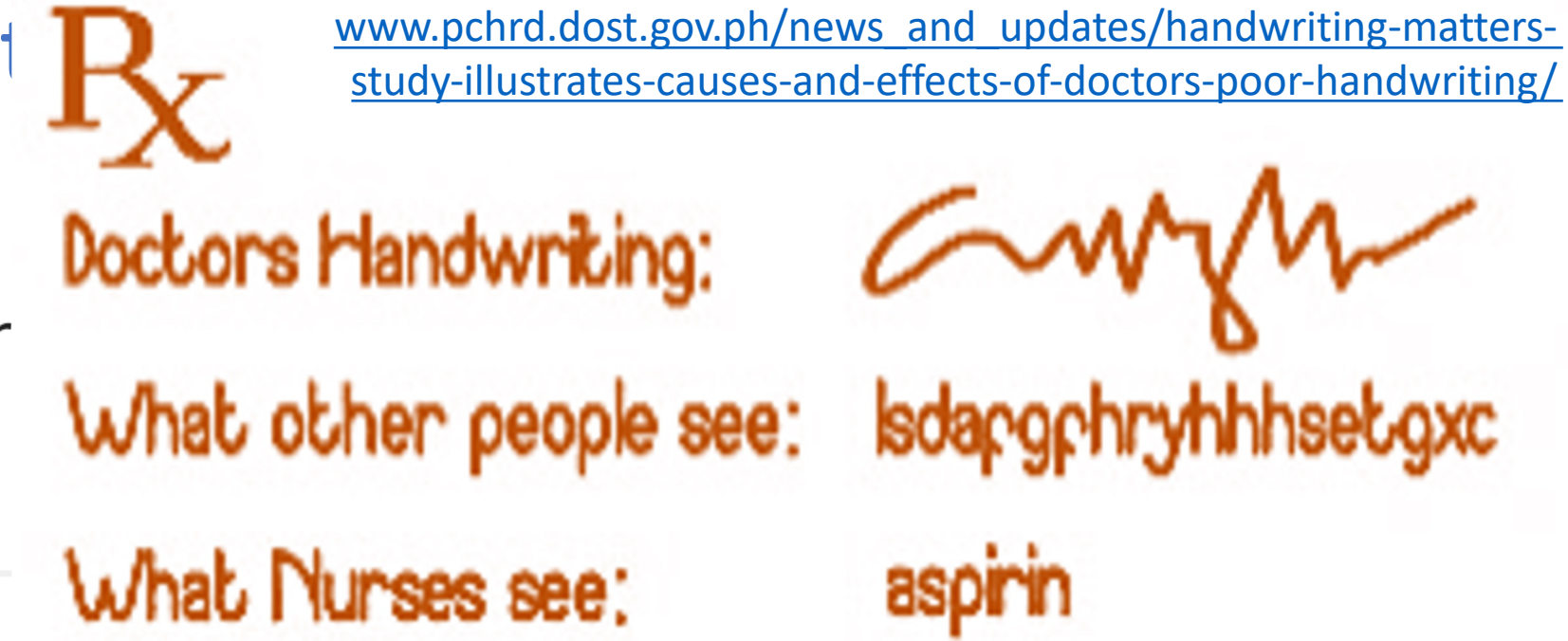
www.reddit.com/r/ENGLISH/comments/1h6dbs6/help_reading_doctors_prescription/#lightbox

Beyond the Wild (Out

www.pchrd.dost.gov.ph/news_and_updates/handwriting-matters-study-illustrates-causes-and-effects-of-doctors-poor-handwriting/

- Philippines

Pr

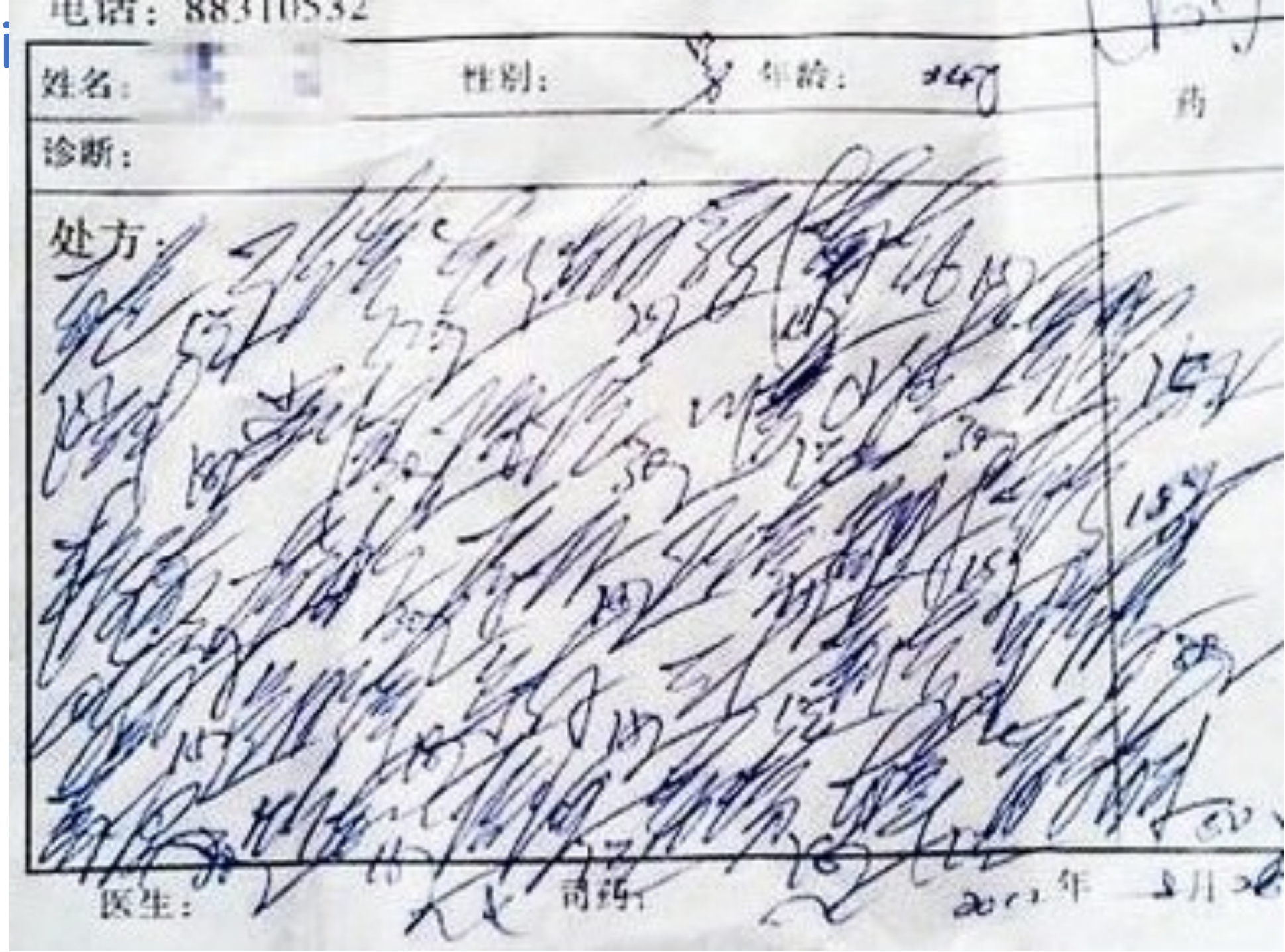


Assessment of the Legibility of the Handwriting in Medical Prescriptions of Doctors from Public and Private Hospitals in Quezon City, Philippines ☆

Art Angelo P. Cerio, Nikole Andrei Louise B. Mallare, Rafael Marco S. Tolentino

Beyond the Wi

- China



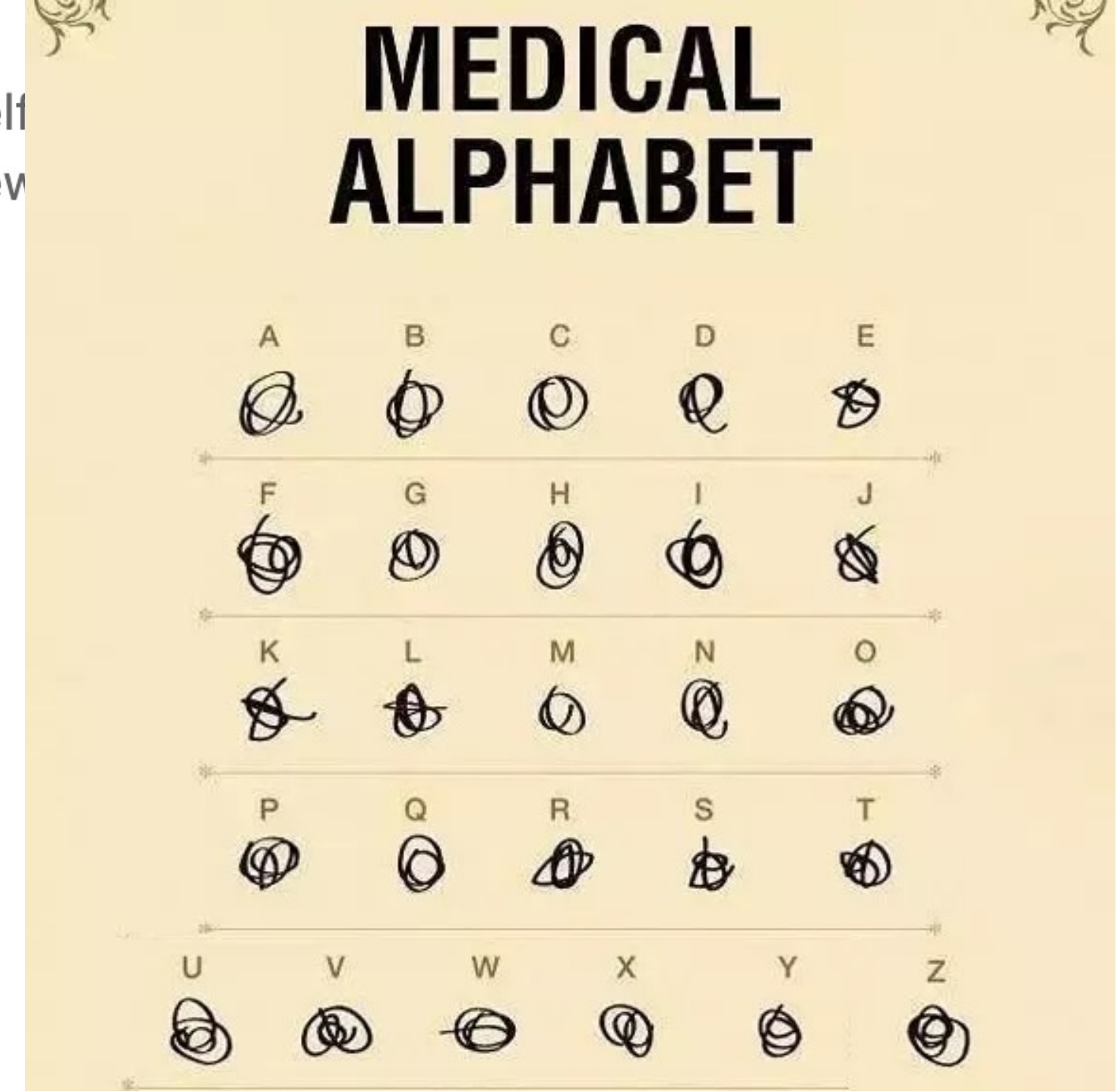
www.ecns.cn/cns-wire/2016-03-23/detail-ifytmzvw2133442.shtml

William Owen

Former Physician (M.D.) at Self

answers and **2.2M** answer view

- www.quora.com/Why-do-doctors-write-in-such-a-way-that-it-is-difficult-to-read-for-a-normal-person



Beyon

EVOLUTION OF HANDWRITING

COLLEGE:
The quick brown fox jumps over the lazy dog.

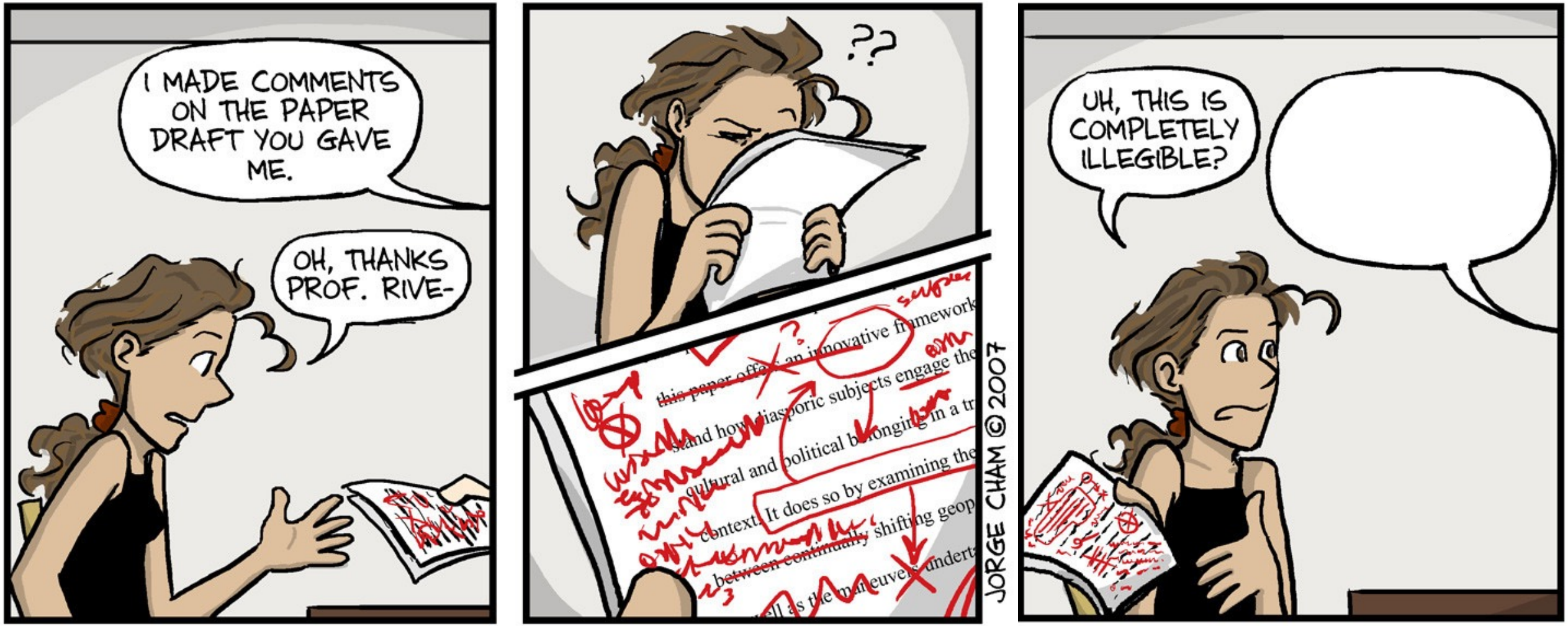
MEDICAL SCHOOL:
The quick brown fox jumps over the lazy dog.

RESIDENCY:
The quick brown fox jumps over lazy dog.
Wisdom 1-2 tab PO q 4 hrs PRN pain

ATTENDING:
See resident note.

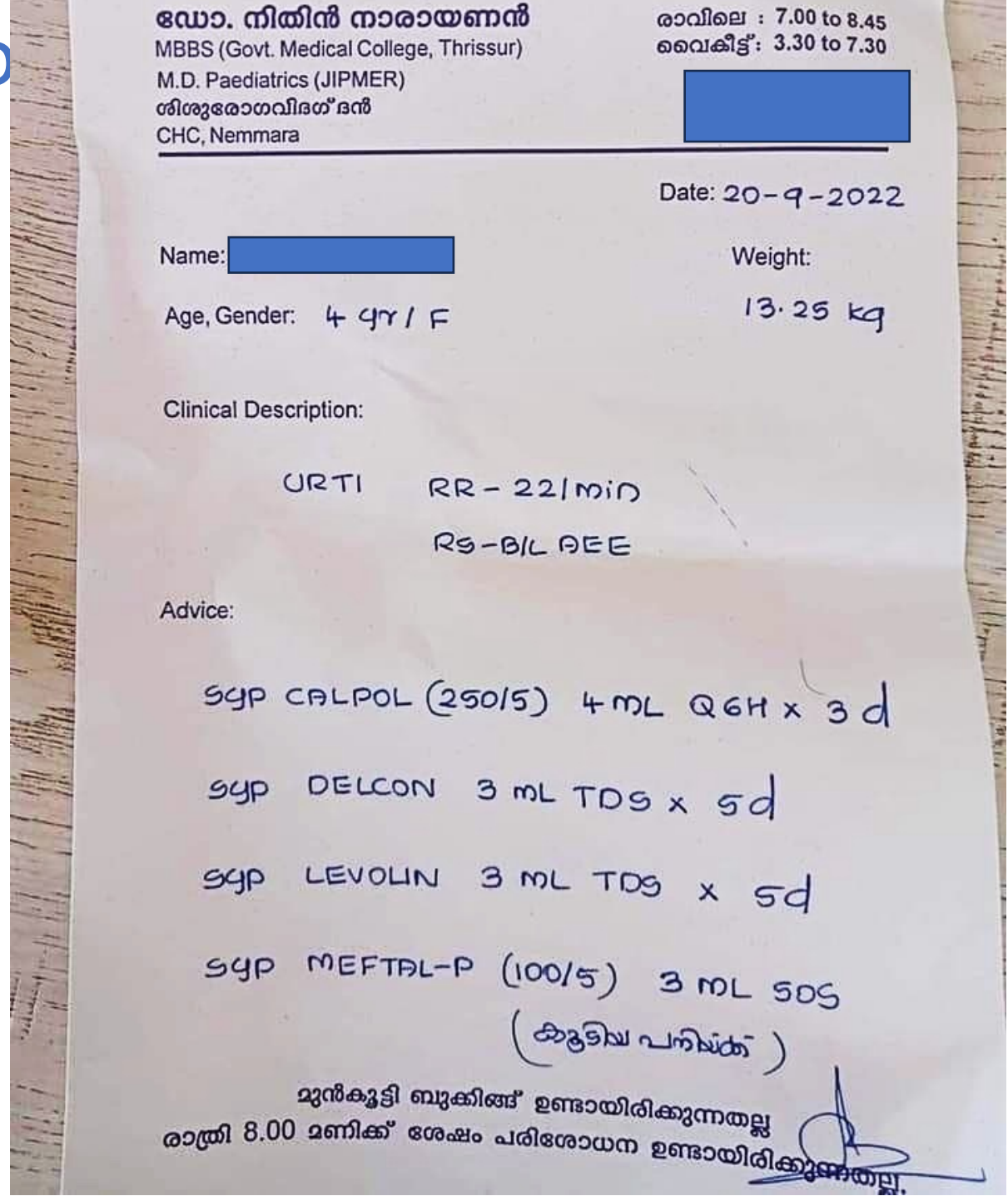
Beyond the Wild (Out of the World)

- PhD Comics



Beyond the Wild (Out of the Wo

- www.livemint.com/news/kerala-doctor-s-neat-handwriting-goes-viral-11664381338471.html



ഡോ. നിതിൻ നാരായണൻ
MBBS (Govt. Medical College, Thrissur)
M.D. Paediatrics (JIPMER)
ശിശുരോഗവിദഗ്ദൻ
CHC, Nemmara

രാവിലെ : 7.00 to 8.45
വൈകീട്ട്: 3.30 to 7.30

[redacted]

Date: 20-9-2022

Name: [redacted]

Weight: 13.25 kg

Age, Gender: 4 yr / F

Clinical Description:

URTI RR - 22/min
RS - B/L AEE

Advice:

Syp CALPOL (250/5) 4 ml Q6H x 3 d

Syp DELCON 3 ml TDS x 5d

Syp LEVOLIN 3 ml TDS x 5d

Syp MEFTAL-P (100/5) 3 ml SOS
(കുടിച്ച പനിക്ക്)

മുൻകൂട്ടി ബുക്കിങ്ങ് ഉണ്ടായിരിക്കുന്നതല്ല
രാത്രി 8.00 മണിക്ക് ശേഷം പരിശോധന ഉണ്ടായിരിക്കുന്നതല്ല.